

Supplementary Information

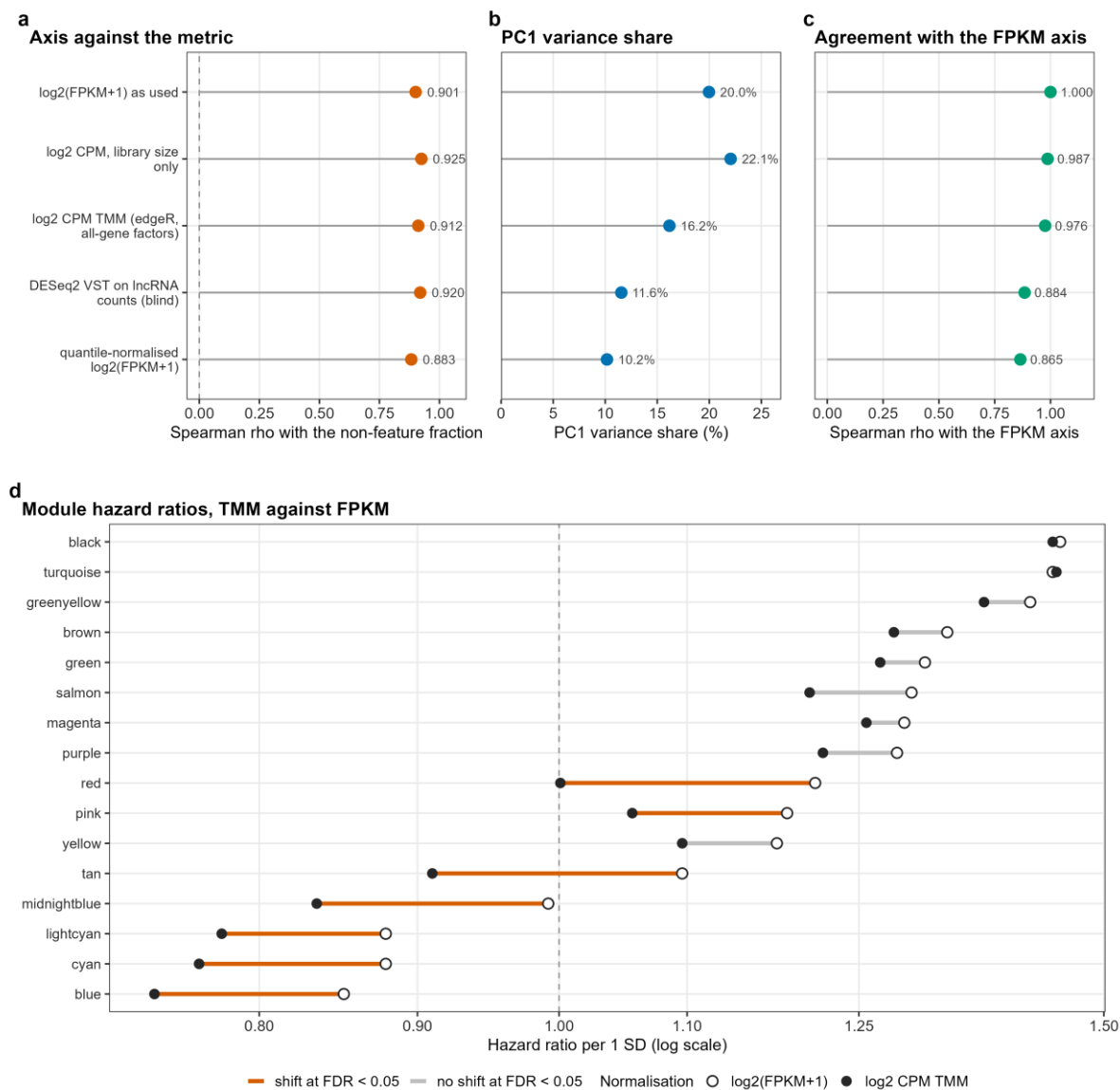
Non-exonic read content reorganises lncRNA co-expression networks in renal and other cancers

Gehad Youssef and Namshik Han

Supplementary Figures 1 to 13, Supplementary Tables 1 to 22 and the descriptions of Supplementary Data 1 to 5 follow the order of citation in the main article. Each value comes from the results file named in the note beneath its table. Superscript numbers refer to the main article's reference list, and Gene Ontology term names are quoted as published. A double hyphen in a table cell means not applicable. The non-feature read fraction is the percentage of reads that STAR assigns to no annotated feature, out of the unmapped, multimapping, non-feature, ambiguous and gene-assigned reads in its gene-counts summary. Hazard ratios are per standard deviation of the named quantity, with 95% confidence intervals. The clinical model comprises age, sex, T category, nodal status, M1 and ordinal grade. The augmented model adds the ESTIMATE stromal and immune scores and the three STAR metrics (non-feature fraction, multimapping fraction and log10 assigned reads). The principal specification for module models enters the module eigengene with the augmented covariates. Adjustment for the alignment metrics means residualisation of the expression matrix on the three STAR metrics before module definition. Benjamini-Hochberg control applies within the family stated in each note. The completed TRIPOD+AI checklist accompanies the article as a separate file.

Supplementary Figures

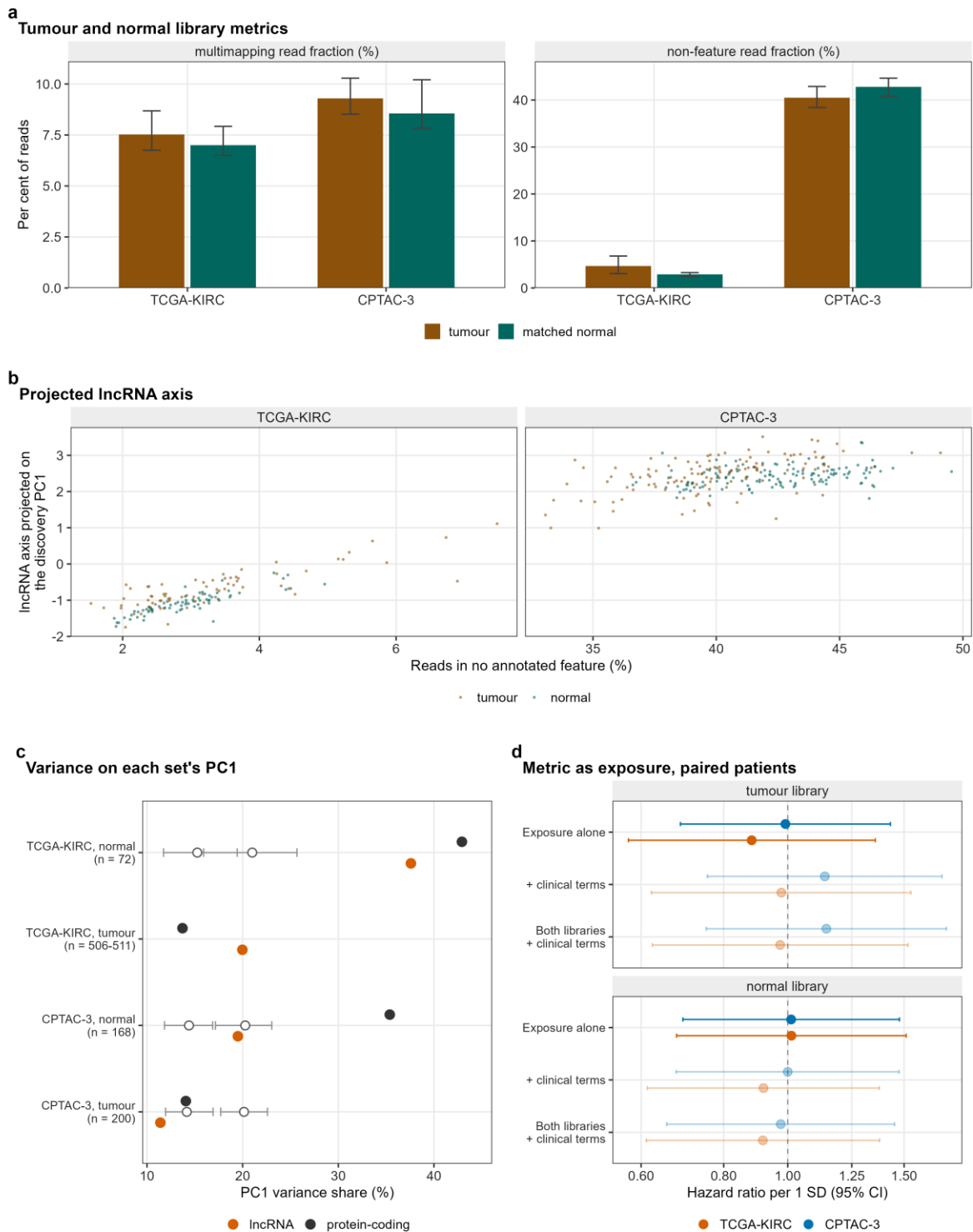
24 **Supplementary Fig. 1 | The axis under five normalisations.**



25 **a** Spearman correlation of the leading lncRNA principal component with the non-feature fraction
26 under five normalisations of the same 3,442 genes and 511 libraries. The five arms are
27 log2(FPKM + 1) as used throughout, log2 counts per million with library size only or with
28 edgeR TMM factors, a blind DESeq2 variance-stabilising transform, and quantile-normalised
29 log2(FPKM + 1). **b** PC1 variance share. Each PC1 is oriented against the log2(FPKM + 1) axis.
30 **c** Spearman correlation of each arm's PC1 with the FPKM axis. Quantile normalisation removes

the per-sample mean, leaving that arm's sign anchor indeterminate, so it is quoted as a magnitude. **d** Paired patient bootstrap difference in log hazard ratio between the TMM and FPKM scales for the 16 lncRNA modules (2,000 resamples, 504 patients, 163 deaths), with the same modules, patients and covariates on both scales. With Benjamini-Hochberg control within each arm, 7 of 16 differences reach a false discovery rate below 0.05 on the observed matrix and 0 of 16 on the residualised matrix; before multiplicity control, 9 and 2 intervals exclude zero. Source tables: 29_normalisation_pc1.tsv, 29_normalisation_module_contrast.tsv.

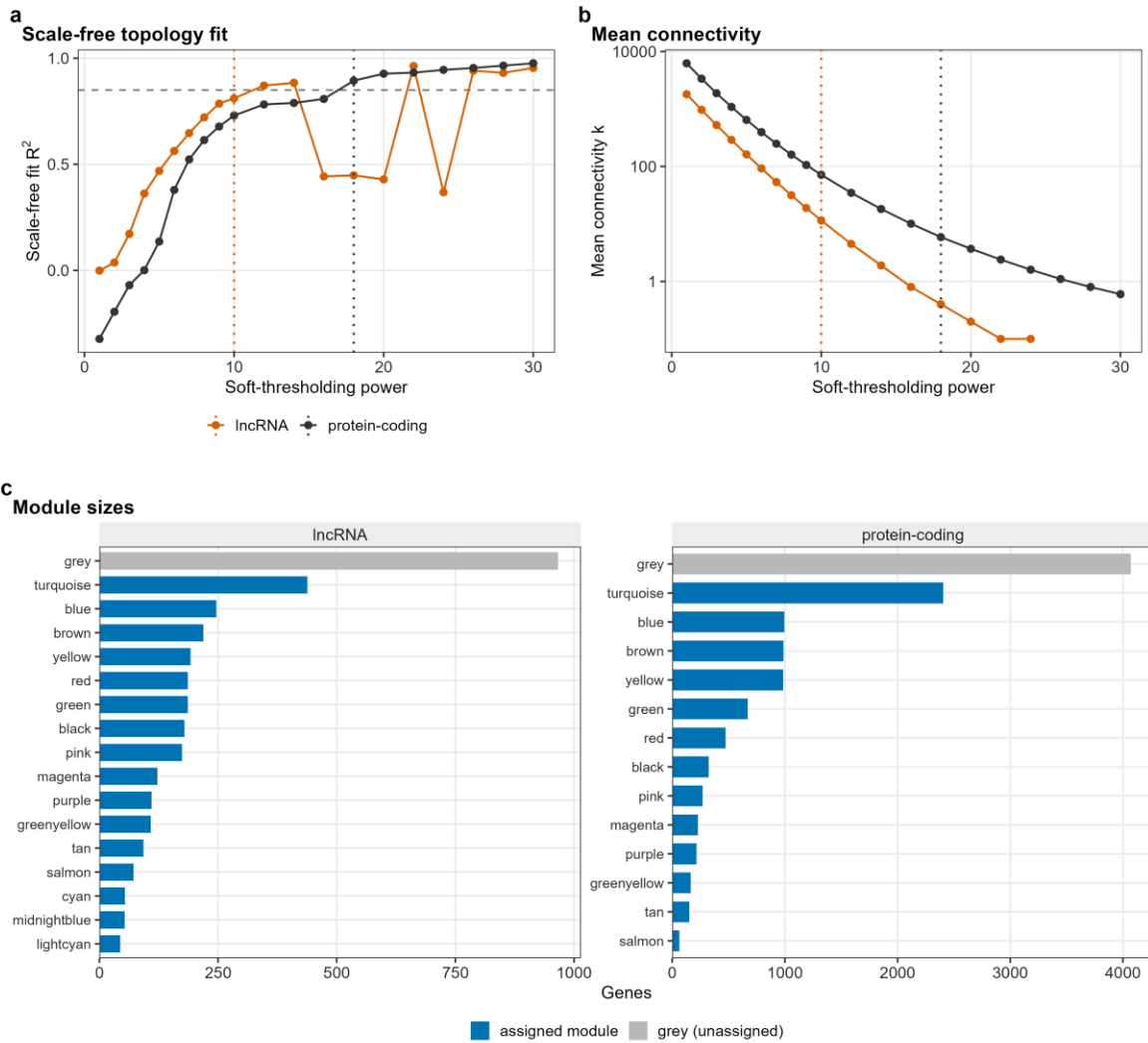
Supplementary Fig. 2 | Matched normal-kidney libraries.



40 **a** Median and interquartile range of the multimapping and non-feature fractions in tumour and
 41 matched normal libraries of TCGA-KIRC (528 tumour, 72 normal) and CPTAC-3 (200 tumour,

168 normal). All four unpaired rank-sum tests reach a false discovery rate below 0.05, but the unpaired contrast mixes tissue with plate. In TCGA-KIRC the non-feature difference falls from 1.73 percentage points unpaired to 0.16 on the 68 plate-matched normal libraries (p 0.049), to 0.15 for the normals of the analysed tumours on shared plates (144 tumour and 67 normal libraries, p 0.057), and to 0.15 within patient (71 pairs, signed-rank p 0.10). **b** Projected lncRNA axis (the discovery lncRNA PC1 direction applied to each sample set) against the non-feature fraction in the paired libraries, faceted by cohort and coloured by tissue type: rho 0.77 in TCGA-KIRC tumours and 0.86 in their normals (71 pairs), and 0.54 and 0.35 in CPTAC-3 (129 pairs). **c** PC1 variance share of each sample set's own lncRNA and protein-coding matrix against a size-matched discovery baseline from 200 resampling draws (open grey points and bars, the median and range of the draws, available for the six non-discovery rows). The baseline range is not a confidence interval. Three of the eight observed shares lie above it. A share carries no sign, and the orientation of three of the eight components is a convention. **d** Forest plot of the non-feature fraction (linear scale, per standard deviation), measured on the tumour or the normal library, as exposure for overall survival in the paired patients under three models. Faded rows have an events-per-variable ratio below 5, as does every adjusted model in the panel (71 patients and 26 events in TCGA-KIRC, 129 and 28 in CPTAC-3). Source tables: 25_tumour_vs_normal_distribution.tsv, 25_paired_scores_per_sample.tsv, 25_normal_axis_projection.tsv, 25_normal_metric_survival.tsv.

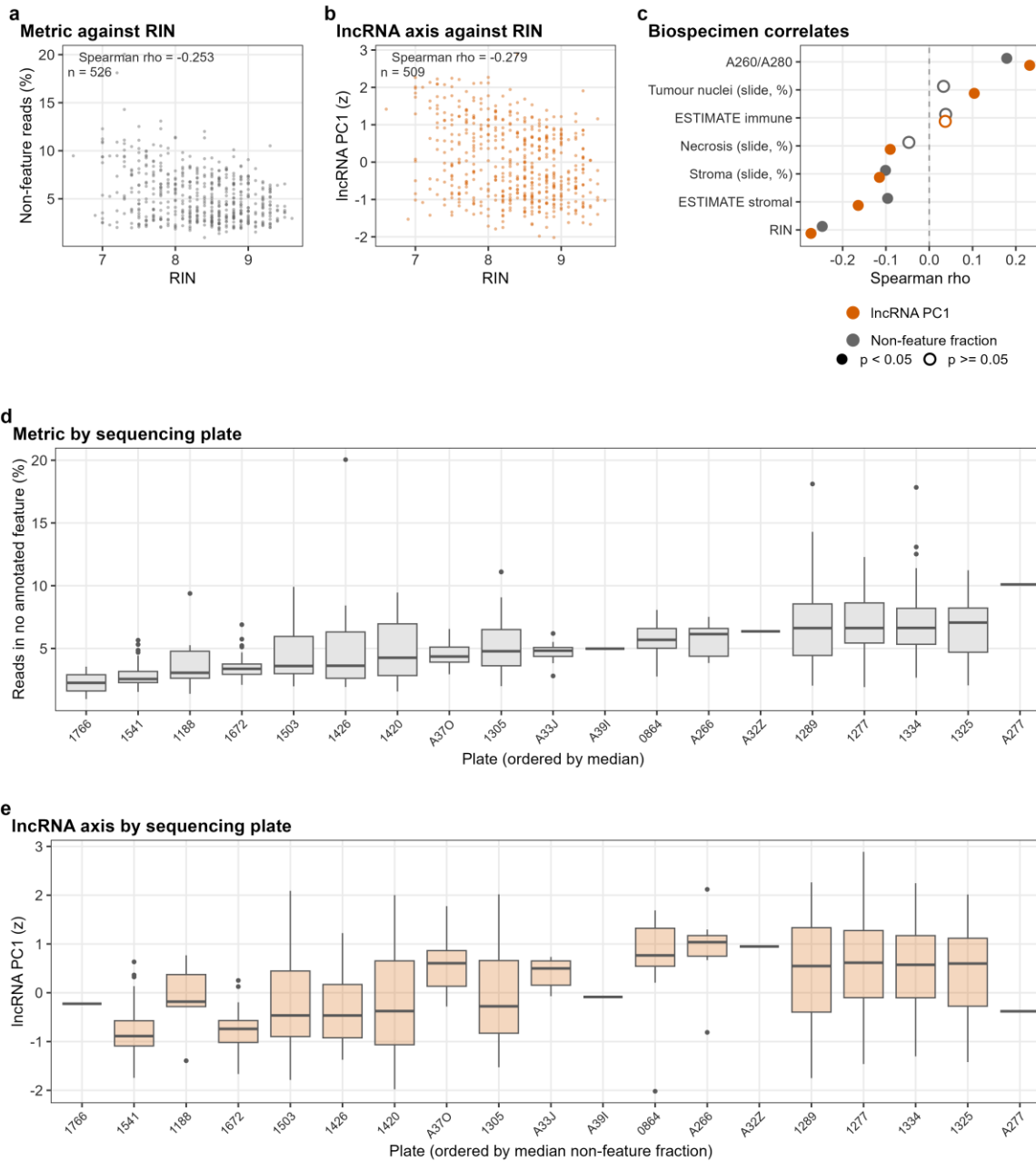
Supplementary Fig. 3 | Network construction diagnostics.



a Signed scale-free topology fit R -squared against soft-thresholding power for the protein-coding (12,000 genes, 506 samples) and lncRNA (3,442 genes, 511 samples) networks. The horizontal line marks the target of 0.85 and the vertical lines the chosen powers, 18 and 10, at which the signed fit is 0.894 and 0.811. The protein-coding power is the lowest reaching the target. The outcome-blind parameter sweep (Methods) fixed the lncRNA power at 10, where the automatic rule would have given 12. **b** Mean connectivity against power on a logarithmic axis, with the same chosen-power lines. **c** Module sizes, with unassigned (grey) genes shown separately. The protein-coding network resolves 13 non-grey modules, leaving 4,069 of 12,000 genes

70 unassigned, and the lncRNA network 16, leaving 966 of 3,442. Source tables:
71 02_lncRNA_soft_threshold.tsv, 02_mRNA_soft_threshold.tsv,
72 02_network_summary.tsv, 02_lncRNA_module_sizes.tsv,
73 02_mRNA_module_sizes.tsv.

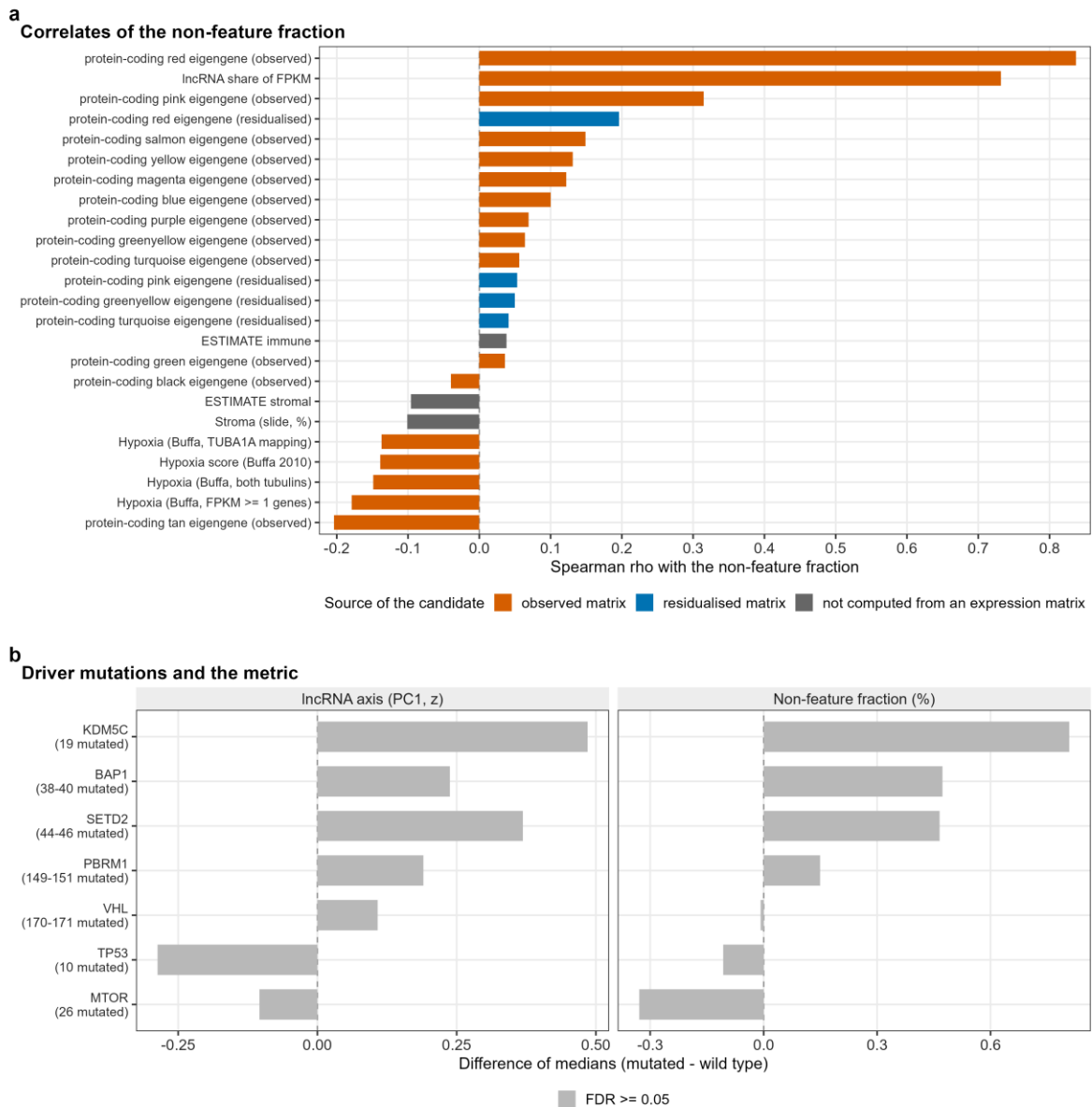
74 **Supplementary Fig. 4 | RNA integrity number and sequencing plate.**



75 **a** Non-feature read fraction against RNA integrity number in TCGA-KIRC, with the integrity
 76 number from the Biospecimen Core Resource biospecimen record. One library with an RNA
 77 integrity number recorded as 0, 6.6 units below the next lowest, is treated as a coded missing
 78 value and excluded from panels **a** and **b**, so their sample size is one below the correlation table's.
 79 **b** Observed IncRNA PC1, the axis of Fig. 2a, against RNA integrity number, with the same

80 exclusion. **c** Spearman correlation of the non-feature fraction and the lncRNA axis with seven
81 biospecimen measures: RNA integrity number, A260/A280 ratio, slide percent necrosis, tumour
82 nuclei and stromal cells, and the ESTIMATE stromal and immune scores (507 to 528 patients).
83 Filled symbols mark the 10 of 14 tests with unadjusted p below 0.05. **d** Non-feature fraction by
84 sequencing plate, ordered by median, for 528 libraries on 19 plates (1 to 59 per plate). **e** lncRNA
85 axis by plate, in the same order, for the 511 libraries with an axis score. Biospecimen and batch
86 metadata exist only for TCGA-KIRC. Source tables: `23_biospecimen_kirc.tsv`,
87 `23_axis_vs_biospecimen.tsv`, `27_axis_scores_per_sample.tsv`.

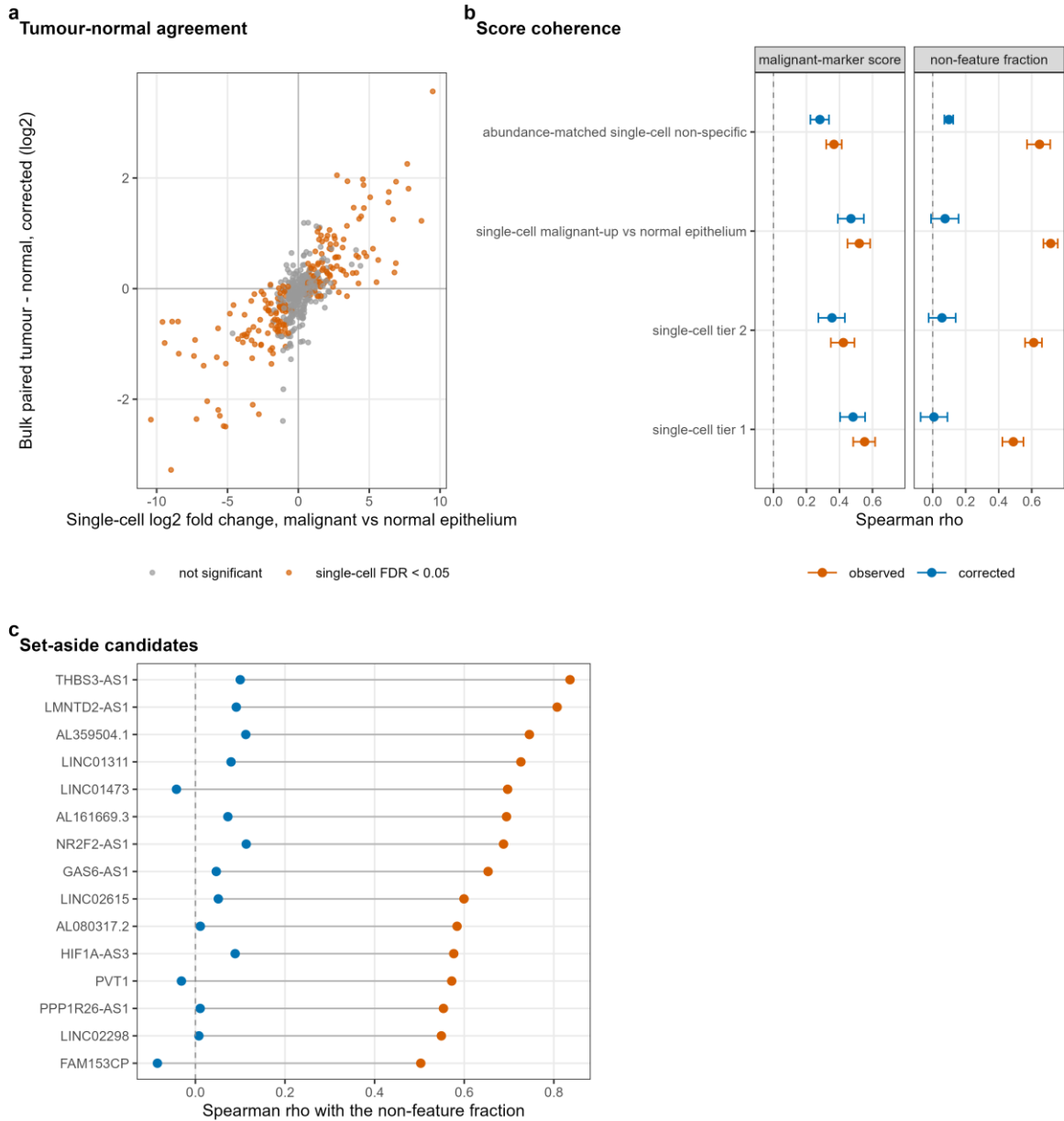
88 **Supplementary Fig. 5 | What the non-feature read fraction correlates with.**



a The 24 strongest candidate correlates of the non-feature fraction in TCGA-KIRC (506 to 528 patients), coloured as eigengenes of the observed or the residualised expression matrix, or quantities computed outside it. Observed eigengenes cannot be read as biology, because the metric was measured on the same matrix. The annotated biological candidates are the hypoxia metagene, the proliferation score, the ESTIMATE stromal and immune scores and the slide tumour-nuclei percentage. **b** Difference of medians in the non-feature fraction and the lncRNA

95 axis between mutated and wild-type patients for seven recurrent clear-cell drivers. None of the
96 14 tests (7 genes by 2 outcomes) reaches a 5% false discovery rate, on 358 to 368 patients per
97 test and as few as 10 mutated patients. Testing is conditional on the 368 of 528 patients with an
98 open-access masked mutation file, whose median non-feature fraction is lower than that of the
99 others. Source tables: 27_metric_biology_correlations.tsv,
100 28_noFeature_vs_mutations.tsv.

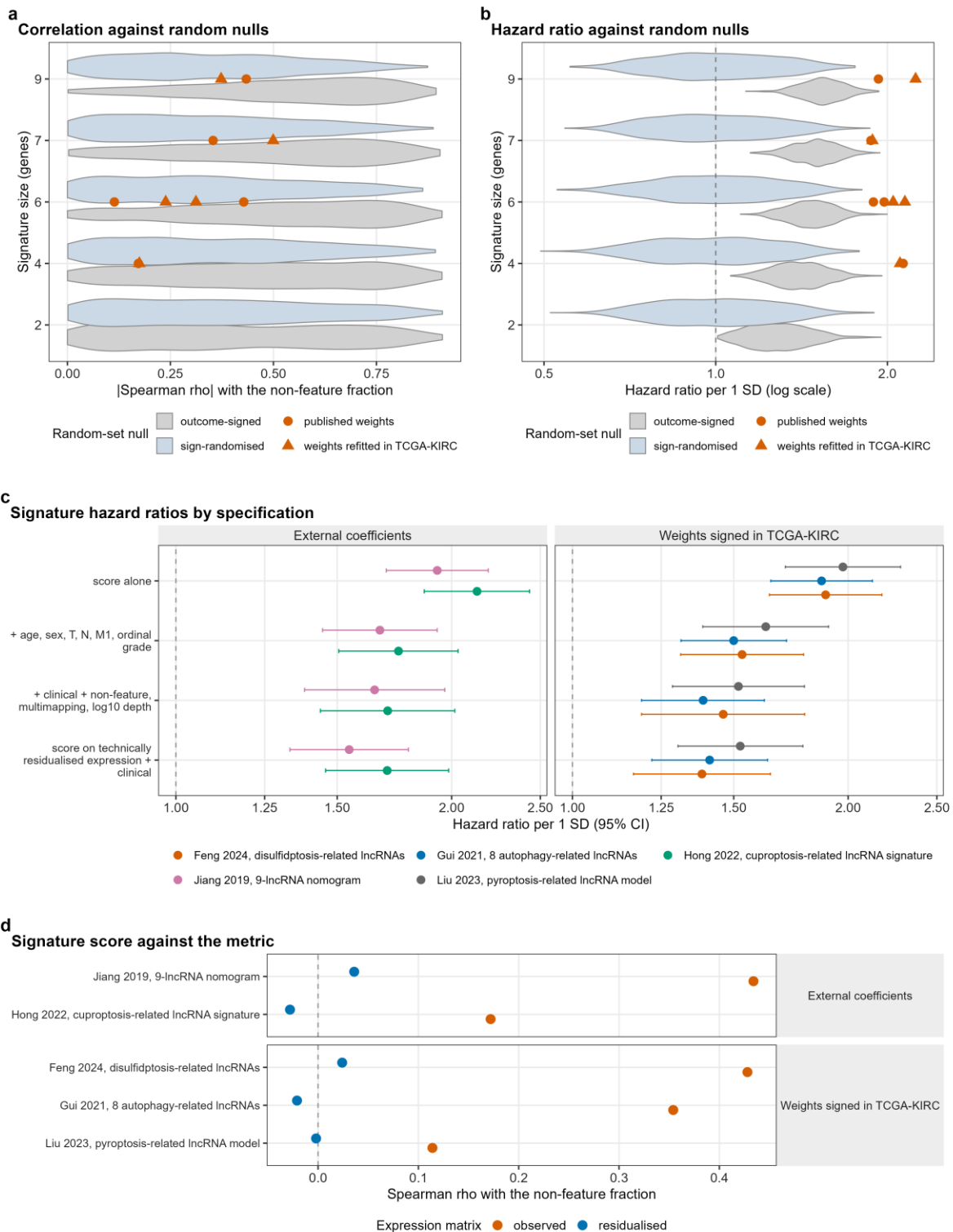
101 **Supplementary Fig. 6 | Single-cell ground truth for the correction.**



a Corrected bulk paired tumour-minus-normal difference against the single-cell fold change of malignant against normal epithelial cells, for 507 lncRNAs in 72 TCGA-KIRC patients. Orange, single-cell false discovery rate below 0.05. Agreement is Spearman rho 0.775 observed and 0.771 corrected, with sign agreement of 92.5% and 92.0% among single-cell-significant lncRNAs. **b** Spearman correlation of lncRNA scores in 532 tumours with the malignant-marker score and the non-feature fraction, observed (orange) and corrected (blue), with 95% bootstrap

108 intervals. The marker score itself correlates with the metric at 0.22. The null row is 1,000
109 abundance-matched sets of single-cell non-specific lncRNAs. Correlation with the metric falls by
110 construction, and the informative quantity is marker coherence relative to the null. **c** Correlation
111 with the non-feature fraction, before and after correction, of the 15 single-cell candidates set
112 aside in bulk as technically sensitive. Source tables: 42_sc_truth_concordance.tsv,
113 42_sc_truth_score_coherence.tsv, 42_sc_truth_technical_candidates.tsv.
114 ---

115 **Supplementary Fig. 7 | Published lncRNA signatures against two random-signature**
116 **nulls.**

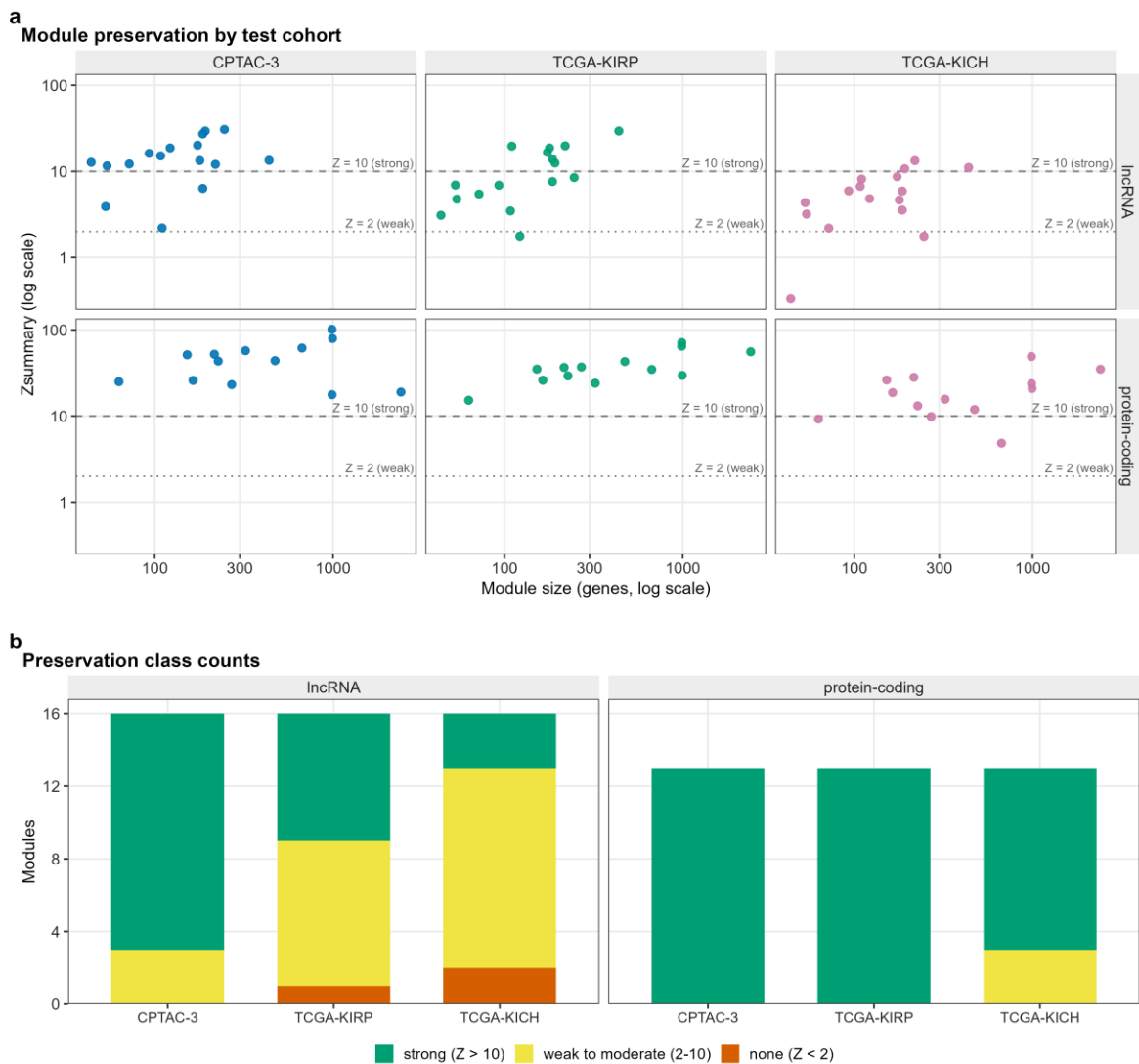


117 **a** Absolute Spearman correlation with the non-feature fraction (violins) for 1,000 random gene
 118 sets at each of five set sizes, under two weightings of the same draws: outcome-signed after

119 Venet and colleagues¹¹, and sign-randomised. The two nulls are the same 5,000 gene sets signed
 120 two ways. Points are the five primary-tier signatures, each with its published weights and with
 121 weights refitted in TCGA-KIRC; the Yu signature is held back for incomplete mapping. Against
 122 the sign-randomised null the signatures sit at percentiles 25.4 to 65.3 for the two with external
 123 coefficients and 19.1 to 64.9 for the three signed in sample, and against the outcome-signed null
 124 at 15.9 to 30.9 and 8.6 to 36.7. **b** Unadjusted hazard ratio per standard deviation (violins) for the
 125 same draws and weightings in 528 patients with 173 deaths, with the signatures overlaid. The
 126 sign-randomised null has a median hazard ratio of 1.00, whereas 95% of outcome-signed sets
 127 reach p below 0.05 by construction. The signatures sit at percentiles 99.8 to 100.0 (external
 128 coefficients) and 99.7 to 99.9 (signed in sample) of the outcome-signed null. **c** Hazard ratio per
 129 standard deviation (95% confidence interval) for the five primary-tier signatures under four
 130 specifications, grouped by weight provenance (520 to 528 patients, 172 to 173 deaths); all 20
 131 intervals exclude 1. Adding the three STAR metrics changes the clinically adjusted log hazard
 132 ratio by -4.9% to -2.6% with external coefficients and -19.0% to -11.1% with weights signed in
 133 sample; residualising the expression instead changes it by -15.0% to -5.1% and -23.6% to -
 134 13.2%. **d** Spearman correlation of each signature score with the non-feature fraction (528
 135 patients), observed against residualised expression, by weight provenance: 0.172 to 0.434
 136 observed with external coefficients and 0.114 to 0.428 signed in sample, and below 0.04 in
 137 absolute value for every signature after residualisation. Outcome-signing itself induces
 138 correlation with the metric in the null, so outcome-signed percentiles cannot show a signature to
 139 be less correlated with the metric than chance. Source tables:
 140 26_random_signature_null.tsv, 26_published_signatures_models.tsv,

26_published_signatures_correlations.tsv,
 26_headline_ranges_by_provenance.tsv.

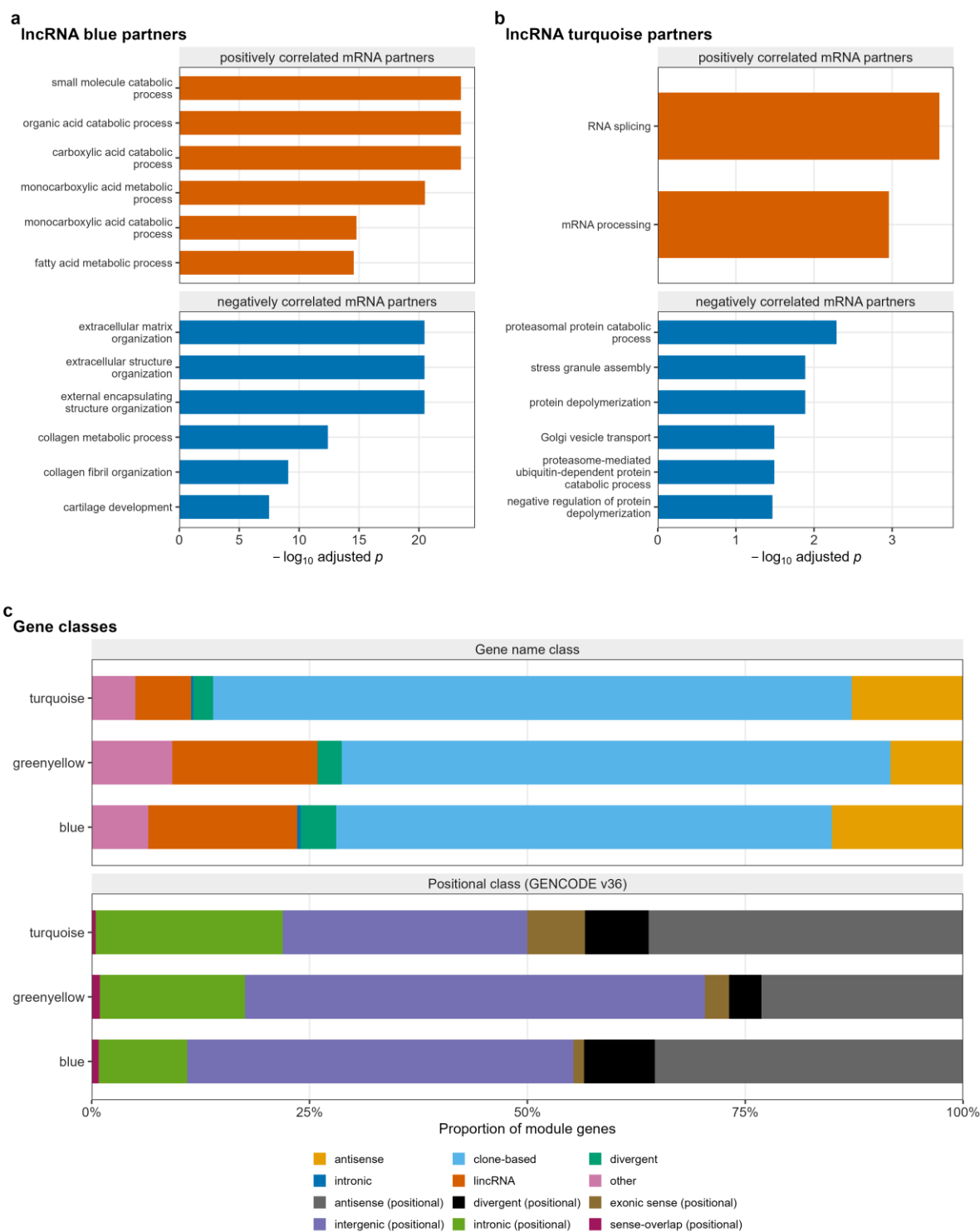
Supplementary Fig. 8 | Module preservation of the discovery networks in the three external cohorts.



a WGCNA Zsummary against discovery module size by network and test cohort, for 29 modules and 87 module-cohort pairs, with lines at the conventional bands of Zsummary 2 and 10. Module size is the true discovery size, because the preservation function caps reported size at 1,000,

148 which affects one module. The reference is the residualised TCGA-KIRC matrix used to define
149 the modules, and each test network is residualised on its own cohort's STAR metrics. The
150 lncRNA network used 100 permutations, the protein-coding network 30. **b** Stacked counts of
151 modules strongly, weakly to moderately or not preserved, per cohort and network. Strongly
152 preserved lncRNA modules number 13 of 16 in CPTAC-3, 7 of 16 in TCGA-KIRP and 3 of 16
153 in TCGA-KICH, and protein-coding modules 13 of 13, 13 of 13 and 10 of 13. The gold and grey
154 rows of the underlying tables are WGCNA internal reference sets, not modules. Source tables:
155 24_module_preservation_lncRNA.tsv, 24_module_preservation_mRNA.tsv,
156 24_module_preservation_summary.tsv, 02_lncRNA_module_sizes.tsv,
157 02_mRNA_module_sizes.tsv.

158 **Supplementary Fig. 9 | Partner Gene Ontology terms and gene classes of the**
159 **prognostic lncRNA modules.**



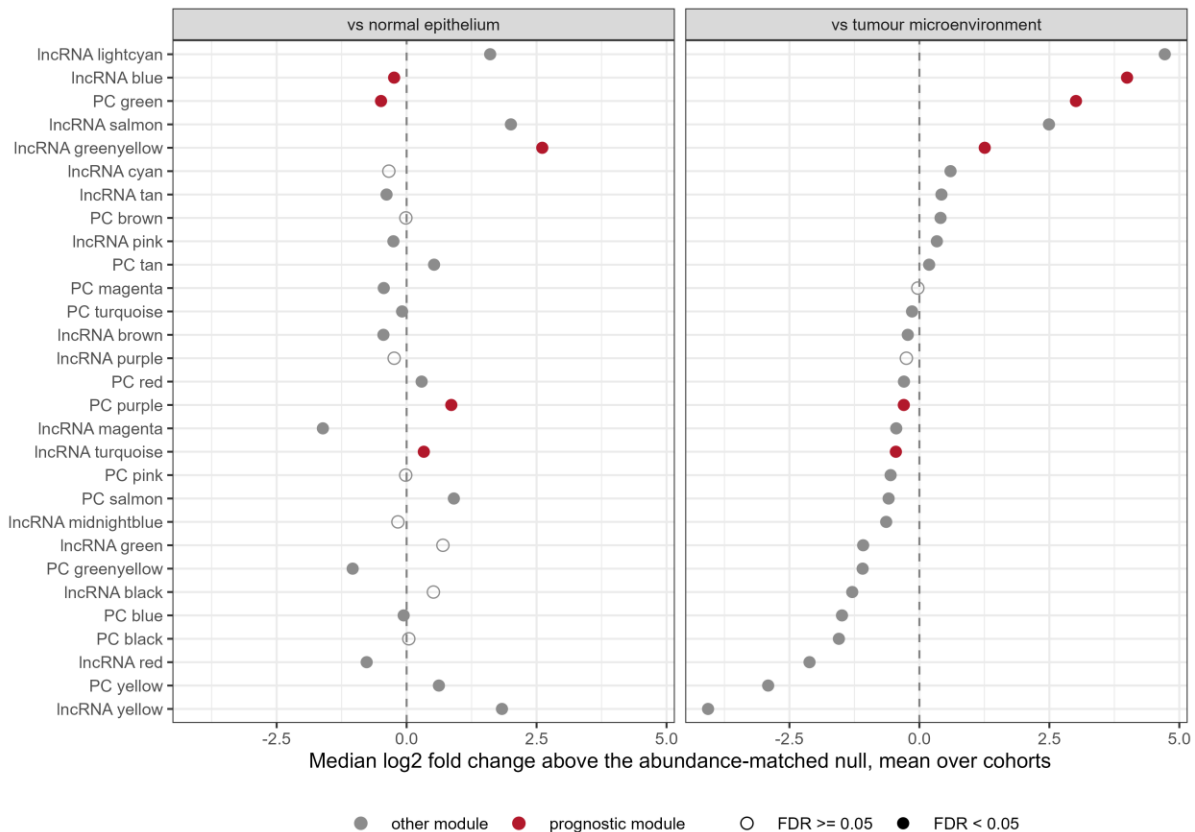
a, b Gene Ontology biological-process terms of the protein-coding partners of the lncRNA blue

(a) and turquoise **(b)** modules, drawn as in Fig. 6c for greenyellow. Partners are split by the sign

of their correlation with the module eigengene, the 300 most strongly correlated of each sign are tested separately against the 12,000 network protein-coding genes, and up to six terms with the smallest Benjamini-Hochberg adjusted p value are shown per sign. Turquoise has two enriched terms among its positive partners. Because partners are selected by correlation, the adjusted p values rank terms and are not calibrated tests. Under the principal specification the hazard ratio per standard deviation is 0.71 (0.60 to 0.83, false discovery rate 4.8e-04) for blue and 1.24 (1.06 to 1.45, 0.039) for turquoise. **c** Gene-class composition, as stacked proportions, of the three prognostic lncRNA modules, which hold 246 genes (blue), 108 (greenyellow) and 438 (turquoise), under two annotation schemes. The name-based classes are antisense, clone-based unnamed, divergent, intronic, named long intergenic and other. The positional classes, relative to GENCODE v36 protein-coding genes, are exonic sense, antisense, intronic, sense-overlapping, divergent and intergenic. Source tables:

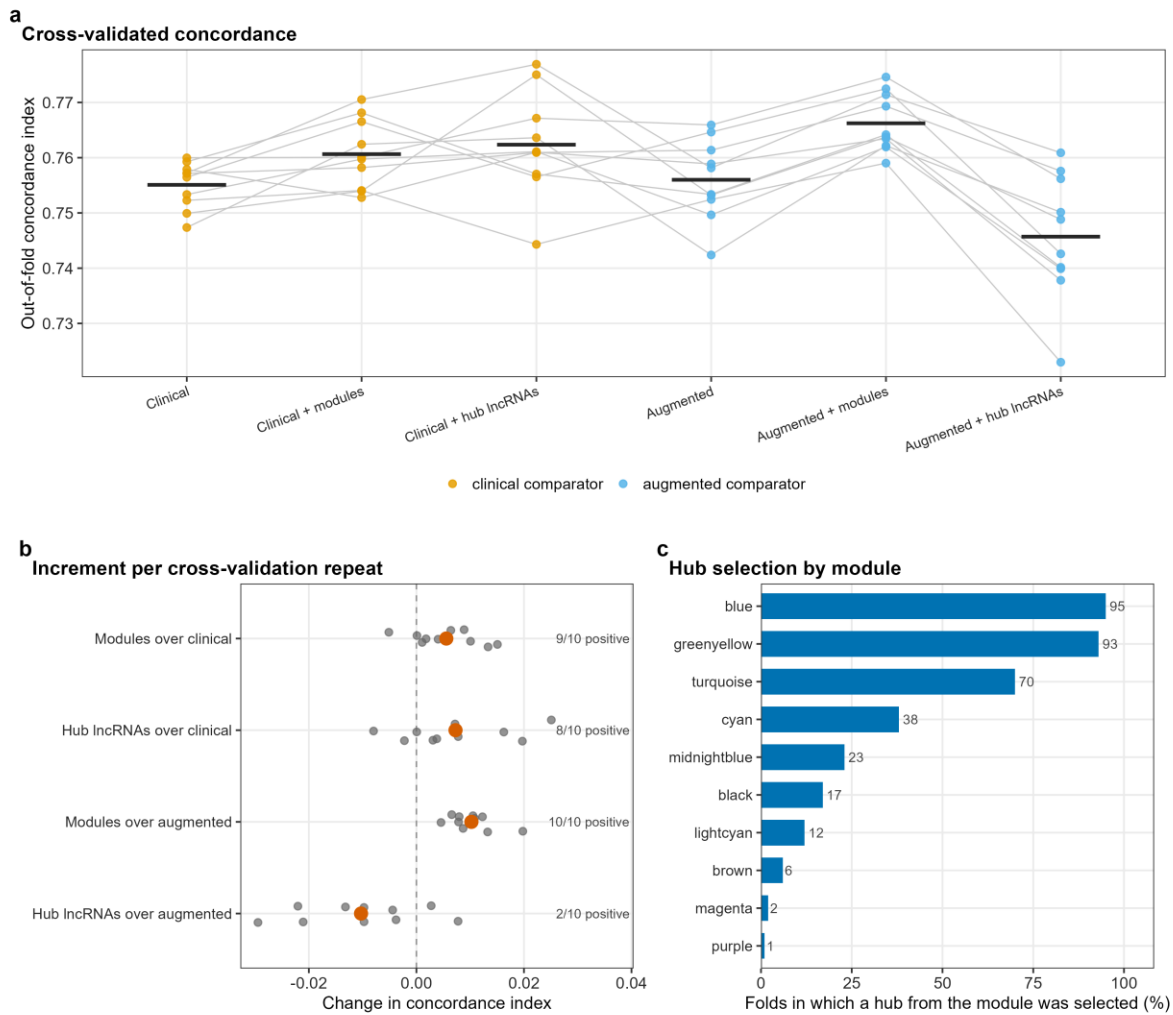
`10_lncRNA_guilt_by_association_GO_signed.tsv` (**a**, **b**),
`10_lncRNA_module_composition.tsv` and `28_module_composition_by_class.tsv`
(c).

Supplementary Fig. 10 | Single-cell localisation of the 29 discovery modules.



For each module, the median log2 fold change of its members minus the mean of 2,000 abundance-matched random gene sets, averaged over cohorts, for malignant cells against normal kidney epithelium (three cohorts, left) and against the rest of the tumour microenvironment (four cohorts, right). Positive values mean higher in malignant cells than a matched random set. Red points, the five prognostic modules. Filled symbols, a within-contrast false discovery rate below 0.05 by Stouffer combination of per-cohort z scores, which overstates precision because genes within a module are correlated. Per-cohort values are in Supplementary Table 19. Source table: 36_sc_module_localisation_combined.tsv.

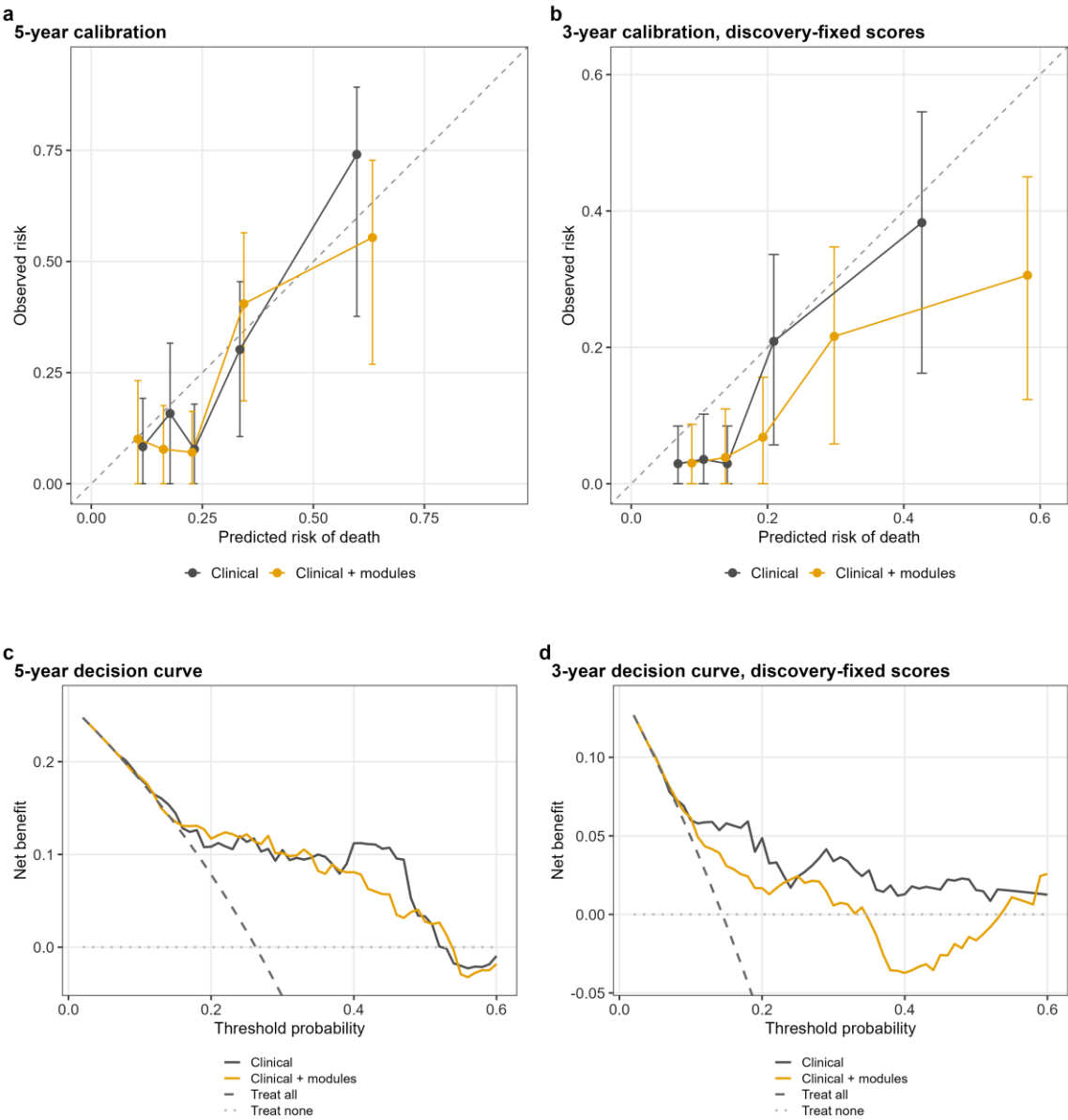
Supplementary Fig. 11 | Cross-validation of hub lncRNAs selected inside each fold.



a Out-of-fold concordance index of six models over ten repeats of ten-fold cross-validation in 493 discovery patients with 159 events. The arms are the clinical and augmented models, each alone, with the 29 module eigengenes or with up to 200 hub lncRNAs added under an elastic-net penalty. Lines join the same repeat, and bars are the mean over repeats. **b** Per-repeat increment for four comparisons, with the number of repeats in which it is positive. Grey points are single repeats and red points the mean over the ten repeats. **c** Percentage of the 100 training folds (10 repeats by 10 folds, bar labels giving fold counts) in which a hub lncRNA from each module entered the penalised model, 95 of 100 for lncRNA blue, 93 for greenyellow and 70 for turquoise. Because the candidate pool is chosen inside each training fold, the panel shows the

198 stability of feature selection, not a fixed signature. Source tables:
199 05_cv_cindex_per_repeat.tsv, 05_hub_module_selection_frequency.tsv.

200 **Supplementary Fig. 12 | Calibration and clinical utility, sensitivity analyses.**



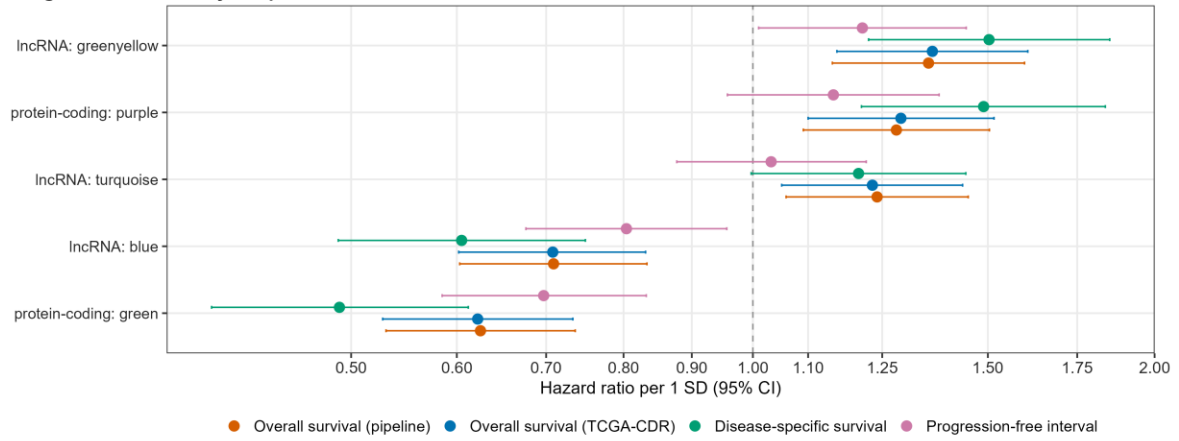
201 All panels are CPTAC-3, and error bars in **a** and **b** are Kaplan-Meier 95% confidence intervals. **a**

202 Calibration at five years under cohort standardisation, by quintile of predicted risk, with the

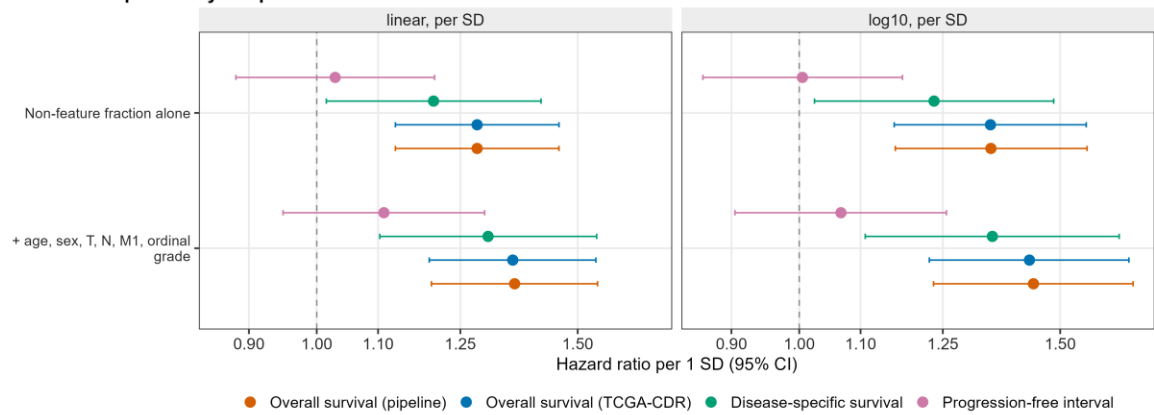
identity line. With twenty-three patients at risk at five years, this is the least stable estimate reported. The calibration slope is 0.97 (0.60 to 1.34) for the clinical model and 0.89 (0.58 to 1.20) with modules, and observed over expected risk 0.90 and 0.89. **b** Calibration at three years, the primary horizon, under discovery-fixed standardisation, with 95 patients at risk. Under this rule the combined model over-predicts risk about two-fold: the slope is 0.97 (0.61 to 1.34) for the clinical model and 0.89 (0.58 to 1.20) with modules, and observed over expected risk 0.76 and 0.55. **c** Decision curve at five years, cohort standardisation. The median change in net benefit over thresholds 0.10 to 0.40 is +0.003, positive at 19 of 31 thresholds. **d** Decision curve at three years, discovery-fixed standardisation, with a median change of -0.027, positive at 3 of 31 thresholds. In **c** and **d** the treat-all curve runs below the panel. All decision curves use the baseline hazard fixed in discovery, and none re-estimates it in CPTAC-3. Source tables: 12_calibration_bins.tsv, 12_calibration_summary.tsv, 12_validation_decision_curve.tsv.

Supplementary Fig. 13 | Endpoint sensitivity.

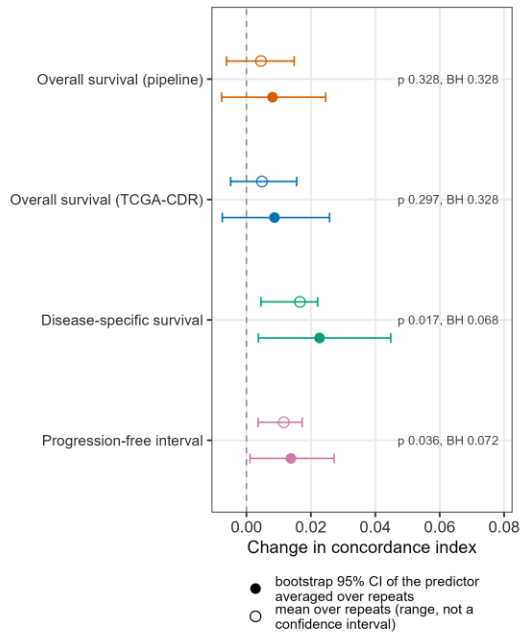
a Prognostic modules by endpoint



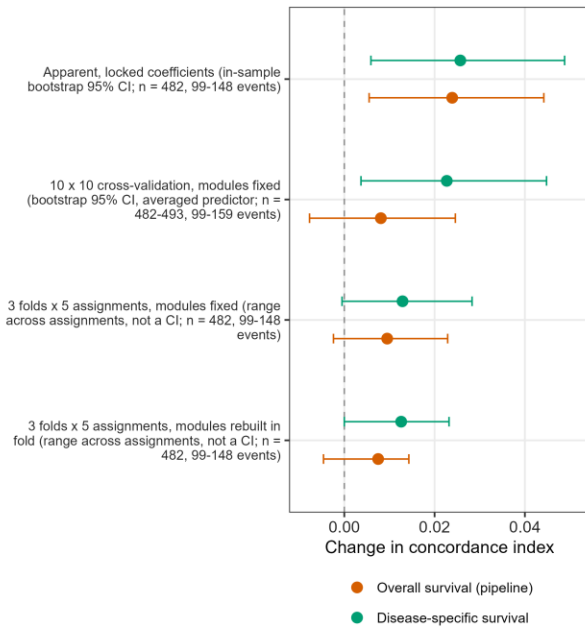
b Metric as exposure by endpoint



c Increment over the clinical model



d Modules rebuilt inside each fold



218 Events are deaths for overall survival, cancer-specific deaths for disease-specific survival, and
 219 progression or death for the progression-free interval. **a** The five prognostic modules under four
 220 endpoints in TCGA-KIRC (principal specification): overall survival from the pipeline and the
 221 Clinical Data Resource (502 to 504 patients, 163 to 165 deaths), disease-specific survival (491 to
 222 493, 102 to 103) and the progression-free interval (500 to 502, 149 to 151). Sixteen of the 20
 223 estimates reach a false discovery rate below 0.05. **b** The non-feature fraction as exposure under
 224 the same endpoints, on linear and log10 scales, with two models per scale (520 patients and 172
 225 events for overall survival, 509 and 109 for disease-specific survival, 518 and 157 for the
 226 progression-free interval). The interval excludes 1 in all 12 overall and disease-specific estimates
 227 and in none of the 4 progression-free estimates. **c** Cross-validated increment in the concordance
 228 index from adding the 29 module eigengenes to the clinical model, by endpoint, in ten repeats of
 229 ten-fold cross-validation (493 patients and 159 events for overall survival, 482 and 99 for
 230 disease-specific survival, 491 and 146 for the progression-free interval). The interval excludes
 231 zero for disease-specific survival and the progression-free interval, and none of the four survives
 232 Benjamini-Hochberg adjustment over the endpoints (smallest adjusted p 0.068). The mean over
 233 the ten repeats is shown beside the bootstrap estimate, the increment of the repeat-averaged out-
 234 of-fold predictor, and is the smaller of the two for all four endpoints; only the bootstrap estimate
 235 carries a confidence interval. **d** Fold-wise network rebuild for disease-specific and overall
 236 survival against matched fixed-module arms, with the ten-by-ten cross-validation rows for scale.
 237 Only the three-fold rows are matched arms, sharing patients, folds and inner splits and differing
 238 only in how modules are defined. Rebuilding costs +0.0003 for disease-specific and +0.0020 for
 239 overall survival. For disease-specific survival both three-fold rows (+0.0129 fixed, +0.0126
 240 rebuilt) sit below the ten-by-ten value (+0.0227), which the matched arm attributes to the coarser

cross-validation. Ranges span five fold assignments and are not confidence intervals. Source tables: 30_endpoint_modules.tsv, 30_endpoint_exposure.tsv, 30_endpoint_cv_increment.tsv, 30_endpoint_foldwise.tsv.

Supplementary Tables

Supplementary Table 1 | Driver mutations and the positional classes of the network lncRNAs

(a) Coverage of the discovery cohort by open-access masked mutation files

Quantity	Value
Patients with a masked MAF	368 of 528 (69.7%)
Primary-tumour MAF files used	411
Deaths among covered patients	94
Death rate, covered against uncovered	25.5% against 49.4%, chi-square p 1.4e-07
Overall survival by MAF availability	log-rank p 5.1e-05
M1, covered against uncovered	12.8% against 21.2%, p 0.019
T3 or T4, covered against uncovered	31.5% against 46.2%, p 0.0017
Median non-feature fraction, covered against uncovered	3.9% against 6.0%, Wilcoxon p 5.9e-07
Coverage of the top decile of the metric	52.8%
Coverage of the bottom decile of the metric	86.8%
Median non-synonymous variants per covered patient	45.5 (IQR 33 to 59)

Note. Source: 28_mutation_coverage.tsv. Mutation-file availability is associated with death, stage and the exposure itself. Any result in this subset is therefore conditional, and the absence of a genotype association is not an unconditional null.

(b) Driver frequencies and association with the metric and the axis

Gene	Mutated of 368 (%)	Median metric, mutated against wild type (%)	p	FDR
VHL	171 (46.5)	3.9 against 3.9	0.68	0.69
PBRM1	151 (41.0)	4.1 against 3.9	0.39	0.69
SETD2	46 (12.5)	4.4 against 3.9	0.54	0.69

Gene	Mutated of 368 (%)	Median metric, mutated against wild type (%)	p	FDR
BAP1	40 (10.9)	4.4 against 3.9	0.29	0.69
MTOR	26 (7.1)	3.7 against 4.1	0.45	0.69
KDM5C	19 (5.2)	4.7 against 3.9	0.20	0.69
TP53	10 (2.7)	3.8 against 3.9	0.69	0.69
Non-synonymous burden --		Spearman rho -0.07	0.19	0.69

Note. Source: 28_mutation_frequency.tsv, 28_noFeature_vs_mutations.tsv.

Tested against the lncRNA axis on 358 patients, the same seven genes give raw p values of 0.065 to 0.88 (smallest false discovery rate 0.29). The largest difference of medians across the 16 tests, for KDM5C on 19 mutated patients, is 0.8 percentage points in the metric and 0.49 standard deviations in the axis, and neither is significant. Without an interval on the difference of medians, a tight null cannot be told from an uninformative one. The finding is no association at a 5% false discovery rate in the 368 patients with a masked MAF, which does not show that genotype fails to explain the metric.

(c) The metric as exposure with genotype in the model

Model	HR	95% CI	p	Events per variable
Metric + clinical model, covered subset	1.31	1.10 to 1.57	0.0029	13.3
+ BAP1	1.29	1.07 to 1.55	0.0077	11.6
+ PBRM1	1.31	1.10 to 1.56	0.0029	11.6
+ SETD2	1.34	1.11 to 1.61	0.0019	11.6
+ BAP1 + PBRM1 + SETD2	1.32	1.09 to 1.60	0.0044	9.3
+ all seven driver genes	1.33	1.09 to 1.62	0.0042	6.6
+ non-synonymous burden	1.31	1.10 to 1.56	0.0026	11.6
+ all seven drivers + burden	1.34	1.10 to 1.63	0.0032	6.2

Note. TCGA-KIRC, 361 patients and 93 events, overall survival, hazard ratio per standard deviation of the metric. Source: 28_mutation_survival_models.tsv. Proportional-hazards p values across the eight models range from 0.036 to 0.058. The largest model has 6.2 events per

variable, and its driver coefficients should not be interpreted alone. In the same file, TP53 has a clinically adjusted hazard ratio of 3.63 (1.53 to 8.61) on 10 mutated patients, and mutation burden 1.41 (1.10 to 1.80) per standard deviation of $\log_{10}(\text{count})$. VHL violates proportional hazards in all three of its models.

(d) Positional classes of the 3,442 analysed lncRNAs and their correlation with the metric

Positional class	Genes	Share of the set (%)	Median absolute rho with the metric	Fraction above 0.3 (%)	Mean observed PC1 loading	Median gene length (kb)
Intronic	698	20.3	0.53	70.8	0.01512	1.01
Exonic sense	126	3.7	0.52	73.8	0.01566	11.60
Sense-overlapping	19	0.6	0.49	68.4	0.01179	28.00
Antisense	1,162	33.8	0.43	62.8	0.01261	8.16
Intergenic	1,122	32.6	0.25	43.5	0.01114	3.66
Divergent	315	9.2	0.25	43.2	0.01015	2.90
All	3,442	100.0	0.37	56.8	0.01252	3.55

Note. Classes are assigned per gene against GENCODE v36 protein-coding genes in a fixed priority order, so a lncRNA meeting several relations is counted once. Per-gene correlations use 511 libraries. Source: `28_lncRNA_positional_classes.tsv`, `28_quality_correlation_by_class.tsv`. The highest-signal subclass, intronic sense, has 317 genes, a median absolute rho of 0.59 and 78.2% above 0.3. No gene in any class falls below -0.3, so the correlation is positive wherever present. Positional class differs from Ensembl name class. Of the 1,122 positionally intergenic lncRNAs, 740 have clone-based names and only 264 are named lincRNAs.

(e) Tests of the class effect

Quantity	Test	df	Statistic	p
Absolute rho with the metric, across the six classes	Kruskal-Wallis	5	263.31	7.7e-55
Signed rho with the metric	Kruskal-Wallis	5	265.65	2.4e-55
Observed PC1 loading	Kruskal-Wallis	5	109.97	4.2e-22

Quantity	Test	df	Statistic	p
Absolute rho, class entered after mean expression and sequential ANOVA log10 gene length		5	F 54.80	6.2e-55
PC1 loading, same adjustment	sequential ANOVA	5	F 18.29	5.8e-18
PC1 loading against per-gene rho with the metric	Spearman	--	rho 0.78	reported as 0
PC1 score against the metric, sample level	Spearman	--	rho 0.90	1.4e-186

278 *Note.* 3,442 genes and 511 libraries. Source: `28_class_comparison_tests.tsv`. The class
279 effect survives adjustment for expression level and gene length, so it is not a detectability
280 artefact. It remains correlational, and residual confounding by other gene properties is not
281 excluded.

282 (f) Positional composition of the lncRNA modules

Module	Genes	Most enriched class (enrichment)	Goodness-of-fit p	FDR
green	186	intronic (2.0-fold)	1.0e-06	4.3e-06
pink	174	divergent (3.4-fold)	1.0e-06	4.3e-06
black	179	exonic sense (2.1-fold)	1.0e-06	4.3e-06
brown	219	divergent (2.6-fold)	1.0e-06	4.3e-06
cyan	54	divergent (3.4-fold)	3.7e-04	7.9e-04
tan	93	divergent (2.6-fold)	2.6e-04	7.9e-04
yellow	192	sense-overlapping (2.8-fold)	3.3e-04	7.9e-04
blue	246	intergenic (1.4-fold)	3.6e-04	7.9e-04
purple	110	intronic (1.5-fold)	4.5e-04	8.5e-04
lightcyan	44	intergenic (2.0-fold)	0.0032	0.0046
greenyellow	108	intergenic (1.6-fold)	0.0032	0.0046
turquoise	438	exonic sense (1.8-fold)	0.0089	0.012
magenta	122	divergent (2.0-fold)	0.0095	0.012
red	186	intergenic (1.3-fold)	0.030	0.034
salmon	72	sense-overlapping (2.5-fold)	0.073	0.078
midnightblue	53	sense-overlapping (3.4-fold)	0.24	0.24

283 *Note.* Enrichment is relative to the class proportions of all 3,442 analysed lncRNAs. Source:
284 `28_module_composition_by_class.tsv`. Over 16 modules and 6 classes on the 2,476 non-

grey genes, the global test gives a chi-square of 558.24, with a simulated p at the floor of 1e-06 over one million replicates. Five module p values also sit on that floor and are upper bounds. Only midnightblue and salmon show no significant class deviation.

Supplementary Table 2 | The lncRNA axis under five normalisations

(a) The leading lncRNA principal component under five normalisations

Normalisation	PC1 variance share	rho with the non-feature fraction	p	rho with the cached FPKM axis	Absolute Pearson r with the per-sample mean	Per-gene median absolute rho
log2(FPKM + 1), as used	0.1997	0.90	1.4e-186	1.00	0.988	0.37
log2 counts per million, library size only	0.2205	0.93	2.1e-215	0.99	0.985	0.37
log2 counts per million, edgeR TMM	0.1617	0.91	1.4e-199	0.98	0.956	0.25
DESeq2 variance-stabilising transform, blind	0.1155	0.92	4.7e-209	0.88	0.744	0.26
Quantile-normalised log2(FPKM + 1)	0.1017	0.88 in magnitude	1.4e-169	0.87	0.084	0.28

Note. The same 3,442 genes and 511 libraries in every row. Source: 29_normalisation_pc1.tsv. Raw counts re-read from the local files matched the STAR summary with a maximum absolute difference of 0 reads and 0 percentage points. Quantile normalisation removes the per-sample mean, leaving the quantile row's sign anchor indeterminate, so it is quoted as a magnitude. Per-sample median centring, whose sign is determinate, flips the sign and leaves the magnitude at 0.89.

(b) Size factors and compartment read shares against the metric

Quantity	Median	rho with the non-feature fraction	p	rho with the FPKM axis
edgeR TMM normalisation factor	0.9991	0.82	3.1e-129	0.81
TMM effective library size	5.815e7	0.38	1.7e-19	0.31

Quantity	Median	rho with the non-feature fraction	p	rho with the FPKM axis
DESeq2 median-of-ratios size factor	1.045	0.13	0.0032	0.08
DESeq2 size factor relative to library size	1.030	0.46	3.4e-29	0.49
Assigned reads	5.854e7	-0.06	0.15	-0.13
lncRNA read share, all 16,901 lncRNA genes	0.01107	0.87	2.2e-166	0.90
lncRNA read share, the 3,442 analysed lncRNAs	0.00995	0.85	5.0e-151	0.90
Protein-coding read share	0.9671	-0.32	3.8e-14	-0.34

297 *Note.* TCGA-KIRC, 528 patients. Source: 29_size_factors_vs_quality.tsv. The protein-
298 coding read share and the FPKM-against-CPM per-sample scale are exact reciprocals, one
299 measurement reported twice, so only one is given.

300 (c) Module-level agreement between the FPKM and TMM scales

Quantity	Value
Modules significant at within-network FDR 0.05, observed arm, FPKM against TMM	11 against 11
The same, with the STAR metrics as covariates	2 against 5
The same, residualised arm	7 against 5
Lost, gained and robust on adjustment, FPKM against TMM	8, 4, 3 against 6, 0, 5
Median bootstrap difference in log HR, TMM minus FPKM, observed against residualised	-0.073 against -0.023
Intervals excluding zero, observed against residualised	9 of 16 against 2 of 16
Surviving Benjamini-Hochberg within arm, observed against residualised	7 of 16 against 0 of 16
Modules whose FPKM and TMM intervals overlap	16 of 16
TMM point estimate inside the FPKM interval	14 of 16
Modules with the same significance call, observed, plus STAR, residualised	8, 13 and 14 of 16
Median eigengene correlation between scales, observed against residualised	0.930 against 0.951
Stability across 50 bootstrap seeds	7 modules survive BH in 50 of 50 seeds

301 The seven modules whose effect shifts detectably between scales on the observed matrix:

Module	Bootstrap difference in log HR	95% CI
red	-0.189	-0.291 to -0.075
tan	-0.186	-0.282 to -0.097
midnightblue	-0.173	-0.276 to -0.085
blue	-0.141	-0.231 to -0.059
cyan	-0.139	-0.235 to -0.049
lightcyan	-0.123	-0.204 to -0.036
pink	-0.115	-0.198 to -0.048

Note. TCGA-KIRC, 504 patients and 163 events, 16 lncRNA modules, paired patient bootstrap over 2,000 resamples (128,064 Cox fits, 5 convergence warnings). Source: 29_normalisation_agreement.tsv, 29_normalisation_module_summary.tsv, 29_normalisation_module_contrast.tsv, 29_bootstrap_seed_stability.tsv. Two further modules, yellow (-0.070) and salmon (-0.075), have intervals excluding zero but fail Benjamini-Hochberg control (false discovery rates 0.076 and 0.085). Only the adjusted counts should be quoted. The bootstrap resamples patients only, holding module definitions, rotations, TMM factors and residualisation coefficients at their full-cohort values, so it compares two measurement scales, not two pipelines.

Supplementary Table 3 | Matched normal-kidney libraries

(a) Libraries read and pairs formed

Quantity	TCGA-KIRC	CPTAC-3
Normal STAR-Counts files in the file map; present on disk	72; 72	168; 168
Excluded	0	0
Normal libraries analysed	72	168
Tumour-normal pairs with survival	71	129
Deaths among the paired patients	26	28
Pairs on the same sequencing plate	67 of 71 (94.4%)	not applicable
Normal libraries with a plate recorded	72	0

Note. Source: 25_normal_consort.tsv. All 168 CPTAC-3 normal libraries exceed the discovery non-feature cap of 30% as written, and, as for the CPTAC-3 tumours, none is excluded on that ground. No TCGA-KIRP or TCGA-KICH normal libraries are held locally. The 32 and 25 files at the GDC were not downloaded or analysed, so no subtype normal control exists (25_subtype_normal_availability.tsv).

(b) The axis in normal kidney

Sample set	Matrix	n	Projected axis, rho with the metric (p)	Own PC1 variance share (size-matched discovery baseline)	Own PC1, rho with the metric (p)
TCGA-KIRC normal	lncRNA	72	0.86 (8.9e-22)	0.376 (0.210, 0.159 to 0.257)	0.33 (0.0042)
TCGA-KIRC normal	protein-coding	72	0.09 (0.48)	0.429 (0.153, 0.117 to 0.194)	0.30 in magnitude (0.010)
CPTAC-3 normal	lncRNA	168	0.43 (4.1e-09)	0.195 (0.203, 0.172 to 0.230)	0.23 in magnitude (0.0030)
CPTAC-3 normal	protein-coding	168	0.39 (2.2e-07)	0.354 (0.144, 0.118 to 0.169)	0.26 in magnitude (8.2e-04)

Note. Source: 25_normal_axis_projection.tsv. The projected axis is the discovery lncRNA or protein-coding PC1 direction applied to the sample set, and the own component is the set's own leading principal component. The two differ in the same samples, and lncRNA specificity holds for the projected axis but not for the own component. The size-matched baseline is the median of 200 resampling draws from the discovery matrix at the same sample size, with their minimum and maximum in brackets, and is not a confidence interval. Three of the eight own-component rows in this file carry a sign set by convention, not by the data, because the primary anchor is weak or the anchors disagree, and are quoted as magnitudes. Supplementary Table 4 gives the full six-sample-set comparison on the projected quantity.

(c) Library metrics and axis scores, tumour against matched normal, TCGA-KIRC

Quantity	Tumour median	Normal median	Unpaired difference (p)	Plate-matched difference (p)	Within-patient difference (signed-rank p)
Non-feature fraction, %	4.6	2.9	1.73 (1.6e-14)	0.15 (0.057)	0.15 (0.099)
Multimapping fraction, %	7.5	7.0	0.48 (0.011)	-0.06 (0.18)	-0.14 (0.069)
Unmapped fraction, %	3.3	2.7	0.53 (0.085)	0.95 (6.0e-04)	0.22 (0.0094)
Ambiguous fraction, %	8.5	8.2	0.35 (7.0e-08)	0.37 (5.5e-05)	0.37 (0.0033)
log10 assigned reads	7.767	7.804	-0.037 (0.0015)	-0.030 (0.0050)	-0.020 (0.11)
Projected lncRNA axis	-0.117	-1.094	0.977 (1.1e-20)	0.397 (1.0e-08)	0.372 (6.6e-06)
Projected protein-coding axis	0.104	0.707	-0.603 (9.1e-08)	-0.439 (1.4e-04)	-0.370 (3.7e-04)
Mean lncRNA expression	0.718	0.617	0.102 (2.7e-10)	0.013 (0.10)	0.009 (0.20)

Note. Source: 25_tumour_vs_normal_distribution.tsv, rows for the normals of analysed-tumour patients. Unpaired comparisons use 528 tumour and 71 normal libraries (rank-sum test), plate-matched comparisons the 144 tumour and 67 normal libraries on plates both sides occupy, and within-patient comparisons the 71 pairs (signed-rank test). Medians of the four percentage metrics are given to one decimal and the three difference columns to two, because several plate-matched and within-patient differences fall below 0.1 percentage points. The file flags every TCGA-KIRC row as confounded with plate, since the tumours span 19 plates and the normals 5. For the non-feature fraction the unpaired difference is about eleven times the plate-matched one and must never be quoted alone. In CPTAC-3 the direction reverses, with 40.5% in 200 tumours against 42.5% in 129 paired normals, a within-patient difference of -2.22 points (signed-rank p 1.6e-10).

(d) Whether the metric is shared between a patient's tumour and normal library: the primary test

Quantity	Role	Spearman rho, tumour against normal	p	FDR	Outcome
Non-feature fraction	primary	-0.02	0.88	0.88	null

Quantity	Role	Spearman rho, tumour against normal	p	FDR	Outcome
Projected lncRNA axis	primary	-0.17	0.16	0.27	null
Multimapping fraction	companion	-0.05	0.68	0.85	null
log10 assigned reads	companion	0.40	4.8e-04	0.0024	shared
Projected protein-coding axis	intended negative control	0.37	0.0013	0.0033	shared

Note. TCGA-KIRC, 71 pairs. Source: 25_paired_tumour_normal.tsv. In CPTAC-3 (129 pairs) the values are 0.17 (FDR 0.15) for the metric, 0.11 (FDR 0.30) for the lncRNA axis and 0.24 (FDR 0.033) for the protein-coding axis. This test failed to discriminate and supports neither reading. It is also weak. Plate explains 7.8% of the variance in the metric within the paired libraries against 30.9% across all 600 analysed libraries, so the pairs carry little of the plate variation that drives the metric elsewhere. Nor is the intended negative control clean. In the discovery tumours the projected protein-coding axis tracks the multimapping fraction (rho - 0.52), so its sharing within patient is evidence of a patient-level library component, not of tumour biology.

(e) Variance in the metric and in the axis explained by plate and tissue type, by library set

Library set	Quantity	Factor	Libraries	Variance explained (%)	p
All analysed libraries	Non-feature fraction	plate	600	30.9	1.1e-38
Tumour libraries only	Non-feature fraction	plate	528	26.6	6.5e-27
Normal libraries only	Non-feature fraction	plate	72	7.2	0.16
Paired patients	Non-feature fraction	plate	142	7.8	0.010
Paired patients	Non-feature fraction	tissue type given patient	142	6.1 partial	0.037
All analysed libraries	Projected lncRNA axis	plate	600	31.2	3.4e-39
Tumour libraries only	Projected lncRNA axis	plate	528	25.6	1.4e-25

Library set	Quantity	Factor	Libraries	Variance explained p (%)	
Normal libraries only	Projected lncRNA axis	plate	72	3.9	0.43
Paired patients	Projected lncRNA axis	plate	142	0.8	0.79
Paired patients	Projected lncRNA axis	tissue type given patient	142	27.2 partial	2.7e-06

Note. TCGA-KIRC. Source: 25_paired_variance_partition.tsv. Sixty-eight of the 72 normal libraries sit on 3 of the 5 plates they span, against 19 plates for the 528 tumours, so the plate partition is a tumour-side result absent within the normals. CPTAC-3 lacks plate and site metadata and has no equivalent rows.

(f) Prognostic value of the normal library's metric

Exposure library	Exposure	Model	HR	95% CI	p
Normal	Non-feature fraction	alone	1.01	0.68 to 1.51	0.95
Normal	Non-feature fraction	+ clinical model	0.92	0.61 to 1.38	0.68
Tumour	Non-feature fraction	alone	0.88	0.57 to 1.36	0.57
Tumour	Non-feature fraction	+ clinical model	0.98	0.62 to 1.54	0.92
Normal	Projected lncRNA axis	alone	1.00	0.67 to 1.47	0.98
Normal	Projected lncRNA axis	+ clinical model	0.95	0.65 to 1.40	0.80
Tumour	Projected lncRNA axis	alone	0.83	0.54 to 1.28	0.40

Note. TCGA-KIRC, 71 paired patients and 26 events. Source: 25_normal_metric_survival.tsv. Every adjusted model falls below the events-per-variable threshold of 5 (ratios 3.2 to 3.7), and the smallest false discovery rate in the 32-row table is 0.995 in TCGA-KIRC and 0.999 in CPTAC-3. The tumour metric is also null in this 71-patient subset, so the panel reflects low power and does not contradict the discovery result. Two of the 32 rows, both for the tumour axis alone, violate proportional hazards.

Supplementary Table 4 | The projected lncRNA axis across all six sample sets

Sample set	Tissue	Library protocol	n	Projected lncRNA axis, rho with the metric (p)	Projected protein-coding axis, rho with the metric (p)	Projected protein-coding axis, rho with the multimapping fraction
TCGA-KIRC	primary tumour	poly(A)	528	0.90 (4.4e-194)	0.06 (0.18)	-0.52
TCGA-KIRP	primary tumour	poly(A)	269	0.89 (2.6e-92)	0.18 (0.0034)	-0.45
TCGA-KIRC	matched normal	poly(A)	72	0.86 (8.9e-22)	0.09 (0.48)	0.12
TCGA-KICH	primary tumour	poly(A)	65	0.59 (1.8e-07)	0.29 (0.018)	-0.66
CPTAC-3	primary tumour	ribo-depleted total RNA	200	0.55 (7.4e-17)	0.29 (2.6e-05)	-0.05
CPTAC-3	matched normal	ribo-depleted total RNA	168	0.43 (4.1e-09)	0.39 (2.2e-07)	-0.51

364 Each cohort's own leading component, a different quantity in the same samples:

Sample set	Matrix	n	Own PC1 variance share	Size-matched discovery baseline, median (minimum to maximum)	Own PC1, rho with the metric (p)
TCGA-KIRC tumour	lncRNA	511	0.200	not applicable	0.90 (1.4e-186)
TCGA-KIRP tumour	lncRNA	269	0.267	0.202 (0.181 to 0.219)	0.83 (7.3e-69)
TCGA-KIRC normal	lncRNA	72	0.376	0.210 (0.159 to 0.257)	0.33 (0.0042)
TCGA-KICH tumour	lncRNA	65	0.145	0.208 (0.165 to 0.269)	-0.13 (0.32)
CPTAC-3 tumour	lncRNA	200	0.114	0.201 (0.177 to 0.226)	0.20 (0.0041)
CPTAC-3 normal	lncRNA	168	0.195	0.203 (0.172 to 0.230)	0.23 in magnitude (0.0030)

365 *Note.* The projected axis is the discovery lncRNA or protein-coding PC1 direction, defined by its
366 gene loadings and the discovery gene means, applied to each sample set. The own component is
367 the set's own leading principal component. Projected and own-component rows are different
368 quantities and are not pooled. Source: 25_normal_axis_projection.tsv and
369 35_axis_projection_subtypes.tsv. Every projected row has an interpretable sign,

inherited from the fixed discovery rotation. Of the own-component rows, three of the eight in the first file are not sign-interpretable and are quoted as magnitudes, and none of the four in the second is flagged.

On the projected quantity the axis is strong in every poly(A) sample set and materially weaker, though far from absent, under ribo-depleted total RNA. The separation is graded, not categorical, with the weakest poly(A) set, TCGA-KICH at 0.59, beside the strongest ribo-depleted set, CPTAC-3 tumour at 0.55. TCGA-KICH has 65 patients over a narrow exposure range, with a median non-feature fraction of 3.1% (interquartile range 2.6% to 3.5%) against 4.6% (3.1% to 6.8%) in TCGA-KIRC. Restricted range attenuates a correlation, so 0.59 is a lower bound. The projected value of 0.59 does not contradict this cohort's own component of -0.13. Its own lncRNA PC1 carries 14.5% of variance, below the size-matched discovery baseline of 20.8%, so no dominant leading component exists for the metric-linked direction to occupy. The direction is present without being first, which is why a fixed projected direction is used throughout. The lncRNA specificity itself depends on protocol. Under poly(A) selection the projected lncRNA and protein-coding axes differ sharply, at 0.90 against 0.06 in the discovery tumours and 0.86 against 0.09 in the matched normals. Under ribo-depletion the difference nearly vanishes, at 0.43 against 0.39 in the CPTAC-3 normals. A protocol that retains unprocessed RNA raises intronic content for every gene, not only for lncRNAs overlapping protein-coding loci, so the effect is no longer lncRNA-specific.

Supplementary Table 5 | The lncRNA network built with and without adjustment for the alignment metrics

Comparison	Genes	Adjusted Rand index	Modules, unadjusted	Modules, adjusted
all genes, grey as a class	3,442	0.062	3	16
genes assigned in both networks	1,722	0.047	3	16

392 Fate of each adjusted module in the unadjusted network (dominant unadjusted module:

393 turquoise, 1,982 genes):

Adjusted module	Genes	In the unadjusted dominant module (%)	In another unadjusted module (%)	Unassigned in the unadjusted network (%)
black	179	100	0	0
green	186	99.5	0.5	0
turquoise	438	99.5	0	0.5
salmon	72	84.7	0	15.3
pink	174	66.7	0.6	32.8
purple	110	66.4	0	33.6
brown	219	61.6	0	38.4
tan	93	53.8	1.1	45.2
magenta	122	40.2	0.8	59
blue	246	36.6	0	63.4
midnightblue	53	34	1.9	64.2
yellow	192	32.8	49.5	17.7
red	186	25.3	47.3	27.4
cyan	54	18.5	1.9	79.6
lightcyan	44	13.6	0	86.4
greenyellow	108	13	0.9	86.1

394 Positional class of the lncRNAs in the unadjusted dominant module:

Positional class	Network lncRNAs	In the dominant module	%	Fisher p	FDR
sense overlapping	19	14	73.7	0.17	0.17
intronic	698	499	71.5	2.9e-17	8.7e-17
exonic sense	126	88	69.8	0.0044	0.0065
antisense	1162	732	63	4.3e-06	8.6e-06
divergent	315	159	50.5	0.0084	0.01
intergenic	1122	490	43.7	3e-30	1.8e-29

Note. Both networks used the same 511 discovery libraries and parameters, the unadjusted network on observed $\log_2(\text{FPKM} + 1)$ and the adjusted network on expression residualised on the three STAR metrics. Mean intramodular connectivity over assigned genes is 70.5 unadjusted and 3.3 adjusted, each on its own matrix (23_unadjusted_lncRNA_network_summary.tsv). Adjusted and unadjusted module colours do not correspond. The false discovery rate is controlled across the six positional classes. The dominant module holds 57.6% of all network lncRNAs. Source tables: 37_network_partition_agreement.tsv, 37_adjusted_module_fate.tsv, 37_dominant_module_positional_class.tsv.

Supplementary Table 6 | The lncRNA axis, network reorganisation and prognosis of the metric across TCGA

(a) Primary tumours, one row per project, with eligible projects (at least 100 retained tumours, TCGA-LAML excluded) entering the counts in the main text

Project	Eligible	Tumors	Median non-feature %	Median A-PC1 rho	Median lncRNA-PC1 rho	Median Protein-PC1 rho	Median Intronic intergenic	Median Platelet R-squared of metric	Median Spread of MA D of metric	Median Largest observed module	Median Largest adjusted module	Median Largest I - module eigengene	Median Largest Rul e C HR, plate-stratified (95% CI)	Deaths	
GBM	yes	256	9.1	0.95	0.87	not met	0.44	0.61	1.11	0.65	0.15	0.04	0.94	met 1.26 (1.00 to 1.59)	203
UVM	no	80	1.8	0.95	0.83	not met	0.21	--	--	--	--	--	--	--	--
KIRC	yes	532	4.6	0.91	0.09	met	0.28	0.27	0.60	0.57	0.13	0.06	0.91	met 1.13 (0.96 to 1.33)	174

Project	Eligible	Tumors	Median non-feature %	Ancillary PC1 rho	IncRNA coding PC1 rho	Protein AIC	Intergenic	Platelet minus R-squared of metric (log metric)	Spread of D of metric	Largest observed module	Largest adjusted module	AR	Largest module eigengene	Rul	Metric HR, plate-stratified (95% CI)	Deaths
SKCM	yes	103	2.8	0.89	0.65	not met	0.17	0.03	0.47	0.55	0.20	0.09	0.93	met	1.47 (0.98 to 2.20)	27
LUSD	yes	517	4.4	0.88	0.20	met	0.22	0.39	0.39	0.44	0.16	0.21	0.91	not met	0.84 (0.68 to 1.04)	176
LUSC	yes	500	4.1	0.86	-0.19	met	0.22	0.40	0.48	0.25	0.18	0.36	0.72	not met	1.02 (0.87 to 1.21)	205
KIRP	yes	290	4.1	0.83	0.71	not met	0.15	0.36	0.49	0.55	0.19	0.14	0.80	met	1.38 (1.00 to 1.89)	39
OV	yes	426	5.2	0.82	0.52	met	0.24	0.26	0.41	0.29	0.15	0.28	0.94	not met	0.86 (0.74 to 1.01)	261
STAD	yes	410	7.1	0.81	0.06	met	0.24	0.29	0.51	0.17	0.15	0.37	0.77	not met	1.11 (0.90 to 1.37)	145
READ	yes	166	3.6	0.80	0.15	met	0.15	0.12	0.35	0.20	0.13	0.19	0.82	not met	0.73 (0.41 to 1.30)	22
PAAD	yes	178	3.8	0.80	0.23	met	0.19	0.21	0.37	0.19	0.17	0.31	0.95	not met	1.02 (0.79 to 1.33)	93
COAD	yes	446	3.4	0.79	0.07	met	0.17	0.17	0.34	0.18	0.16	0.26	0.81	not met	0.89 (0.70 to 1.14)	91
SARC	yes	259	2.9	0.73	0.63	not met	0.06	0.13	0.39	0.24	0.15	0.30	0.74	not met	0.99 (0.80 to 1.23)	98
HNSC	yes	520	2.5	0.72	0.48	not met	0.13	0.13	0.41	0.20	0.16	0.34	0.63	not met	0.82 (0.69 to 0.99)	186
LGG	yes	516	4.5	0.68	0.05	met	0.11	0.26	0.29	0.18	0.10	0.32	0.73	not met	1.19 (0.94 to 1.51)	117
BLC A	yes	406	3.5	0.68	-0.54	not met	0.07	0.10	0.39	0.23	0.19	0.35	0.63	not met	0.90 (0.74 to 1.09)	176

Project	Eligible	Tumors	Median non-feature %	IncRNA A PC1 rho	Protein-coding PC1 rho	Relative A c intergenic	Intronic minus	Platelet R-squared of metric (e)	Spread (MA D of log metric e)	Largest observed module	Largest adjusted module	Inter-module eigengene r	Largest module eigengene r	Rul C HR, plate-stratified (95% CI)	Deaths
BRC A	yes	1094	3.9	0.61	0.44	not met	0.11	0.11	0.34	0.16	0.14	0.35	0.84	not met (0.61 to 1.00)	129
UCS	no	57	3.3	0.60	-0.06	met	0.04	--	--	--	--	--	--	--	--
DLB C	no	48	3.1	0.58	0.66	not met	0.07	0.13	--	--	--	--	--	--	--
THC A	yes	505	3.3	0.57	-0.28	not met	0.10	0.14	0.33	0.22	0.20	0.29	0.75	not met (0.74 to 1.93)	16
ESC A	yes	183	8.4	0.55	0.28	not met	0.25	0.32	0.45	0.40	0.28	0.31	0.86	not met (0.68 to 1.19)	64
PCPG	yes	179	3.3	0.52	0.42	not met	0.05	0.00	0.26	0.16	0.15	0.25	0.49	not met (0.83 to 3.72)	6
KICH	no	66	3.1	0.52	-0.27	not met	0.05	0.02	--	--	--	--	--	--	--
LAM L	no	63	27.7	0.50	0.33	not met	0.11	0.11	--	--	--	--	--	--	--
CESC	yes	304	3.5	0.49	0.22	not met	0.08	0.17	0.31	0.19	0.22	0.40	0.93	not met (0.69 to 1.18)	70
UCE C	yes	539	4.1	0.45	0.09	not met	0.07	0.16	0.39	0.20	0.11	0.52	0.50	not met (1.00 to 1.49)	91
CHOL	no	35	4.6	0.43	0.27	not met	0.05	--	--	--	--	--	--	--	--
TGC T	yes	150	4.8	0.42	-0.39	not met	0.04	0.03	0.25	0.19	0.20	0.39	-0.17	not met (0.19 to 2.38)	3
ACC	no	79	3.0	0.40	0.40	not met	0.06	--	--	--	--	--	--	--	--
MES O	no	87	3.2	0.36	0.09	not met	0.08	0.02	--	--	--	--	--	--	--
PRA D	yes	497	3.8	0.33	0.25	not met	0.09	0.29	0.31	0.19	0.18	0.26	0.32	not met (0.68 to 2.97)	10
LIHC	yes	371	2.3	0.26	0.39	not met	0.08	0.15	0.36	0.21	0.14	0.34	0.61	not met (0.81 to 1.24)	116

Project	Eligible	Tumours	Median non-feature %	IncRNA A PC1 rho	Protein-coding PC1 rho	Rule A thresholds met	Intronic intergenic	Platelet R-squared of metric	Spread of D of metric	Largest observed module	Largest adjusted module	AR I - module eigengene	Largest module eigengene	Rule C HR, plate-stratified (95% CI)	Deaths
THYM	yes	120	3.8	-0.42	0.57	not met	0.05	0.03	0.31	0.20	0.24	0.50	0.39	not met (0.36 to 1.56)	9

409 (b) Solid-tissue normals with at least 20 retained libraries (descriptive)

Project	Normals	IncRNA PC1 rho	Protein-coding PC1 rho	Rule A thresholds met
LUAD	59	0.95	0.66	no
LUSC	51	0.90	0.20	yes
KICH	25	0.84	0.86	no
KIRP	32	0.82	0.73	no
STAD	36	0.76	-0.62	no
BRCA	113	0.71	0.49	no
THCA	59	0.61	-0.03	yes
UCEC	35	0.51	0.45	no
KIRC	72	0.33	0.29	no
PRAD	52	0.26	0.12	no
HNSC	44	0.20	0.24	no
COAD	41	0.14	-0.10	no
LIHC	50	-0.16	0.37	no

410 (c) Random-effects meta-analysis of the metric's hazard ratio for overall survival

Model	Cancer types	Pooled HR (95% CI)	p	I-squared (%)	Prediction interval	HR above 1	p below 0.05
primary (plate stratum)	20	1.00 (0.93 to 1.08)	0.98	50	0.79 to 1.26	10	4
secondary (no plate stratum)	20	1.01 (0.94 to 1.08)	0.83	64	0.79 to 1.29	10	4

411 (d) Post hoc dose-response: within-cohort spread of the metric against the rule quantities,

412 Spearman across 25 cancer types

Spread measure	Outcome	rho	p	FDR		
mad_log_metric		rho	of the lncRNA PC1 with the metric	0.73	3.1e-05	0.00061
mad_log_metric	lncRNA minus protein- coding	rho		0.32	0.12	0.14
mad_log_metric	largest observed module, share of genes	0.69	0.00016	0.0016		
mad_log_metric	adjusted Rand index, observed vs adjusted	-0.33	0.1	0.13		
mad_log_metric	metric variance explained by plate	0.53	0.0062	0.014		
cv_metric		rho	of the lncRNA PC1 with the metric	0.64	0.00053	0.0025
cv_metric	lncRNA minus protein- coding	rho		0.39	0.057	0.081
cv_metric	largest observed module, share of genes	0.50	0.01	0.017		
cv_metric	adjusted Rand index, observed vs adjusted	-0.20	0.33	0.37		
cv_metric	metric variance explained by plate	0.53	0.0062	0.014		
iqr_metric		rho	of the lncRNA PC1 with the metric	0.64	0.00063	0.0025
iqr_metric	lncRNA minus protein- coding	rho		0.41	0.041	0.063
iqr_metric	largest observed module, share of genes	0.53	0.0069	0.014		
iqr_metric	adjusted Rand index, observed vs adjusted	-0.20	0.35	0.37		
iqr_metric	metric variance explained by plate	0.65	0.00046	0.0025		
sd_metric		rho	of the lncRNA PC1 with the metric	0.63	0.00081	0.0027
sd_metric	lncRNA minus protein- coding	rho		0.52	0.0076	0.014
sd_metric	largest observed module, share of genes	0.38	0.06	0.081		
sd_metric	adjusted Rand index, observed vs adjusted	-0.09	0.66	0.66		
sd_metric	metric variance explained by plate	0.62	0.001	0.0029		

Note. Decision rules were fixed in the pipeline configuration, committed and hashed (38_decision_rules_lock.json) before any non-renal project was analysed. The renal projects had been analysed during discovery, so the thresholds were set with those results known. Excluding TCGA-KIRC and TCGA-KIRP, Rule A is met in 8 of 23 projects and Rule C in 2 of 23. The rule letters follow the three measurements of the analysis plan; the second (B), the gene-level and positional-class comparison, had no decision threshold and is reported descriptively. Panel (d) was specified afterwards and is exploratory. Rule A requires a lncRNA PC1 $|\rho|$ of at least 0.5 with the non-feature fraction and at least 0.3 above the protein-coding PC1. Rule C requires the largest observed lncRNA module to hold at least 40% of genes, with an eigengene $|r|$ of at least 0.7 with the metric and an adjusted Rand index below 0.2. Overall survival from the TCGA Clinical Data Resource, censored at ten years, is adjusted for age, sex (omitted in single-sex cancers) and stage, and stratified by sequencing plate, pooling plates with fewer than 10 patients. Stage is AJCC pathologic where recorded for at least 70% of patients, otherwise clinical, otherwise omitted. Projects with at least 20 deaths enter the meta-analysis. Each project uses its own expressed genes. Source tables: 38_pancancer_cohorts.tsv, 39_pancancer_axis.tsv, 39_pancancer_class_gradient.tsv, 40_pancancer_network.tsv, 41_pancancer_metric_survival.tsv, 41_pancancer_metric_meta.tsv, 43_pancancer_dose_response.tsv, 43_pancancer_dose_response_projects.tsv.

Supplementary Table 7 | The lncRNA axis and the non-feature read fraction: extended models, batch structure and biospecimen correlates

(a) Nested Cox models for the leading lncRNA axis, hazard ratio per standard deviation of the axis, overall survival

Model	HR	95% CI	p	C	Proportional-hazards p
Axis alone	1.31	1.12 to 1.53	5.3e-04	0.555	0.14
+ age, sex, T, N, M1, ordinal grade	1.34	1.15 to 1.57	1.9e-04	0.773	0.11
+ ESTIMATE stromal and immune scores	1.36	1.16 to 1.60	2.3e-04	0.773	0.10
+ non-feature fraction, multimapping fraction, log10 assigned reads	1.08	0.81 to 1.43	0.61	0.777	0.093
+ RIN	1.34	1.13 to 1.58	5.5e-04	0.773	0.089
+ RIN + slide necrosis	1.39	1.17 to 1.64	1.5e-04	0.777	0.11
+ RIN + slide necrosis + the three STAR metrics	1.12	0.84 to 1.50	0.44	0.779	0.099
+ RIN + slide necrosis + the three STAR metrics, stratified by sequencing plate	0.90	0.62 to 1.29	0.55	0.760	0.57

Note. TCGA-KIRC, 503 patients and 163 events in every row, those with a recorded RNA integrity number and complete covariates. Source: 23_axis_nested_extended.tsv. The association survives adding RIN and slide necrosis, falls only when the three STAR metrics enter, and disappears with stratification by sequencing plate.

(b) The non-feature read fraction as the exposure, hazard ratio per standard deviation, overall survival

Model	Linear HR (95% CI)	p	log10 HR (95% CI)	p
Non-feature fraction alone	1.28 (1.11 to 1.47)	5.9e-04	1.32 (1.13 to 1.54)	5.1e-04
+ age, sex, T, N, M1, ordinal grade	1.36 (1.18 to 1.56)	2.0e-05	1.41 (1.20 to 1.66)	2.9e-05
+ ESTIMATE stromal and immune scores	1.37 (1.18 to 1.58)	2.0e-05	1.43 (1.21 to 1.69)	2.4e-05

Model	Linear HR (95% CI)	p	log10 HR (95% CI)	p
+ multimapping fraction, log10 assigned reads	1.36 (1.17 to 1.58)	4.5e-05	1.42 (1.20 to 1.68)	4.5e-05
+ ESTIMATE, stratified by tissue source site	1.44 (1.22 to 1.69)	1.3e-05	1.48 (1.24 to 1.78)	2.0e-05
+ RIN	1.35 (1.17 to 1.56)	6.0e-05	1.41 (1.20 to 1.67)	4.6e-05
+ RIN + slide necrosis	1.37 (1.19 to 1.59)	2.5e-05	1.44 (1.22 to 1.71)	1.9e-05
+ RIN + slide necrosis + multimapping, log10 assigned reads	1.37 (1.18 to 1.59)	4.7e-05	1.44 (1.21 to 1.70)	3.0e-05
+ RIN + slide necrosis + multimapping, log10 assigned reads, stratified by plate	1.23 (1.02 to 1.50)	0.035	1.23 (1.00 to 1.51)	0.049

Note. TCGA-KIRC, 503 patients and 163 events in every row, restricted to patients with a recorded RNA integrity number so that the RIN and plate rows are nested. Source: 23_noFeature_exposure_extended.tsv. The headline figures in the main text use the wider complete-case set of 520 patients and 172 events (07_noFeature_as_exposure.tsv, reproduced in Supplementary Table 8c), so confidence limits differ slightly for the same named model. The exposure is right-skewed, so every model is reported on both scales. Unlike the axis in panel (a), the exposure survives stratification by sequencing plate.

(c) Spearman correlations of the metric and of the axis with biospecimen measures

Measure	Non-feature fraction, rho (p, n)	lncRNA axis, rho (p, n)
RNA integrity number	-0.25 (9.4e-09, 527)	-0.27 (3.4e-10, 510)
A260/A280 ratio	0.18 (3.6e-05, 528)	0.23 (1.1e-07, 511)
Slide percent necrosis, mean over slides	-0.05 (0.28, 528)	-0.09 (0.042, 511)
Slide percent tumour nuclei, mean over slides	0.03 (0.45, 528)	0.10 (0.019, 511)
Slide percent stromal cells, mean over slides	-0.10 (0.021, 524)	-0.12 (0.0094, 507)
ESTIMATE stromal score	-0.10 (0.028, 528)	-0.16 (2.0e-04, 511)
ESTIMATE immune score	0.04 (0.39, 528)	0.04 (0.40, 511)

Note. TCGA-KIRC only. No equivalent metadata exist for CPTAC-3, TCGA-KIRP or TCGA-KICH. Source: `23_axis_vs_biospecimen.tsv`. In the same file, the multimapping fraction correlates with RIN at ρ -0.19 (p 1.3e-05), with slide necrosis at 0.12 (p 0.0078) and with the ESTIMATE stromal score at -0.21 (p 1.5e-06), and log10 assigned reads correlates with RIN at 0.12 (p 0.0051). RIN was recorded for 527 of 528 patients. One of those 527, recorded as 0, is a coded missing value, leaving 526 usable, although the regression of panel (e) uses all 527. Slide necrosis, with a median of 0 and an interquartile range of 0 to 0, is uninformative.

(d) Variance in the metric and in the axis explained by batch variables

Quantity	Factor	Levels modelled	Variance explained (%)	ANOVA p	Kruskal-Wallis p	n
Non-feature fraction	Sequencing plate	15	26.6	6.5e-27	4.0e-28	528
Non-feature fraction	BCR batch	15	24.6	4.2e-24	1.0e-25	528
Non-feature fraction	Tissue source site	11	11.1	2.3e-09	8.3e-11	528
lncRNA axis	Sequencing plate	15	26.2	2.1e-25	9.3e-22	511
lncRNA axis	BCR batch	15	26.0	4.3e-25	1.6e-21	511
lncRNA axis	Tissue source site	11	11.6	1.6e-09	3.1e-09	511

Note. One-way analysis of variance in TCGA-KIRC. Source: `23_batch_variance.tsv`. Nineteen sequencing plates and 15 BCR batches are recorded. Plate, batch and site levels with fewer than ten patients are pooled into one "other" stratum, giving 15, 15 and 11 modelled levels. Across the 14 plates with at least ten patients the median non-feature fraction ranges from 2.6% to 7.1%.

(e) Determinants of the non-feature read fraction

Term	df	Variance explained alone (%)	Partial variance explained (%)	p alone	p partial
RNA integrity number	1	6.9	5.4	9.9e-10	1.5e-07
Sequencing plate	14	26.6	24.0	7.4e-27	9.4e-23

Term	df	Variance explained alone (%)	Partial variance explained (%)	p alone	p partial
Tissue source site	10	11.1	7.8	2.3e-09	1.0e-05
RIN + plate + site jointly	25	36.2	--	4.1e-35	--

Note. TCGA-KIRC, 527 patients with a recorded RNA integrity number. Source: 23_noFeature_determinants.tsv. Partial variance explained is conditional on the other two terms. Plate, batch and RIN proxy unrecorded extraction, library-preparation and run variables, so 36.2% is a lower bound on the technical share.

Supplementary Table 8 | Disease-specific and progression-free endpoints

(a) Endpoint coverage in the discovery cohort

Endpoint	Patients scoreable	Events after ten-year censoring	Unscoreable	Median follow-up (days)	Events per variable at 12 parameters
Overall survival, pipeline	528	173	0	1,657	14.4
Overall survival, Clinical Data Resource	528	173	0	1,666	14.4
Disease-specific survival	517	109	11	1,495	9.1
Progression-free interval	526	158	2	1,471	13.2
Disease-free interval	114	14	414	1,525	1.2, not analysed

Note. Source: 30_cdr_coverage_and_os_concordance.tsv. The Clinical Data Resource overall-survival endpoint matches the pipeline endpoint on all 528 event indicators, with times correlated at Spearman 0.998. Six patients differ in time only, 5 from benign data-freeze differences and 1 flagged as not explained by a later freeze. All 11 patients unscorable for disease-specific survival are pipeline deaths, and both unscorable for the progression-free interval carry an event. Event-indicator dependence (ϕ) is 0.77 between overall and disease-specific survival, 0.52 between overall survival and the progression-free interval, and 0.79 between the latter two. The four endpoints are therefore not independent, and adjustment across

them is crude in both directions. The Clinical Data Resource labels disease-specific survival approximate for this disease because cause of death is unverified. No patient is marked redacted. These are sensitivity endpoints, not independent tests.

(b) The five prognostic modules under three endpoints, principal specification

Module	Overall survival	Disease-specific survival	Progression-free interval
Protein-coding green	0.62 (0.53 to 0.74)	0.49 (0.39 to 0.61)	0.70 (0.59 to 0.83)
Protein-coding purple	1.28 (1.09 to 1.50)	1.49 (1.21 to 1.84)	1.15 (0.96 to 1.38), p 0.14
lncRNA blue	0.71 (0.60 to 0.83)	0.61 (0.49 to 0.75)	0.80 (0.68 to 0.96)
lncRNA greenyellow	1.35 (1.15 to 1.60)	1.50 (1.22 to 1.85)	1.21 (1.01 to 1.45), FDR 0.065
lncRNA turquoise	1.24 (1.06 to 1.45)	1.20 (1.00 to 1.44), p 0.054	1.03 (0.88 to 1.22), p 0.70

Note. Hazard ratio per standard deviation of the module eigengene. Sample sizes and events for protein-coding modules are 502 and 165 (overall survival), 491 and 103 (disease-specific survival) and 500 and 151 (progression-free interval), and for lncRNA modules 504 and 163, 493 and 102, and 502 and 149. Source: `30_endpoint_modules.tsv`. The false discovery rate is controlled across the five modules within each endpoint, not across the 20 module-by-endpoint tests. Unmarked cells are below 0.05. Sixteen of the 20 tests reach a 5% false discovery rate. The Clinical Data Resource overall-survival column agrees with the pipeline column to within 0.01 throughout.

(c) The non-feature read fraction as exposure, by endpoint

Endpoint	n / events	Alone	+ clinical model	log10 scale, + clinical model
Overall survival, pipeline	520 / 172	1.28 (1.13 to 1.46), p 1.2e-04	1.36 (1.20 to 1.55), p 3.0e-06	1.44 (1.23 to 1.68)
Overall survival, Clinical Data Resource	520 / 172	1.28 (1.13 to 1.46)	1.36 (1.19 to 1.54), p 4.0e-06	1.43 (1.22 to 1.67)
Disease-specific survival	509 / 109	1.20 (1.02 to 1.42), p 0.032	1.31 (1.10 to 1.55), p 0.0020	1.35 (1.11 to 1.64)
Progression-free interval	518 / 157	1.03 (0.88 to 1.20), p 0.71	1.11 (0.95 to 1.30), p 0.19	1.07 (0.91 to 1.26), p 0.44

(d) The projected lncRNA axis, nested sequence, by endpoint

Endpoint	n / events	Alone	+ clinical model	+ ESTIMATE	+ the three STAR metrics
Overall survival, pipeline	504 / 163	1.31 (1.13 to 1.53)	1.34 (1.15 to 1.57)	1.36 (1.16 to 1.60)	1.08 (0.81 to 1.43), p 0.61
Overall survival, Clinical Data Resource	504 / 163	1.31	1.33	1.35	1.07 (0.81 to 1.41), p 0.66
Disease-specific survival	493 / 102	1.20 (0.98 to 1.45), p 0.073	1.28 (1.04 to 1.56), p 0.017	1.30 (1.05 to 1.61)	0.99 (0.70 to 1.39), p 0.94
Progression-free interval	502 / 149	0.97 (0.82 to 1.15), p 0.73	1.01 (0.85 to 1.19), p 0.92	1.09 (0.91 to 1.31)	0.91 (0.67 to 1.24), p 0.57

Note for panels (c) and (d). Source: `30_endpoint_exposure.tsv`, `30_endpoint_axis_nested.tsv`. The metric and the axis are prognostic for overall and disease-specific survival but not for progression. The axis's variance-inflation factor reaches 4.5 once the three STAR metrics enter, so its collapse to 1.08 reflects collinearity as much as adjustment.

(e) Cross-validated increment in the concordance index from adding the 29 module eigengenes to the clinical model

Endpoint	n	Events	C, clinical	C, clinical plus eigengenes	Mean over repeats	Bootstrap increment	95% CI	p	BH over the four endpoints
Overall survival, pipeline	493	159	0.7551	0.7596	0.0045	0.0081	-0.0077 to 0.33 0.0246		0.33
Overall survival, Clinical Data Resource	493	159	0.7540	0.7588	0.0048	0.0087	-0.0075 to 0.30 0.0258		0.33
Disease-specific survival	482	99	0.8230	0.8396	0.0166	0.0227	0.0037 to 0.017 0.0448		0.068
Progression-free interval	491	146	0.8007	0.8123	0.0116	0.0138	0.0011 to 0.036 0.0272		0.072

Note. Ten repeats of ten-fold cross-validation, with module definitions fixed on the full cohort and the penalty tuned inside each training fold. Source: `30_endpoint_cv_increment.tsv`.

The mean over repeats has no interval. The confidence interval and p value belong only to the bootstrap increment, the increment of the repeat-averaged out-of-fold predictor. Neither positive endpoint survives Benjamini-Hochberg adjustment over the four endpoints. Across 200 bootstrap seeds the lower bound ranges from 0.0010 to 0.0046 for disease-specific survival, excluding zero in 200 of 200, and from 0.0004 to 0.0025 for the progression-free interval, also 200 of 200, against 0 of 200 for overall survival.

(f) The locked model of the main analysis re-scored against each endpoint, apparent and in sample

Endpoint	Comparator	n	Events	Modules retained	C	C	Increment	95% CI	p	BH over the eight rows
Overall survival, pipeline	clinical	493	159	9	0.7662	0.7867	0.0205	0.0022 to 0.0400	0.026	0.052
Overall survival, Clinical Data Resource	clinical	493	159	9	0.7649	0.7858	0.0209	0.0024 to 0.0407	0.024	0.052
Disease-specific survival	clinical	482	99	9	0.8218	0.8475	0.0257	0.0059 to 0.0488	0.010	0.052
Progression-free interval	clinical	491	146	9	0.7924	0.8021	0.0098	-0.0123 to 0.0319	0.40	0.45
Overall survival, pipeline	augmented	493	159	5	0.7750	0.7916	0.0165	-0.0005 to 0.0347	0.056	0.075
Overall survival, Clinical Data Resource	augmented	493	159	5	0.7739	0.7910	0.0171	-0.0003 to 0.0356	0.052	0.075
Disease-specific survival	augmented	482	99	5	0.8218	0.8458	0.0240	0.0051 to 0.0475	0.016	0.052
Progression-free interval	augmented	491	146	5	0.7850	0.7925	0.0074	-0.0139 to 0.0293	0.52	0.52

Note. Source: `30_endpoint_locked_model.tsv`. Every row is apparent. The coefficients were fitted on the overall survival of exactly these patients, whose disease-specific deaths are a subset of the deaths used in fitting, as the file's own evaluation column states. On the same 482 patients the apparent overall-survival increment is 0.0239, essentially identical to the disease-specific figure, so the locked-model rows carry no information about differences between endpoints. The cross-validated rows of panel (e) do.

(g) Rebuilding both networks inside every training fold, disease-specific survival

Analysis	Endpoint	n	Events	C clinical	C combined	Increment	SD	Range	Assignments
Ten by ten cross-validation, modules fixed	disease-specific	482	99	0.8265	0.8493	0.0227	0.0050	0.0037 to 0.0448 (bootstrap)	10 repeats
Three folds by five assignments, modules fixed	disease-specific	482	99	0.8141	0.8271	0.0129	0.0124	-0.0005 to 0.0283	5
Three folds by five assignments, modules rebuilt inside each fold	disease-specific	482	99	0.8141	0.8268	0.0126	0.0083	0.0000 to 0.0232	5
Three folds by five assignments, modules fixed	overall survival	482	148	0.7403	0.7498	0.0095	0.0096	-0.0024 to 0.0229	5
Three folds by five assignments, modules rebuilt inside each fold	overall survival	482	148	0.7403	0.7478	0.0075	0.0075	-0.0046 to 0.0143	5

Note. Source: `30_endpoint_foldwise.tsv`. The fixed-module three-fold arm shares folds and inner splits with the rebuilt arm, so the two differ only in module definition. Rebuilding both networks and all loadings inside every training fold costs 0.0003 for disease-specific and 0.0020 for overall survival, so the positive disease-specific increment is not an artefact of defining modules on the full cohort. The matched fixed-module arm shows that the drop from 0.0227 to about 0.013 reflects the coarser three-fold cross-validation (two-thirds of the data for training, no

averaging over repeats). Rebuilt training-fold networks resolve 11 to 19 protein-coding and 11 to 17 lncRNA modules, giving 24 to 35 module eigengenes against the 29 fixed ones. The range across five fold assignments is not a confidence interval. CPTAC-3 lacks the disease-specific and progression-free endpoints, so neither positive result is externally replicated.

Supplementary Table 9 | What the non-feature read fraction correlates with, and what attenuates it

(a) Candidate correlates of the metric and of the axis

Candidate	Matrix	rho with the metric (FDR)	rho with the lncRNA axis (FDR)	Interpretable as biology
Protein-coding red eigengene	observed	0.84 (3.5e-133)	0.92 (1.5e-204)	no
Protein-coding red eigengene	residualised	0.20 (1.2e-04)	0.42 (6.8e-22)	yes
lncRNA share of total FPKM	observed	0.73 (7.1e-89)	0.74 (7.7e-89)	no
Protein-coding pink eigengene	observed	0.32 (2.8e-12)	0.22 (8.3e-06)	no
Protein-coding tan eigengene	observed	-0.20 (1.7e-05)	--	no
Hypoxia metagene, Buffa 2010 ⁵⁸	observed	-0.14 (0.0056)	-0.12 (0.019)	yes
Proliferation score, 10 genes	observed	-0.01 (0.91)	-0.10 (0.031)	yes
ESTIMATE stromal score	--	-0.10 (0.056)	-0.16 (8.1e-04)	yes
ESTIMATE immune score	--	0.04 (0.60)	0.04 (0.40)	yes
ESTIMATE score	--	-0.02 (0.67)	-0.06 (0.22)	yes
Slide percent tumour nuclei	--	0.03 (0.60)	0.10 (0.031)	yes
Slide percent stromal cells	--	-0.10 (0.055)	-0.12 (0.019)	yes

Note. TCGA-KIRC. Eigengene rows use 506 patients against the metric and 497 against the axis, other rows 528 or 511. Source: 27_metric_biology_correlations.tsv. Benjamini-Hochberg families are declared per matrix. Eigengenes of the observed expression matrix cannot

be read as biology, because the metric was measured on that matrix. All hypoxia sensitivity variants run negative: rho -0.18 (p 3.6e-05) for the metagene restricted to genes with a median FPKM of at least 1, and -0.14 and -0.15 for the two alternative tubulin mappings. A higher non-feature fraction thus goes with a lower hypoxia score, which is evidence against the hypoxia reading.

(b) Which protein-coding genes carry the metric's signal

Module	Genes	Median per-gene rho with the metric	Fraction with above 0.3 in absolute value	Eigengene rho, observed	Eigengene rho, residualised	Top GO term (adjusted p)
red	473	0.68	94.5%	0.84	0.20	RNA splicing (4.5e-15)
pink	270	0.22	37.0%	0.32	0.05	none
tan	152	-0.16	2.6%	-0.20	-0.03	none
salmon	63	0.06	11.1%	0.15	0.01	none
yellow	986	0.05	14.2%	0.13	0.00	none
magenta	227	0.03	17.6%	0.12	0.02	none
blue	994	0.04	12.1%	0.10	-0.01	none
purple	216	0.01	15.3%	0.07	0.01	chromosome segregation (1.5e-44)
greenyellow	164	0.01	16.5%	0.06	0.05	none
turquoise	2,405	0.01	11.7%	0.06	0.04	none
black	323	-0.04	1.9%	-0.04	-0.03	none
green	670	-0.01	7.5%	0.04	0.02	small molecule catabolic process (4.1e-38)
brown	988	-0.05	15.8%	-0.01	-0.01	none

Note. Per-gene Spearman correlation with the metric on the observed matrix, 506 samples. Source: 27_module_gene_level_metric_correlation.tsv, GO terms from 06_GO_enrichment_top10_per_module.tsv. The only protein-coding module tracking the metric is the RNA-splicing and mRNA-processing module, as a shift towards unspliced and

543 intronic reads predicts. This identifies the affected genes, not how much of the metric is
 544 biological.

545 **(c) Variance in the metric explained by candidate blocks, with negative controls**

Block	Kind	Interpretable as biology	df	Variance explained (%)	Partial variance explained given the other block (%)
Annotated biology: hypoxia, proliferation, ESTIMATE stromal and immune, slide tumour nuclei	biological	yes	5	6.1	5.0
Technical: RIN, sequencing plate, tissue source site	technical	no, but interpretable as technical	24	36.8	36.0
Observed expression space, all terms	upper bound, not identifiable as biology	no	18	77.9	71.5
13 observed protein-coding eigengenes alone	upper bound	no	13	77.3	71.1
Observed protein-coding red eigengene alone	upper bound	no	1	64.7	60.4
Negative control: 13 eigengenes of random gene sets of the same sizes, observed matrix	negative control	no	13	61.0	54.3
Negative control: 13 leading principal components of the observed matrix	negative control	no	13	80.9	76.6
13 residualised protein-coding eigengenes	residualised expression space	no	13	0.01	6.9
Negative control: 13 random gene-set eigengenes, residualised matrix	negative control	no	13	0.01	2.9
Annotated biology plus observed eigengenes excluding red	upper bound	no	17	40.6	41.4

546 *Note.* TCGA-KIRC, 501 complete cases, linear scale. Source:
 547 `27_noFeature_variance_partition.tsv`. Only the annotated-biology and technical
 548 blocks are interpretable shares. Values are given to one decimal, or to two for the two
 549 residualised rows, which would otherwise round to zero. Thirteen random gene sets with no

550 biological content reach 61.0% and thirteen unsupervised components 80.9%, more than the
 551 block containing the real eigengenes. On the residualised matrix the same eigengenes explain
 552 0.01%. These rows measure how much of the metric the observed expression matrix recovers, a
 553 property of the shared alignment and FPKM normalisation, not a biological cause. The random-
 554 gene-set row is one draw. Over 20 draws the mean is 60.8% (range 51.7% to 70.8%). The 77.9%
 555 and 77.3% rows are references for the negative controls, not estimates of any biological share.
 556 Plate, batch and RIN only proxy unrecorded pre-analytical variables, so 36.8% is a lower bound
 557 on the technical share and 6.1% an upper bound on the annotated-biological share, as far as these
 558 data can measure.

559 **(d) What attenuates the hazard of the metric, and what does not**

Model added to the base	Interpretable	HR	95% CI	p	Standard-error inflation
Base: metric per SD + clinical model	--	1.37	1.20 to 1.57	4.8e-06	1.00
+ hypoxia metagene	yes	1.38	1.20 to 1.59	4.2e-06	1.02
+ proliferation score	yes	1.37	1.19 to 1.56	5.8e-06	1.00
+ ESTIMATE stromal and immune	yes	1.38	1.20 to 1.58	6.4e-06	1.03
+ slide percent tumour nuclei	yes	1.37	1.20 to 1.57	6.0e-06	1.01
+ RIN	yes	1.35	1.18 to 1.55	1.7e-05	1.02
+ sequencing plate stratum	yes	1.24	1.04 to 1.48	0.018	1.32
+ residualised protein-coding red eigengene	yes	1.40	1.21 to 1.63	5.9e-06	1.09
+ all 13 residualised eigengenes	yes	1.39	1.19 to 1.62	2.3e-05	1.14
+ every interpretable candidate jointly, residualised	yes	1.23	1.01 to 1.50	0.043	1.49
+ observed protein-coding red eigengene	no, collinearity diagnostic	1.14	0.89 to 1.45	0.31	1.83

Model added to the base	Interpretable	HR	95% CI	p	Standard-error inflation
+ all 13 observed eigengenes	no, collinearity diagnostic	1.13	0.85 to 1.49	0.40	2.07
+ lncRNA share of total FPKM	no, collinearity diagnostic	1.27	1.04 to 1.54	0.018	1.45

Note. TCGA-KIRC, 501 patients and 165 events, overall survival, hazard ratio per standard deviation of the metric on the linear scale. Source: `27_metric_attenuation.tsv`. Rows marked as collinearity diagnostics add a covariate that is largely the exposure itself, measured through the expression matrix. For the lncRNA axis (492 patients, 159 events), the same sequence gives a base hazard ratio of 1.36 (1.17 to 1.59), falling to 1.13 (0.94 to 1.36, p 0.18) with stratification by plate and rising to 1.46 (1.22 to 1.75) with all 13 residualised eigengenes.

(e) Collinearity diagnostics for the two rows that appear to attenuate

Candidate	Pearson r with the exposure	Variance inflation	Correlation of the orthogonal part with the exposure	Synthetic-proxy null, median HR (2.5 to 97.5 centile)	Contains the observed value
Observed protein-coding red eigengene	0.80	2.89	0.99	1.37 (1.08 to 1.71)	yes, observed 1.14
lncRNA share of total FPKM	0.72	2.14	--	1.37 (1.15 to 1.62)	yes, observed 1.27

Note. TCGA-KIRC, 501 patients and 165 events, 300 synthetic draws. Source: `27_collinearity_diagnostics.tsv`. A synthetic covariate correlating 0.80 with the metric but carrying no survival information reproduces the apparent attenuation. The five residualised eigengenes, by comparison, correlate with the exposure between -0.003 and 0.003, with variance inflation of at most 1.02.

(f) The metric against clinical, batch and biospecimen strata

Variable	Test	n	Summary	p
RIN tertile	Kruskal-Wallis	527	medians 6.0%, 4.3%, 4.1%	4.7e-07

Variable	Test	n	Summary	p
Sequencing plate, 15 pooled levels	Kruskal-Wallis	528	medians 2.6% to 7.1%	4.0e-28
BCR batch, 14 levels	Kruskal-Wallis	527	medians 2.6% to 6.9%	3.5e-26
Sex	Kruskal-Wallis	528	5.0% in women, 4.5% in men	0.21
Age tertile	Kruskal-Wallis	528	4.7%, 4.6%, 4.3%	0.74
Stage group	Wilcoxon	525	4.5% in stage I to II, 5.1% in III to IV	0.35
Grade group	Wilcoxon	520	4.5% in G1 to G2, 4.7% in G3 to G4	0.97
M1	Wilcoxon	528	4.6% M0, 4.8% M1	0.71
T category	Spearman	528	rho 0.06	0.21
Age	Spearman	528	rho -0.04	0.41
Ordinal grade	Spearman	520	rho -0.03	0.57

Note. TCGA-KIRC. Source: 27_noFeature_reverse_check.tsv, which reuses the clinical rows of 07_noFeature_vs_clinical.tsv unchanged. The smallest p for any clinical variable is 0.21, whereas RIN, plate and batch each associate with the metric at p below 1e-06. The batch row drops the single-patient "other" stratum kept in Supplementary Table 7(d), so it has 14 levels on 527 patients, not 15 on 528.

Supplementary Table 10 | Single-cell ground truth for the correction, TCGA-KIRC

(a) Agreement of the bulk tumour-minus-normal difference with the single-cell malignant-versus-normal-epithelium fold change

Contrast	Correction	IncRNAs	Single-cell significant	rho observed	rho corrected	Change (95% interval)	Sign agreement observed	Sign agreement corrected
all pairs	tumour fitted	507	213	0.775	0.771	-0.004 (-0.008 to -0.001)	92.5%	92.0%
all pairs	joint fitted	507	213	0.775	0.770	-0.005 (-0.009 to -0.002)	92.5%	92.0%
shared-plate pairs	tumour fitted	507	213	0.763	0.760	-0.003 (-0.006 to 0.000)	92.0%	92.5%

Contrast	Correction	lncRNAs	Single-cell significant	rho observed	rho corrected	Change (95% interval)	Sign agreement observed	Sign agreement corrected
shared-plate pairs	joint fitted	507	213	0.763	0.760	-0.003 (-0.006 to -0.000)	92.0%	92.5%
unpaired, all tumours vs all normals (added after review)	tumour fitted	507	213	0.773	0.774	0.001 (-0.013 to 0.014)	93.9%	91.5%
unpaired, all tumours vs all normals (added after review)	joint fitted	507	213	0.773	0.776	0.002 (-0.011 to 0.016)	93.9%	91.5%

581 Tumour-minus-normal difference in the correction covariates, in pooled standard-deviation units.
582 The unpaired contrast carries the between-library difference in alignment metrics that the
583 correction can remove, the paired contrast very little of it.

Covariate	Unpaired difference	Mean paired difference
pct_noFeature	0.86	0.13
pct_multimapping	0.30	-0.24
log_depth	-0.37	-0.21

584 *(b) lncRNA scores in 532 tumours against the malignant-marker score and the non-feature*
585 *fraction*

Gene set	Genes	Correction	rho with marker score (95% interval)	rho with non-feature fraction (95% interval)
single-cell tier 1 (tumour-specific)	38	observed	0.55 (0.48 to 0.62)	0.49 (0.43 to 0.55)
single-cell tier 1 (tumour-specific)	38	tumour fitted	0.48 (0.41 to 0.55)	0.01 (-0.07 to 0.09)
single-cell tier 1 (tumour-specific)	38	joint fitted	0.48 (0.40 to 0.56)	0.01 (-0.08 to 0.09)
single-cell tier 2 (malignant-compartment)	77	observed	0.42 (0.35 to 0.49)	0.61 (0.56 to 0.66)
single-cell tier 2 (malignant-compartment)	77	tumour fitted	0.35 (0.27 to 0.43)	0.06 (-0.02 to 0.14)
single-cell tier 2 (malignant-compartment)	77	joint fitted	0.35 (0.27 to 0.43)	0.05 (-0.03 to 0.13)

Gene set	Genes	Correction	rho with marker score (95% interval)	rho with non-feature fraction (95% interval)
single-cell malignant-up vs normal epithelium (meta FDR < 0.05)	106	observed	0.52 (0.45 to 0.59)	0.71 (0.66 to 0.76)
single-cell malignant-up vs normal epithelium (meta FDR < 0.05)	106	tumour fitted	0.47 (0.39 to 0.54)	0.07 (-0.00 to 0.15)
single-cell malignant-up vs normal epithelium (meta FDR < 0.05)	106	joint fitted	0.47 (0.39 to 0.54)	0.07 (-0.01 to 0.15)
abundance-matched single-cell non-specific (null, 1000 draws)	38	observed	0.37 (0.32 to 0.41)	0.64 (0.57 to 0.71)
abundance-matched single-cell non-specific (null, 1000 draws)	38	tumour fitted	0.28 (0.22 to 0.34)	0.10 (0.07 to 0.12)
abundance-matched single-cell non-specific (null, 1000 draws)	38	joint fitted	0.28 (0.22 to 0.33)	0.08 (0.06 to 0.11)

586 (c) The 15 single-cell candidates set aside in bulk as technically sensitive

lncRNA	Tier	Single-cell log2FC vs normal epithelium	rho with metric, observed	rho with metric, corrected	rho with marker score, observed	rho with marker score, corrected	Bulk tumour minus normal, observed	Bulk tumour minus normal, corrected
THBS3-AS1	tier2	--	0.84	0.10	0.27	0.14	--	--
LMNTD2-AS1	tier2	--	0.81	0.09	0.35	0.26	--	--
LINC01473	tier2	--	0.70	-0.04	0.25	0.14	--	--
LINC01311	tier2	--	0.73	0.08	0.32	0.22	--	--
AL359504.1	tier2	--	0.75	0.11	0.33	0.21	--	--
AL161669.3	tier2	--	0.69	0.07	0.40	0.34	--	--
NR2F2-AS1	tier1	1.42	0.69	0.11	0.40	0.34	-0.10	-0.14
GAS6-AS1	tier2	--	0.65	0.05	0.33	0.19	--	--
PVT1	tier1	5.64	0.57	-0.03	0.37	0.26	0.54	0.52
LINC02615	tier1	3.01	0.60	0.05	0.22	0.09	0.37	0.34
LINC02298	tier2	--	0.55	0.01	0.50	0.43	--	--
AL080317.2	tier2	--	0.58	0.01	0.21	0.06	--	--
PPP1R26-AS1	tier2	--	0.55	0.01	0.24	0.12	--	--
HIF1A-AS3	tier1	6.38	0.58	0.09	0.62	0.59	1.89	1.75
FAM153CP	tier1	5.52	0.50	-0.08	0.33	0.17	0.16	0.12

587 *Note.* 532 tumours, 72 normals and 72 tumour-normal pairs (68 on a shared plate) from the pan-
588 cancer build, which retains the 4 patients excluded from discovery for survival time. Of 4,088
589 lncRNAs expressed in either group, 507 have a single-cell estimate. The malignant-marker score
590 (CA9, NDUFA4L2, ANGPTL4, VEGFA, EGLN3) itself correlates with the non-feature fraction
591 at Spearman 0.22. The null row is 1,000 abundance-matched sets of single-cell non-specific
592 lncRNAs (median, and 2.5th to 97.5th percentiles). Contrary to the pre-recorded expectation,
593 these sets are not neutral. The single-cell candidates were locked before any bulk step. Source
594 tables: 42_sc_truth_concordance.tsv, 42_sc_truth_score_coherence.tsv,
595 42_sc_truth_technical_candidates.tsv, 42_sc_truth_gene_level.tsv.
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597 **Supplementary Table 11 | Residualisation against RUV-III with pseudo-replicates,**
598 **TCGA-KIRC**

599 *(a) The leading components and per-gene correlation with the non-feature fraction*

Version	lncRNA PC1 rho with metric	Protein-coding PC1 rho with metric	Plate R-squared of lncRNA PC1	Median per-gene abs rho, lncRNAs
observed	0.91	0.08	0.26	0.38
residualised	0.06	0.03	0.06	0.04
RUV-III (k = 3)	0.05	0.01	0.05	0.05
RUV-III (k = 1)	0.10	0.19	0.04	0.10
RUV-III (k = 5)	0.02	-0.01	0.05	0.05

600 *(b) The lncRNA network on the 3,442 discovery lncRNAs and 511 tumours, discovery parameters*

Version	Modules	Largest module	Unassigned	ARI vs observed	ARI vs residualised	ARI vs RUV-III
observed	3	57.6%	36.9%	1.00	0.07	0.07
residualised	14	13.5%	29.5%	0.07	1.00	0.23
RUV-III (k = 3)	10	14.1%	45.1%	0.07	0.23	1.00

(c) Single-cell ground truth: coherence of the 38 tumour-specific lncRNAs with the malignant-marker score

Version	rho with marker score	Null median (500 abundance-matched sets)	Excess over null	rho with metric
observed	0.55	0.36	0.20	0.49
residualised	0.48	0.26	0.22	0.01
RUV-III (k = 3)	0.28	0.17	0.11	0.03

(d) The five prognostic modules, fixed discovery loadings, overall survival (Clinical Data Resource) adjusted for age, sex and stage

Version	Module	Patients	Deaths	HR per SD (95% CI)	p
observed	protein-coding green	525	172	0.65 (0.56 to 0.75)	7.2e-09
observed	protein-coding purple	525	172	1.45 (1.27 to 1.66)	4.6e-08
observed	lncRNA blue	525	172	0.77 (0.66 to 0.90)	0.0014
observed	lncRNA greenyellow	525	172	1.44 (1.26 to 1.65)	1.5e-07
observed	lncRNA turquoise	525	172	1.42 (1.24 to 1.63)	3.2e-07
residualised	protein-coding green	525	172	0.66 (0.57 to 0.76)	2.7e-08
residualised	protein-coding purple	525	172	1.44 (1.26 to 1.65)	8.8e-08
residualised	lncRNA blue	525	172	0.70 (0.60 to 0.81)	2.8e-06
residualised	lncRNA greenyellow	525	172	1.35 (1.17 to 1.56)	3.2e-05
residualised	lncRNA turquoise	525	172	1.32 (1.13 to 1.53)	0.00034
RUV-III (k = 3)	protein-coding green	525	172	0.64 (0.55 to 0.74)	2.2e-09
RUV-III (k = 3)	protein-coding purple	525	172	1.48 (1.30 to 1.69)	3.3e-09
RUV-III (k = 3)	lncRNA blue	525	172	0.64 (0.55 to 0.74)	8.7e-09
RUV-III (k = 3)	lncRNA greenyellow	525	172	1.37 (1.19 to 1.58)	1.8e-05
RUV-III (k = 3)	lncRNA turquoise	525	172	1.33 (1.13 to 1.56)	0.00055

Note. Post hoc sensitivity analysis of 528 TCGA-KIRC tumours with a ccA/ccB class from the pan-cancer build. RUV-III (ruv package) removed k unwanted factors using 12 replicate sets of 66 pseudo-samples (plate, and the extremes of the non-feature fraction and of depth, within ccA/ccB by stage groups) and 1,000 negative-control genes. The single-cell null uses 500 draws,

609 so its values differ slightly from Supplementary Table 10. Source tables:
610 44_prps_design.tsv, 44_prps_axis.tsv, 44_prps_network.tsv,
611 44_prps_sc_truth.tsv, 44_prps_module_hr.tsv.
612 ---

613 **Supplementary Table 12 | Module associations with and without adjustment for the**
614 **alignment metrics, all 29 modules**

615 **(a) Clinical covariate set**

Network	Module	HR without adjustment (FDR)	HR with STAR metrics as covariates (FDR)	HR with adjustment for the alignment metrics (FDR)	Status
lncRNA	blue	0.85 (0.096)	0.73 (0.0091)	0.73 (0.0015)	gained
lncRNA	greenyellow	1.42 (8.6e-05)	1.34 (0.0091)	1.31 (0.0092)	robust
lncRNA	turquoise	1.44 (1.3e-05)	1.47 (0.056)	1.26 (0.024)	robust
lncRNA	cyan	0.88 (0.19)	0.78 (0.089)	0.78 (0.031)	gained
lncRNA	black	1.45 (1.7e-05)	1.49 (0.089)	1.28 (0.042)	robust
lncRNA	lightcyan	0.88 (0.18)	0.82 (0.089)	0.81 (0.042)	gained
lncRNA	midnightblue	0.99 (0.93)	0.81 (0.13)	0.81 (0.043)	gained
lncRNA	brown	1.34 (1.9e-04)	1.22 (0.10)	1.17 (0.079)	lost
lncRNA	tan	1.10 (0.33)	0.90 (0.45)	0.87 (0.20)	not significant
lncRNA	magenta	1.29 (0.0011)	1.16 (0.19)	1.14 (0.22)	lost
lncRNA	purple	1.29 (0.0013)	1.15 (0.26)	1.12 (0.22)	lost
lncRNA	green	1.31 (0.0013)	1.07 (0.82)	0.98 (0.90)	lost
lncRNA	pink	1.19 (0.049)	1.01 (0.94)	0.99 (0.90)	lost
lncRNA	red	1.21 (0.047)	1.03 (0.85)	1.01 (0.90)	lost
lncRNA	salmon	1.30 (0.0031)	1.03 (0.85)	0.98 (0.90)	lost
lncRNA	yellow	1.18 (0.047)	0.97 (0.85)	0.97 (0.90)	lost
protein-coding	green	0.67 (1.5e-05)	0.68 (8.5e-05)	0.69 (3.8e-05)	robust
protein-coding	purple	1.27 (0.012)	1.22 (0.071)	1.24 (0.036)	robust
protein-coding	turquoise	0.80 (0.014)	0.78 (0.071)	0.81 (0.036)	robust
protein-coding	pink	0.91 (0.36)	0.80 (0.097)	0.85 (0.10)	not significant
protein-coding	magenta	0.86 (0.14)	0.84 (0.17)	0.87 (0.16)	not significant

Network	Module	HR without adjustment (FDR)	HR with STAR metrics as covariates (FDR)	HR with adjustment for the alignment metrics (FDR)	Status
protein-coding	red	1.45 (2.5e-05)	1.27 (0.18)	1.17 (0.16)	lost
protein-coding	brown	1.18 (0.11)	1.15 (0.18)	1.14 (0.18)	not significant
protein-coding	tan	1.10 (0.36)	1.15 (0.18)	1.15 (0.18)	not significant
protein-coding	blue	0.89 (0.35)	0.86 (0.18)	0.88 (0.22)	not significant
protein-coding	greenyellow	0.90 (0.35)	0.90 (0.36)	0.92 (0.47)	not significant
protein-coding	yellow	0.98 (0.77)	0.92 (0.36)	0.93 (0.47)	not significant
protein-coding	black	1.04 (0.73)	1.05 (0.58)	1.06 (0.50)	not significant
protein-coding	salmon	1.03 (0.77)	0.97 (0.76)	0.99 (0.93)	not significant

616 **(b) Clinical covariate set with the ESTIMATE stromal and immune scores added**

Network	Module	HR without adjustment (FDR)	HR with STAR metrics as covariates (FDR)	HR with adjustment for the alignment metrics (FDR)	Status
lncRNA	blue	0.80 (0.018)	0.68 (5.0e-04)	0.67 (2.1e-05)	robust
lncRNA	greenyellow	1.45 (3.3e-05)	1.38 (0.0026)	1.34 (0.0042)	robust
lncRNA	black	1.47 (9.1e-06)	1.56 (0.061)	1.31 (0.023)	robust
lncRNA	cyan	0.87 (0.17)	0.76 (0.052)	0.76 (0.023)	gained
lncRNA	lightcyan	0.86 (0.11)	0.81 (0.061)	0.79 (0.023)	gained
lncRNA	turquoise	1.50 (8.5e-06)	1.55 (0.039)	1.28 (0.023)	robust
lncRNA	tan	1.06 (0.62)	0.81 (0.11)	0.78 (0.033)	gained
lncRNA	midnightblue	1.00 (0.98)	0.80 (0.11)	0.81 (0.044)	gained
lncRNA	brown	1.39 (2.5e-04)	1.24 (0.11)	1.16 (0.19)	lost
lncRNA	magenta	1.29 (0.0034)	1.13 (0.32)	1.10 (0.44)	lost
lncRNA	purple	1.30 (0.0020)	1.15 (0.29)	1.10 (0.44)	lost
lncRNA	red	1.35 (0.0034)	1.05 (0.77)	1.12 (0.46)	lost
lncRNA	pink	1.17 (0.11)	0.96 (0.77)	0.94 (0.54)	not significant
lncRNA	yellow	1.68 (8.2e-05)	1.44 (0.11)	1.15 (0.54)	lost
lncRNA	salmon	1.28 (0.0047)	0.99 (0.93)	0.96 (0.67)	lost
lncRNA	green	1.33 (9.3e-04)	1.09 (0.64)	0.99 (0.93)	lost
protein-coding	green	0.605 (2.0e-08)	0.611 (2.0e-07)	0.608 (1.4e-08)	robust
protein-coding	black	1.50 (0.0054)	1.33 (0.092)	1.52 (0.0044)	robust
protein-coding	purple	1.30 (0.0051)	1.28 (0.017)	1.27 (0.013)	robust
protein-coding	turquoise	0.81 (0.024)	0.78 (0.065)	0.82 (0.052)	lost

Network	Module	HR without adjustment (FDR)	HR with STAR metrics as covariates (FDR)	HR with adjustment for the alignment metrics (FDR)	Status
protein-coding	pink	0.93 (0.57)	0.80 (0.092)	0.86 (0.13)	not significant
protein-coding	magenta	0.86 (0.21)	0.82 (0.092)	0.88 (0.22)	not significant
protein-coding	red	1.48 (3.1e-05)	1.29 (0.12)	1.16 (0.22)	lost
protein-coding	tan	1.08 (0.57)	1.13 (0.24)	1.14 (0.22)	not significant
protein-coding	brown	1.16 (0.21)	1.16 (0.24)	1.11 (0.39)	not significant
protein-coding	blue	0.93 (0.59)	0.78 (0.12)	0.91 (0.51)	not significant
protein-coding	yellow	1.18 (0.59)	1.09 (0.77)	0.82 (0.51)	not significant
protein-coding	greenyellow	0.94 (0.59)	0.92 (0.56)	0.97 (0.82)	not significant
protein-coding	salmon	1.06 (0.59)	0.97 (0.77)	1.01 (0.92)	not significant

Note for panels (a) and (b). TCGA-KIRC, 504 patients and 163 events for the lncRNA modules and 502 patients and 165 events for the protein-coding modules in every row. Source: 12_technical_adjustment_sensitivity.tsv, with confidence intervals for every cell in 12_technical_adjustment_arms.tsv. Membership is fixed at the definition made on the residualised matrix, and the three arms differ only in how the score is computed and what enters the model. Status is robust (significant in both arms), lost (without adjustment only), gained (with adjustment only) or never significant. Benjamini-Hochberg control is within one network, arm and covariate set, so over 13 or 16 tests, not 29. Under the clinical covariate set, 15 of 29 modules are significant without adjustment, 10 with adjustment for the alignment metrics and 3 with the STAR metrics as covariates. Nine are lost, 4 gained, 6 robust and 10 never significant. Under the clinical plus ESTIMATE set the counts are 16, 11 and 5, with 9 lost, 4 gained, 7 robust and 9 never significant. The protein-coding green row of panel (b) is given to three decimals because the three arms agree to two.

(c) The same contrast with RIN and slide necrosis added to the clinical covariate set

Covariate set	Significant without adjustment	Significant with adjustment	Lost	Gained	Robust	Not significant
Clinical	15 of 29 (11 lncRNA, 4 protein-coding)	10 of 29 (7 lncRNA, 3 protein-coding)	9	4	6	10
Clinical + RIN + slide necrosis	12 of 29 (9 lncRNA, 3 protein-coding)	4 of 29 (3 lncRNA, 1 protein-coding)	9	1	3	16

631 The five prognostic modules under the clinical, RIN and necrosis covariate set:

Module	HR without adjustment (95% CI), FDR	HR with adjustment for the alignment metrics (95% CI), FDR	Status
lncRNA blue	0.87 (0.73 to 1.04), 0.17	0.74 (0.63 to 0.87), 0.0035	gained
lncRNA greenyellow	1.41 (1.20 to 1.66), 1.8e-04	1.30 (1.10 to 1.54), 0.018	robust
lncRNA turquoise	1.47 (1.26 to 1.71), 1.2e-05	1.28 (1.08 to 1.51), 0.019	robust
Protein-coding green	0.69 (0.58 to 0.81), 4.9e-05	0.70 (0.60 to 0.83), 1.8e-04	robust
Protein-coding purple	1.26 (1.07 to 1.48), 0.021	1.24 (1.05 to 1.46), 0.068	lost

632 *Note.* TCGA-KIRC, 503 patients and 163 events for the lncRNA modules and 501 patients and
633 165 events for the protein-coding modules, one fewer in each network than panels (a) and (b)
634 because one case lacks an RNA integrity number. Source:
635 23_technical_adjustment_with_rin.tsv. Adding RIN and slide necrosis to the clinical
636 block reduces the number of apparently prognostic modules before adjustment, and further after
637 it. The direction of the contrast is unchanged.

638 **Supplementary Table 13 | Published TCGA-KIRC lncRNA signatures re-scored**

639 **(a) Provenance and mapping**

Signature	Weight provenance	Tier	Published genes	Scored	In the 3,442-gene production set
Jiang 2019 ⁴ , 9 lncRNA nomogram	published coefficients	primary	9	9	7
Gui 2021 ⁵ , 8 autophagy-related	unit weights signed in these same patients	primary	8	7	6
Yu 2021 ⁶ , m6A	footnote, incomplete mapping	footnote	5	2	2

Signature	Weight provenance	Tier	Published genes	Scored	In the 3,442-gene production set
Hong 2022 ⁷ , cuproptosis	published coefficients	primary	4	4	3
Liu 2023 ⁸ , pyroptosis	unit weights signed in these same patients	primary	6	6	5
Feng 2024 ⁹ , disulfidptosis	unit weights signed in these same patients	primary	6	6	5

Note. Source: 26_signature_provenance_key.tsv, 26_published_signatures_genes.tsv. The six signatures publish 38 genes, of which 36 are in the current annotation and 34 are scored. The two unscored genes have a median FPKM of 0. Neither Ensembl host resolved two names. Hong 2022 and Feng 2024 share AC026401.3, so the six signatures are correlated arms, not independent tests. Gui 2021, Liu 2023 and Feng 2024 publish neither coefficients nor per-gene hazard directions. Their weights are therefore plus or minus one, signed by the univariate Cox coefficient in these same 528 patients, so their p values are invalid and their attenuation is inflated. Yu 2021 retained 2 of its 5 lncRNAs, lost the gene with the largest published coefficient and scores opposite to the published model. Ranges must never be quoted across two provenance classes.

(b) Correlation of each score with the alignment metrics

Signature, published weights	rho with the non-feature fraction (p)	rho with the lncRNA axis	rho with the multimapping fraction	rho with the non-feature fraction after residualisation (p)
Jiang 2019	0.43 (1.1e-25)	0.39	0.30	0.04 (0.41)
Gui 2021	0.35 (5.3e-17)	0.32	0.20	-0.02 (0.62)
Yu 2021	-0.24 (1.8e-08)	-0.34	0.15	0.00 (0.96)
Hong 2022	0.17 (6.9e-05)	0.18	0.33	-0.03 (0.51)
Liu 2023	0.11 (0.0087)	-0.04	0.32	0.00 (0.96)
Feng 2024	0.43 (6.3e-25)	0.53	0.32	0.02 (0.58)

Note. TCGA-KIRC, 528 patients for the non-feature and multimapping fractions and 511 for the lncRNA axis. Source: 26_published_signatures_correlations.tsv. With coefficients

re-estimated in this cohort, correlations with the non-feature fraction range from 0.18 to 0.50. After residualising the expression matrix on the three STAR metrics, all twelve arms lie within 0.04 of zero (p at least 0.41), confirming that the adjustment works. Residualisation does not zero the correlations with the other two quantities. Against the lncRNA axis, the residualised published scores of Liu (-0.18), Feng (0.15) and Yu (-0.14) survive Benjamini-Hochberg control.

(c) Hazard ratios per standard deviation, published weights

Signature	Score alone	+ clinical model	+ clinical model + the three STAR metrics	Residualised score + clinical model
Jiang 2019	1.93 (1.70 to 2.19)	1.67 (1.45 to 1.93)	1.65 (1.38 to 1.97)	1.55 (1.33 to 1.79)
Gui 2021	1.87 (1.65 to 2.13)	1.50 (1.31 to 1.71)	1.39 (1.19 to 1.62)	1.41 (1.22 to 1.63)
Yu 2021	1.63 (1.36 to 1.96)	1.46 (1.21 to 1.77)	1.51 (1.25 to 1.81)	1.57 (1.31 to 1.86)
Hong 2022	2.13 (1.87 to 2.43)	1.75 (1.51 to 2.03)	1.70 (1.44 to 2.02)	1.70 (1.46 to 1.99)
Liu 2023	1.97 (1.71 to 2.28)	1.63 (1.39 to 1.90)	1.52 (1.29 to 1.79)	1.53 (1.30 to 1.79)
Feng 2024	1.89 (1.64 to 2.18)	1.53 (1.31 to 1.79)	1.46 (1.19 to 1.79)	1.39 (1.17 to 1.65)

(d) Hazard ratios per standard deviation, coefficients re-estimated in TCGA-KIRC

Signature	+ clinical model	+ clinical model + the three STAR metrics	Residualised score + clinical model
Jiang 2019	1.88 (1.60 to 2.22)	1.85 (1.53 to 2.24)	1.70 (1.45 to 2.00)
Gui 2021	1.53 (1.35 to 1.74)	1.46 (1.25 to 1.72)	1.46 (1.28 to 1.67)
Yu 2021	1.63 (1.40 to 1.89)	1.52 (1.28 to 1.80)	1.57 (1.33 to 1.85)
Hong 2022	1.74 (1.50 to 2.01)	1.68 (1.43 to 1.98)	1.70 (1.46 to 1.98)
Liu 2023	1.60 (1.39 to 1.85)	1.51 (1.27 to 1.80)	1.52 (1.30 to 1.78)
Feng 2024	1.67 (1.40 to 1.98)	1.59 (1.29 to 1.96)	1.57 (1.30 to 1.89)

Note for panels (c) and (d). Score-alone models use 528 patients and 173 events, adjusted models 520 patients and 172 events. Source: 26_published_signatures_models.tsv. All

24 adjusted estimates remain significant (largest p 2.9e-04). The refit arm is fitted and evaluated on the same patients, so it is optimistic by construction.

(e) Attenuation of the clinically adjusted log hazard ratio

Signature, published weights	Change on adding the three STAR metrics (%)	Change on residualising the expression (%)	Provenance
Jiang 2019	-2.6	-15.0	published coefficients
Hong 2022	-4.9	-5.1	published coefficients
Gui 2021	-19.0	-14.9	signed in sample
Liu 2023	-14.1	-13.2	signed in sample
Feng 2024	-11.1	-23.6	signed in sample
Yu 2021	7.6	17.5	footnote, 2 of 5 genes

Note. Source: 26_published_signatures_models.tsv, 26_headline_ranges_by_provenance.tsv. For the only two signatures whose weights were not re-derived, the log hazard ratio moves by a few per cent and is not abolished. All six models were nonetheless fitted in TCGA-KIRC, so no arm is out of sample relative to the discovery cohort. With coefficients re-estimated in this cohort, the five refit arms move by -12.6% to -2.8% on adding the STAR metrics and by -15.9% to -4.1% on residualising. Yu 2021 moves the other way and must not be pooled into any range.

(f) Apparent increment in the concordance index over the reference model of the same specification

Signature, published weights	Over the clinical model	Over the clinical model plus the STAR metrics	Residualised score over the clinical model
Jiang 2019	0.0215 (0.0014 to 0.0445), p 0.041	0.0105 (-0.0071 to 0.0290), p 0.24	0.0138 (-0.0057 to 0.0350), p 0.17
Gui 2021	0.0182 (-0.0007 to 0.0372), p 0.055	0.0117 (-0.0024 to 0.0260), p 0.11	0.0107 (-0.0061 to 0.0277), p 0.20
Yu 2021	0.0086 (-0.0071 to 0.0245), p 0.30	0.0173 (0.0030 to 0.0330), p 0.022	0.0147 (-0.0032 to 0.0325), p 0.12

Signature, published weights	Over the clinical model	Over the clinical model plus the STAR metrics	Residualised score over the clinical model
Hong 2022	0.0236 (0.0030 to 0.0447), p p 0.024	0.0169 (-0.0018 to 0.0364), p 0.078	0.0163 (-0.0034 to 0.0372), p 0.11
Liu 2023	0.0181 (-0.0025 to 0.0397), p 0.074	0.0144 (-0.0017 to 0.0311), p 0.089	0.0140 (-0.0046 to 0.0328), p 0.14
Feng 2024	0.0182 (0.0002 to 0.0359), p p 0.046	0.0070 (-0.0061 to 0.0200), p 0.30	0.0107 (-0.0047 to 0.0253), p 0.18

Note. TCGA-KIRC, 520 patients and 172 events, paired patient bootstrap over 2,000 resamples.

Source: 26_published_signatures_models.tsv. All increments are apparent and in sample. Each score is evaluated on the patients whose model it entered, and four of the six signatures also derived their weights in this cohort. The modules of this paper have an apparent increment of 0.0205 and a cross-validated increment of 0.0092 (493 patients, 159 events). The relevant comparison is between apparent and cross-validated increments.

(g) Position of each signature in two random-signature nulls

Signature, published weights	Set size	rho with the metric	Percentile of the absolute rho, outcome- signed null	Percentile of the absolute rho, sign- randomised null	Percentile of the univariate HR, outcome-signed null
Liu 2023	6	0.11	8.6	19.1	99.9
Hong 2022	4	0.17	15.9	25.4	100.0
Gui 2021	7	0.35	28.5	55.4	99.7
Jiang 2019	9	0.43	30.9	65.3	99.8
Yu 2021	2	-0.24	31.1	34.5	97.6
Feng 2024	6	0.43	36.7	64.9	99.7

Median absolute correlation with the metric of the two nulls, by set size:

Set size	Outcome-signed	Sign-randomised	Median paired difference	Wilcoxon p
2	0.40	0.36	0.000	3.5e-06
4	0.48	0.34	0.056	5.9e-38
6	0.53	0.33	0.137	1.5e-63
7	0.54	0.32	0.144	2.3e-69
9	0.57	0.33	0.169	3.8e-86

Note. One set of 1,000 random gene sets per size, weighted two ways and paired on the same draws. Source: 26_random_signature_null.tsv. The paired contrast shows directly that outcome-signing itself induces correlation with the metric in the null, so outcome-signed percentiles cannot show that a published signature is less correlated with the metric than chance. Against the sign-randomised null, three of the six published arms sit at or above the median. Percentiles carry Monte-Carlo error of about plus or minus 1.5 points. Neither null matches the magnitude of a published coefficient vector, because both weight every drawn gene plus or minus one.

Supplementary Table 14 | Module hazard ratios under four codings of unassessed nodal and metastasis status

Module	A: NX as N0, MX as M0, M1 reconciled from stage IV	B: NX and MX as their own categories	C: assessed N and M only	D: M1 as coded, no reconciliation
Protein-coding green	0.62 (0.53 to 0.74), p 1.7e-08	0.62 (0.53 to 0.73), p 1.3e-08	0.55 (0.43 to 0.70), p 1.2e-06	0.63 (0.54 to 0.74), p 2.6e-08
Protein-coding purple	1.28 (1.09 to 1.50), p 0.0025	1.30 (1.10 to 1.53), p 0.0018	1.33 (1.03 to 1.72), p 0.027	1.28 (1.09 to 1.50), p 0.0025
lncRNA blue	0.71 (0.60 to 0.83), p 3.0e-05	0.70 (0.60 to 0.83), p 2.3e-05	0.61 (0.48 to 0.78), p 6.3e-05	0.72 (0.61 to 0.84), p 5.9e-05
lncRNA greenyellow	1.35 (1.15 to 1.60), p 3.4e-04	1.36 (1.15 to 1.60), p 2.5e-04	1.45 (1.17 to 1.81), p 7.4e-04	1.37 (1.16 to 1.62), p 1.7e-04
lncRNA turquoise	1.24 (1.06 to 1.45), p 0.0074	1.26 (1.07 to 1.48), p 0.0054	1.48 (1.11 to 1.97), p 0.0082	1.27 (1.08 to 1.48), p 0.0039

Note. TCGA-KIRC, principal specification in every cell. Specification A is the primary analysis and reproduces the discovery column of Table 2 of the main text. Specifications A, B and D use 502 patients and 165 events for the protein-coding modules and 504 patients and 163 events for the lncRNA modules. Specification C uses 240 patients and 87 events for the protein-coding

modules and 241 patients and 87 events for the lncRNA modules. Source: 03_nodal_metastasis_sensitivity.tsv. All p values are nominal. Hazard ratios under specifications B and D lie within 0.020 and 0.026, respectively, of their specification A estimates. Specification C is a differently and more completely staged population, not a matched subset, so its larger effects do not show that the primary coding is conservative. Together, the four codings show that the primary coding did not create the associations.

Supplementary Table 15 | Sex-disaggregated results

(a) Module scores associated with sex in TCGA-KIRC

Module	Point-biserial r, male coded 1	Wilcoxon p	BH over 29 modules	Higher in
lncRNA lightcyan	0.23	4.3e-07	6.2e-06	men
lncRNA greenyellow	0.22	3.6e-07	6.2e-06	men
Protein-coding green	-0.19	1.1e-05	1.0e-04	women
Protein-coding purple	0.10	0.0037	0.027	men
lncRNA blue	-0.12	0.0052	0.028	women
Protein-coding tan	0.12	0.0057	0.028	men
Protein-coding black	0.11	0.0071	0.029	men
Protein-coding red	-0.11	0.012	0.044	women

Note. TCGA-KIRC, 511 patients for the lncRNA modules (178 women, 333 men) and 506 for the protein-coding modules (179 women, 327 men). Source: 33_sex_correlations.tsv. Benjamini-Hochberg is applied to the Wilcoxon p values across the 29 module eigengenes within a cohort. Eight of 29 eigengenes differ by sex, including four of the five prognostic modules, all but lncRNA turquoise (r -0.03, BH 0.78). The counts are 3 of 29 in CPTAC-3 (1 prognostic), 10 of 29 in TCGA-KIRP (3 prognostic) and 1 of 29 in TCGA-KICH (none prognostic).

(b) Alignment metrics by sex in TCGA-KIRC

Quantity	Median, women	Median, men	Hodges-Lehmann shift (95% CI)	Wilcoxon p	BH
Non-feature read fraction, %	5.0	4.5	-0.26 (-0.68 to 0.13)	0.21	0.28
Multimapping fraction, %	7.2	7.7	0.37 (0.13 to 0.62)	0.0027	0.011
log10 assigned reads	7.763	7.770	0.003 (-0.015 to 0.022)	0.71	0.71
lncRNA axis	-0.0018	-0.0100	-0.006 (-0.015 to 0.002)	0.12	0.25

Note. 528 patients (185 women, 343 men) for the STAR metrics and 511 (178, 333) for the axis.

Source: `33_noFeature_by_sex.tsv`. Of the alignment metrics, only the multimapping fraction differs by sex. The axis is in raw loading units in this file, so its effect size is not comparable with the per-standard-deviation axis used elsewhere.

(c) Module by sex interaction, common-covariate model

Cohort and specification	Module	HR in women	HR in men	Interaction HR (95% CI)	Wald p	BH over 5 modules
TCGA-KIRC, principal	Protein-coding green	0.64	0.62	0.96 (0.70 to 1.32)	0.82	0.82
TCGA-KIRC, principal	Protein-coding purple	1.06	1.45	1.36 (1.03 to 1.80)	0.030	0.15
TCGA-KIRC, principal	lncRNA blue	0.62	0.75	1.20 (0.87 to 1.65)	0.26	0.35
TCGA-KIRC, principal	lncRNA greenyellow	1.58	1.29	0.81 (0.59 to 1.12)	0.20	0.35
TCGA-KIRC, principal	lncRNA turquoise	1.06	1.28	1.21 (0.86 to 1.71)	0.28	0.35
CPTAC-3, six- parameter	lncRNA greenyellow	2.33	1.23	0.53 (0.27 to 1.04)	0.065	0.32
CPTAC-3, six- parameter	Protein-coding purple	1.85	1.26	0.68 (0.37 to 1.25)	0.21	0.50
TCGA-KIRP, six- parameter	lncRNA blue	0.39	0.90	2.32 (1.12 to 4.82)	0.024	0.067
TCGA-KIRP, six- parameter	lncRNA turquoise	3.00	0.97	0.32 (0.12 to 0.88)	0.027	0.067
TCGA-KIRP, six- parameter	lncRNA greenyellow	1.72	1.32	0.77 (0.43 to 1.38)	0.38	0.47

Note. The interaction model has one baseline hazard, covariate effects common to both sexes and a module-by-sex product term. Patients and events: TCGA-KIRC 502 and 165 for the protein-

coding modules and 504 and 163 for the lncRNA modules, CPTAC-3 199 and 32, TCGA-KIRP 263 and 40. Source: `33_sex_interaction_models.tsv`. No interaction test survives Benjamini-Hochberg control in any cohort. The family is the five modules of one cohort-and-specification block, not all 20 tests. The smallest raw p over the 20 module interaction tests is 0.024. All five CPTAC-3 rows are below the events-per-variable threshold of 5, at 4.6. The metric exposures show no sex interaction. The non-feature fraction gives 1.41 in women and 1.34 in men (interaction hazard ratio 0.95, 0.73 to 1.24, p 0.71), and the axis 1.34 and 1.35 (1.01, 0.73 to 1.39, p 0.97).

(d) Sex-specific fits, a different estimand

Cohort and specification	Module	HR in women (95% CI), n / events	HR in men (95% CI), n / events	Ratio, men over women (95% CI)	Wald p	BH over 5 modules
TCGA-KIRC, principal	lncRNA greenyellow	1.97 (1.40 to 2.78), 174 / 57	1.20 (0.98 to 1.47), 330 / 106	0.61 (0.41 to 0.91)	0.015	0.074
TCGA-KIRC, principal	lncRNA blue	0.59 (0.43 to 0.79), 174 / 57	0.76 (0.62 to 0.94), 330 / 106	1.30 (0.90 to 1.87)	0.16	0.40
TCGA-KIRC, principal	Protein-coding purple	1.12 (0.85 to 1.47), 176 / 60	1.35 (1.09 to 1.68), 326 / 105	1.21 (0.85 to 1.71)	0.29	0.48
TCGA-KIRC, principal	lncRNA turquoise	1.12 (0.80 to 1.57), 174 / 57	1.25 (1.04 to 1.50), 330 / 106	1.12 (0.76 to 1.64)	0.58	0.72
TCGA-KIRC, principal	Protein-coding green	0.65 (0.49 to 0.85), 176 / 60	0.63 (0.49 to 0.80), 326 / 105	0.97 (0.68 to 1.39)	0.87	0.87

Note. Separate baseline hazards and covariate effects in women and in men. Source: `33_sex_stratified_hr.tsv`, `33_sex_contrast_reconciliation.tsv`. Hazard ratios are per standard deviation of the pooled score. Per within-stratum standard deviation, greenyellow gives 1.76 (1.32 to 2.34) in women and 1.21 (0.98 to 1.49) in men, because the score varies less in women (standard deviations 0.83 and 1.04). The sex-specific fits and the

interaction model of panel (c) estimate different quantities and disagree in 4 of 25 rows, so each hazard ratio must be quoted with its own model's p value. For lncRNA greenyellow the sex-specific fits give a difference at Wald p 0.015 and likelihood-ratio p 0.015, whereas the interaction model on the same patients gives 0.81 at p 0.20. Nothing survives multiplicity in either test. The five modules were chosen by overall survival in these same patients, so the family is post-selection and every false discovery rate is conditional on that selection. Fifteen of the 50 stratified models, all ten CPTAC-3 rows and the five TCGA-KIRP female rows, are below the events-per-variable threshold of 5.

(e) X- and Y-encoded members of the lncRNA network

Set	X or Y encoded	Total	Odds ratio (95% CI)	Fisher p	BH over 17 tests
The analysed lncRNA network	76 (2.2%)	3,442	0.98 (0.75 to 1.27)	0.95	--
lncRNA turquoise	9 (2.1%)	438	0.91 (0.41 to 1.77)	1.00	1.00
lncRNA blue	3 (1.2%)	246	0.53 (0.11 to 1.59)	0.38	0.81
lncRNA greenyellow	2 (1.9%)	108	0.82 (0.10 to 3.06)	1.00	1.00
lncRNA black	9 (5.0%)	179	2.34 (1.04 to 4.59)	0.020	0.20
lncRNA red	0	186	0 (0 to 0.87)	0.024	0.20

Note. Chromosomes are from the GENCODE v36 annotation. The reference set is the GENCODE v36 lncRNA genes outside the tested set, of which 379 of 16,889 (2.2%) are X or Y encoded. Source: 33_xy_lncRNA_counts_by_module.tsv. Sixty-eight of the 76 are on chrX and 8 on chrY. No module survives Benjamini-Hochberg control over the 17 tests. XIST is in the network but unassigned (module membership -0.26), as are 7 of the 8 chrY lncRNAs.

(f) Sensitivity of the sex contrast

Analysis	lncRNA greenyellow, women	lncRNA greenyellow, men	p for the sex difference
Baseline	1.97	1.20	0.015
X- and Y-encoded members deleted, 2 of 108 genes	2.00	1.20	0.012

Analysis	lncRNA greenyellow, women	lncRNA greenyellow, men	p for the sex difference
Recorded sex replaced by the marker call where the two disagree	1.93	1.22	0.023
The five discordant samples dropped	1.96	1.21	0.018

Note. TCGA-KIRC, principal specification. Source: 33_sensitivity_analyses.tsv, 33_sex_label_check.tsv. Deleting the X- and Y-encoded members of each prognostic module (20, 6, 3, 2 and 9 genes) changes the score negligibly, with correlations of 0.9997 to 0.9999 against the full score. Recorded sex was checked against XIST and the mean of eight chrY lncRNAs in the 511 discovery samples of the lncRNA network: 448 concordant, 5 discordant and 58 ambiguous. The ambiguous calls are mostly recorded males with low chrY expression, consistent with loss of chromosome Y in renal tumour tissue, so the discordance rate is best quoted as 5 of the 453 samples the markers call. Marker cut-offs were fixed in advance. All sex-disaggregated analyses are descriptive.

Supplementary Table 16 | Module estimates in chromophobe renal cell carcinoma

Module	HR	95% CI	Nominal p	Zsummary in TCGA-KICH
Protein-coding green	0.88	0.20 to 3.90	0.87	4.83
Protein-coding purple	2.67	1.25 to 5.70	0.011	28.18
lncRNA blue	1.41	0.48 to 4.17	0.53	1.76
lncRNA greenyellow	5.50	1.06 to 28.61	0.043	6.70
lncRNA turquoise	1.77	0.60 to 5.19	0.30	11.10

Note. TCGA-KICH, 65 patients and 9 events. Hazard ratio per standard deviation of the projected module eigengene, from the shared six-parameter cross-cohort model: the eigengene plus age, sex, T category, M1 and the non-feature fraction. Grade is not modelled, being unrecorded for this cohort and inapplicable to chromophobe histology. Source:

11_subtype_module_effects.tsv, preservation column from
 24_module_preservation_lncRNA.tsv and 24_module_preservation_mRNA.tsv.
 Nine events against six parameters give 1.5 events per variable, below the adequacy threshold of
 5, so these estimates are reported for completeness and not interpreted. The greenyellow interval
 spans a 27-fold range, and lncRNA blue is not preserved in this cohort. Undoing the
 reconciliation of stage IV to M1 changes nothing, because no TCGA-KICH patient was
 reconciled. Twelve covariates on nine events separate completely, so the six-parameter set is
 used.

Supplementary Table 17 | Module preservation of the discovery networks in the three external cohorts

(a) Summary

Network	Test cohort	Modules	Strong	Weak to moderate	None	Median Zsummary	Minimum	Maximum	Test samples	Test genes	Permutations
lncRNA	CPTAC-163	13	3	0	13.39	2.20	30.65	200	3,442	100	
lncRNA	TCGA-KIRP	16	7	8	1	8.04	1.77	29.39	269	3,442	100
lncRNA	TCGA-KICH	16	3	11	2	5.36	0.33	13.31	65	3,441	100
Protein-coding	CPTAC-13	13	13	0	0	43.99	17.69	101.66	200	12,000	30
Protein-coding	TCGA-KIRP	13	13	0	0	35.08	15.24	71.33	269	12,000	30
Protein-coding	TCGA-KICH	13	10	3	0	18.72	4.83	49.11	65	12,000	30

(b) lncRNA network, Zsummary per module

Module	Size	CPTAC-3	TCGA-KIRP	TCGA-KICH
blue	246	30.65	8.47	1.76
greenyellow	108	15.14	3.47	6.70

Module	Size	CPTAC-3	TCGA-KIRP	TCGA-KICH
turquoise	438	13.42	29.39	11.10
yellow	192	29.39	12.51	10.71
red	186	27.21	7.61	5.90
pink	174	20.14	16.63	8.66
magenta	122	18.71	1.77	4.82
tan	93	16.18	6.90	5.94
black	179	13.36	18.72	4.65
lightcyan	44	12.74	3.09	0.33
salmon	72	12.19	5.45	2.19
brown	219	12.07	19.80	13.31
cyan	54	11.59	4.76	3.19
green	186	6.34	13.87	3.55
midnightblue	53	3.90	6.93	4.33
purple	110	2.20	19.64	8.14

774 **(c) Protein-coding network, Zsummary per module**

Module	Size	CPTAC-3	TCGA-KIRP	TCGA-KICH
green	670	61.62	34.79	4.83
purple	216	52.20	36.70	28.18
yellow	986	101.66	64.67	49.11
blue	994	79.48	29.65	20.88
black	323	57.44	24.06	15.67
tan	152	51.48	35.08	26.22
red	473	43.99	42.96	11.90
magenta	227	43.50	29.28	13.13
greenyellow	164	25.93	26.05	18.72
salmon	63	25.06	15.24	9.22
pink	270	23.18	37.10	9.85
turquoise	2,405	18.99	55.77	35.00
brown	988	17.69	71.33	23.77

775 *Note.* WGCNA modulePreservation, with the residualised TCGA-KIRC matrix on which
776 the modules were defined as reference and each test network residualised on its own cohort's

777 STAR metrics. Bands: Zsummary above 10 strong, 2 to 10 weak to moderate, below 2 none.
 778 Source: 24_module_preservation_summary.tsv,
 779 24_module_preservation_lncRNA.tsv, 24_module_preservation_mRNA.tsv. The
 780 algorithm sub-sampled the protein-coding turquoise module to 1,000 genes. Z.density and
 781 Z.connectivity are not populated, and there is no per-module preservation p value. Median
 782 ranks (lower is better preserved) for lncRNA blue, greenyellow and turquoise and protein-coding
 783 green and purple are 5, 7, 13, 7 and 3 in CPTAC-3, 14, 17, 5, 12 and 3 in TCGA-KIRP, and 15,
 784 7, 8, 14 and 2 in TCGA-KICH. Band edges are exact, so lncRNA blue at 1.76 in TCGA-KICH is
 785 not preserved, while salmon at 2.19 is weak to moderate.

786 **Supplementary Table 18 | The catabolic protein-coding module as a proximal-tubule**
 787 **programme**

788 **(a) Marker set and module hubs**

Quantity	Value
Proximal-tubule marker genes assembled	21
Of those, flagged core	15
Of those, named as proximal tubule by the Human Protein Atlas single-cell string	18
Of those, inside the protein-coding network	14, all 14 members of the catabolic module
Leave-out markers, not members of that module	7
Markers exclusive to proximal tubule in the Human Protein Atlas	7: LRP2, SLC34A1, SLC22A6, SLC13A3, SLC17A1, SLC7A13, MIOX
Top 20 module hubs whose Human Protein Atlas string names proximal tubule	17
Of those, exclusive to proximal tubule	1, LRP2
Top 20 hubs also in the 21-gene marker set	3: PDZK1, LRP2, GLYAT
Top 20 hubs in ClearCode34	1: GIPC2, a ccA gene

789 *Note.* Source: 34_pt_marker_set.tsv, 34_green_hubs_pt_flag.tsv. The Human
790 Protein Atlas flag is permissive. Of the 17 flagged hubs, 2 name one other cell type and 14
791 appear in lists of 4 to 12 cell types. The flag has no background rate.

792 **(b) Correlations of the module and the marker score**

Feature	Variable	Expression space	Pearson r	Spearman rho	n
Catabolic module eigengene	proximal-tubule marker score, 21 genes	residualised	0.84	0.85	506
Catabolic module eigengene	proximal-tubule marker score, 21 genes	observed	0.86	0.88	506
Catabolic module eigengene	leave-out marker score, 7 non-member genes	residualised	0.29	0.42	506
Catabolic module eigengene	ClearCode34 ccB minus ccA score	residualised	-0.74	-0.70	506
Catabolic module eigengene	slide percent tumour nuclei	residualised	0.21	0.25	506
Catabolic module eigengene	non-feature read fraction	observed	0.02	0.04	506
lncRNA blue	proximal-tubule marker score	residualised	0.75	0.74	511
lncRNA blue	non-feature read fraction	observed	0.44	0.44	511
lncRNA greenyellow	proximal-tubule marker score	residualised	-0.44	-0.52	511
lncRNA greenyellow	non-feature read fraction	observed	0.22	0.25	511
lncRNA turquoise	proximal-tubule marker score	residualised	0.04	0.03	511
lncRNA turquoise	non-feature read fraction	observed	0.85	0.90	511
lncRNA turquoise	non-feature read fraction	residualised	0.00	0.20	511
Proximal-tubule marker score	non-feature read fraction	observed	0.08	0.05	528

793 *Note.* TCGA-KIRC. Source: 34_pt_score_correlations.tsv. The catabolic module and
794 the marker score are nearly the same measurement. The module does not track the alignment
795 metrics, whereas the three prognostic lncRNA modules do. Residualisation removes most but not
796 all of that dependence for lncRNA turquoise, whose residualised eigengene still correlates with
797 the metric at Spearman 0.20 (p 5.3e-06), against 0.029 (p 0.69) in CPTAC-3. In CPTAC-3 the
798 module correlates with the marker score at 0.87 on 199 patients.

799 **(c) Separability of the module and the marker score**

Model, clinical frame	Terms	HR (95% CI), p	n / events	Variance inflation
Marker score, 21 genes	pt_full	0.69 (0.58 to 0.81), p 6.8e-06	520 / 172	1.07
Leave-out marker score, 7 genes	pt_leaveout	0.90 (0.75 to 1.09), p 0.29	520 / 172	1.05
Catabolic module alone	green	0.69 (0.59 to 0.81), p 2.9e-06	502 / 165	1.13
Marker score on the module's analysis set	pt_full	0.70 (0.59 to 0.82), p 2.6e-05	502 / 165	1.10
Module and marker score together	green / pt_full	0.74 (0.55 to 1.00), p 0.046 / 0.92 (0.67 to 1.25), p 0.58	502 / 165	4.05 / 3.98
Module and lncRNA blue together	blue / green	1.18 (0.79 to 1.77), p 0.41 / 0.61 (0.40 to 0.91), p 0.015	493 / 159	7.53 / 7.59
Module and lncRNA turquoise together	turquoise / green	1.34 (1.14 to 1.59), p 5.5e-04 / 0.68 (0.58 to 0.80), p 2.0e-06	493 / 159	1.15 / 1.17
Module and lncRNA greenyellow together	greenyellow / green	1.14 (0.94 to 1.40), p 0.19 / 0.76 (0.63 to 0.91), p 0.0035	493 / 159	1.83 / 1.58

800 **Wald contrasts and nested likelihood-ratio tests:**

Test	Estimate (SE) or statistic	p	Pairwise r
Contrast, module minus marker score, clinical frame	-0.2105 (0.2949)	0.48	0.84
Contrast, module minus marker score, principal frame	-0.5617 (0.3385)	0.097	0.84
Contrast, lncRNA blue minus module, clinical frame	0.6722 (0.4053)	0.097	0.91
Adding the module on top of the marker score, clinical frame	3.801	0.051	--
Adding the marker score on top of the module, clinical frame	0.319	0.57	--
Adding the module on top of the marker score, principal frame	7.637	0.0057	--
Adding the marker score on top of the module, principal frame	0.078	0.78	--
Adding lncRNA turquoise on top of the module	11.509	6.9e-04	--
Adding lncRNA greenyellow on top of the module	1.700	0.19	--
Adding lncRNA blue on top of the module	0.683	0.41	--

801 *Note.* TCGA-KIRC, overall survival, hazard ratio per standard deviation. Source:

802 `34_pt_adjusted_models.tsv`. The module eigengene and the marker score are statistically

803 indistinguishable as predictors. They correlate at 0.84 with variance inflation of about 4, and the

Wald contrast gives p 0.48, so the split of p 0.046 against p 0.58 in the joint model is not evidence of absorption. Whether the module adds to the marker score depends on the specification (nested-test p 0.051 under the clinical frame, p 0.0057 under the principal frame) and remains unresolved. The module and lncRNA blue are worse still, correlating at 0.91 with variance inflation of 7.5 (493 patients, 159 events), so no clean attribution is possible there either. Every CPTAC-3 proximal-tubule model is below the events-per-variable threshold of 5 and is descriptive.

(d) ccA and ccB classification

Quantity	TCGA-KIRC	CPTAC-3
Patients classified	528	200
ccB, n (%)	219 (41.5)	81 (40.5)
Agreement of the two published classifier genes	98.5%	99.0%
Mean catabolic eigengene, ccA against ccB	0.492 against -0.753, p 1.5e-44	0.534 against -0.782, p 3.0e-20
ccB, unadjusted HR	2.33 (1.72 to 3.15), p 4.2e-08, 528 / 173	1.80 (0.89 to 3.61), p 0.10, 199 / 32
ccB + clinical model	1.62 (1.16 to 2.27), p 0.0052, 520 / 172	1.51 (0.74 to 3.11), p 0.26
ccB + clinical model + the catabolic module	1.03 (0.67 to 1.58), p 0.89, with the module at 0.70 (0.58 to 0.84)	0.96 (0.36 to 2.57), p 0.94

Note. Source: 34_ccAB_class_summary.tsv, 34_pt_adjusted_models.tsv.

ClearCode34 is used as a template axis, not as the published nearest-centroid classifier. Of its 34 published genes, 31 are in the protein-coding network and 12 in the catabolic module, all 12 of them ccA genes, so the contrast between the class and the module is partly definitional.

(e) Stromal and immune content of the catabolic module

Module	Genes	ESTIMATE stromal genes in the module	Expected	p	ESTIMATE immune genes	Expected	p
Catabolic (green)	670	4 (0.6%)	6.36	0.89	0	7.48	1.00

Module	Genes	ESTIMATE stromal genes in the module	Expected	p	ESTIMATE immune genes	Expected	p
black	323	28 (8.7%)	3.07	1.5e-19	0	3.61	1.00
yellow	986	40 (4.1%)	9.37	4.6e-16	111 (11.3%)	11.01	6.9e-99
blue	994	28 (2.8%)	9.44	1.2e-07	11 (1.1%)	11.10	0.56
turquoise	2,405	2 (0.1%)	22.85	1.00	4 (0.2%)	26.86	1.00

Note. Hypergeometric test against a universe of the 12,000 network genes, which include 114 of the 141 ESTIMATE stromal genes and 134 of the 141 immune genes. Source: 34_green_estimate_overlap.tsv. The catabolic module carries fewer stromal genes than chance and no immune genes. The stromal and immune programmes sit in other modules.

(f) Normal-kidney admixture in the catabolic module

Cohort	Feature	Expression space	Normal mean in tumour SD units (AUC)	Wilcoxon p	Normals above the tumour maximum	Confounded by the alignment metrics	Direction survives adjustment
TCGA-KIRC	Catabolic module	observed	0.358 (0.587)	0.017	0 of 72	no	yes
TCGA-KIRC	Catabolic module	residualised	0.313 (0.576)	0.036	0 of 72	no	yes
TCGA-KIRC	Proximal-tubule markers in the network	observed	0.744 (0.663)	7.8e-06	6 of 72	no	yes
TCGA-KIRC	lncRNA blue	observed	-0.398 (0.359)	1.1e-04	0 of 72	yes	no
TCGA-KIRC	lncRNA blue	residualised	-0.053 (0.467)	0.37	0 of 72	no	no
TCGA-KIRC	lncRNA greenyellow	observed	-0.970 (0.159)	7.8e-21	0 of 72	yes	yes
TCGA-KIRC	lncRNA greenyellow	residualised	-0.758 (0.259)	3.2e-11	0 of 72	no	yes
TCGA-KIRC	lncRNA turquoise	observed	-0.896 (0.230)	1.1e-13	0 of 72	yes	no
TCGA-KIRC	lncRNA turquoise	residualised	-0.202 (0.441)	0.11	0 of 72	no	no
CPTAC-3	Catabolic module	observed	1.524 (0.902)	2.5e-40	6 of 168	no	yes

Cohort	Feature	Expression space	Normal mean in tumour SD units (AUC)	Wilcoxon p	Normals above the tumour maximum	Confounded by the alignment metrics	Direction survives adjustment
CPTAC-3	Proximal-tubule markers in the network	observed	2.109 (0.950)	5.3e-50	119 of 168	no	yes
CPTAC-3	lncRNA blue	observed	0.666 (0.740)	2.4e-15	0 of 168	yes	yes
CPTAC-3	lncRNA greenyellow	observed	-1.069 (0.124)	1.5e-35	0 of 168	no	yes
CPTAC-3	lncRNA turquoise	observed	0.294 (0.615)	1.5e-04	0 of 168	yes	no

Note. Scores are standardised to the tumour distribution, so tabulated values are mean normal-library scores in tumour standard deviations. Source: `34_normal_admixture.tsv`. In TCGA-KIRC, an entirely normal library sits 0.358 tumour standard deviations above the mean tumour on the catabolic module, on a tumour range of 5.07 standard deviations. No normal library exceeds the tumour maximum, so no mixing fraction can reach the top of the tumour distribution. The same cannot be asserted for CPTAC-3, where the protocol differs and 6 of 168 normals exceed the tumour maximum on the module eigengene and 119 of 168 on the marker score. The like-for-like marker-score figure in TCGA-KIRC is 6 of 72. Five rows, all on the observed matrix, are flagged as confounded by the alignment metrics. After residualisation the apparent opposite direction of lncRNA blue and turquoise in normal kidney disappears and only greenyellow survives. The file also gives the normal fraction needed for a one-standard-deviation shift. That column assumes the score is linear in the normal fraction, which it is not, and is flagged as anti-conservative, so the argument rests on the observed ceiling.

Supplementary Table 19 | Single-cell localisation of the five prognostic modules

Contrast	Network	Module	Cohort	Members tested	Median log2FC	Null mean	z	Empirical p	Members up (%)	Members down (%)
vs normal epithelium	lncRNA	blue	GSE159115	69	-0.11	-0.08	-0.2	0.68	6	3
vs normal epithelium	lncRNA	blue	GSE222703	96	-0.32	0.23	-2.8	0.0015	24	22
vs normal epithelium	lncRNA	blue	Li2022	55	0.04	0.17	-0.6	0.57	16	7
vs normal epithelium	lncRNA	greenyellow	GSE159115	18	1.04	-0.08	3.5	0.0025	11	0
vs normal epithelium	lncRNA	greenyellow	GSE222703	14	4.38	0.40	7.2	0.0005	64	0
vs normal epithelium	lncRNA	greenyellow	Li2022	14	2.86	0.14	6.4	0.0005	50	0
vs normal epithelium	lncRNA	turquoise	GSE159115	60	0.24	-0.10	2.1	0.049	7	0
vs normal epithelium	lncRNA	turquoise	GSE222703	44	0.82	0.34	1.6	0.12	34	2
vs normal epithelium	lncRNA	turquoise	Li2022	33	0.25	0.08	0.6	0.61	18	0
vs normal epithelium	protein- coding	green	GSE159115	640	-0.09	-0.00	-2.4	0.021	3	1
vs normal epithelium	protein- coding	green	GSE222703	633	-0.98	-0.11	-20.4	0.0005	9	42
vs normal epithelium	protein- coding	green	Li2022	627	-0.27	0.24	-12.6	0.0005	9	10
vs normal epithelium	protein- coding	purple	GSE159115	143	0.88	-0.02	9.5	0.0005	10	0
vs normal epithelium	protein- coding	purple	GSE222703	102	0.69	0.07	5.4	0.0005	26	3
vs normal epithelium	protein- coding	purple	Li2022	131	1.26	0.21	10.5	0.0005	24	1
vs microenvironment	lncRNA	blue	GSE159115	118	3.60	0.42	22.4	0.0005	84	0
vs microenvironment	lncRNA	blue	GSE207493	161	3.57	0.45	20.5	0.0005	89	1
vs microenvironment	lncRNA	blue	GSE222703	89	5.04	0.32	15.6	0.0005	81	1
vs microenvironment	lncRNA	blue	Li2022	85	5.33	0.38	16.3	0.0005	78	9
vs microenvironment	lncRNA	greenyellow	GSE159115	49	1.73	0.44	5.6	0.0015	55	4

Contrast	Network	Module	Cohort	Members tested	Median log2FC	Null mean	z	Empirical p	Members up (%)	Members down (%)
vs microenvironment	lncRNA	greenyellow	GSE207493	69	1.51	0.45	4.4	0.0005	65	16
vs microenvironment	lncRNA	greenyellow	GSE222703	21	1.56	0.50	1.4	0.16	43	10
vs microenvironment	lncRNA	greenyellow	Li2022	28	1.89	0.27	3.3	0.011	54	21
vs microenvironment	lncRNA	turquoise	GSE159115	164	0.32	0.41	-0.7	0.49	22	5
vs microenvironment	lncRNA	turquoise	GSE207493	243	-0.04	0.43	-3.7	0.0025	33	31
vs microenvironment	lncRNA	turquoise	GSE222703	83	-0.40	0.29	-2.5	0.0075	8	22
vs microenvironment	lncRNA	turquoise	Li2022	91	-0.38	0.19	-2.1	0.018	14	29
vs microenvironment coding	protein-	green	GSE159115	661	2.29	-0.02	55.2	0.0005	79	1
vs microenvironment coding	protein-	green	GSE207493	668	2.63	-0.01	53.4	0.0005	93	2
vs microenvironment coding	protein-	green	GSE222703	601	3.21	-0.14	46.7	0.0005	81	0
vs microenvironment coding	protein-	green	Li2022	646	3.59	-0.15	50.9	0.0005	85	3
vs microenvironment coding	protein-	purple	GSE159115	216	-0.33	-0.00	-3.8	0.0015	6	25
vs microenvironment coding	protein-	purple	GSE207493	216	-0.31	-0.01	-3.0	0.0025	23	49
vs microenvironment coding	protein-	purple	GSE222703	154	-0.31	-0.08	-1.4	0.15	15	21
vs microenvironment coding	protein-	purple	Li2022	211	-0.52	-0.17	-2.6	0.011	13	31

836 Combined across cohorts, all 29 modules:

Contrast	Network	Module	Prognostic	Cohorts	Mean median log2FC	Stouffer z	FDR	Cohorts above null
vs normal epithelium	lncRNA	greenyellow	yes	3	2.76	9.9	1.3e-22	3
vs normal epithelium	lncRNA	turquoise	yes	3	0.44	2.4	0.022	3

Contrast	Network	Module	Prognostic	Cohorts	Mean median log2FC	Stouffer z	FDR	Cohorts above null
vs normal epithelium	lncRNA	blue	yes	3	-0.13	-2.1	0.0498	0
vs normal epithelium	protein-coding	purple	yes	3	0.94	14.7	5.2e-48	3
vs normal epithelium	protein-coding	green	yes	3	-0.45	-20.4	3.7e-91	0
vs normal epithelium	lncRNA	salmon	no	3	2.20	6.3	5.8e-10	3
vs normal epithelium	lncRNA	lightcyan	no	3	1.70	5.4	1.1e-07	3
vs normal epithelium	lncRNA	yellow	no	3	2.00	5.3	2.1e-07	3
vs normal epithelium	lncRNA	black	no	3	0.58	2.0	0.053	2
vs normal epithelium	lncRNA	green	no	3	0.96	1.8	0.084	3
vs normal epithelium	lncRNA	midnightblue	no	3	-0.08	-0.5	0.65	1
vs normal epithelium	lncRNA	purple	no	3	0.00	-0.7	0.57	1
vs normal epithelium	lncRNA	cyan	no	3	-0.31	-1.2	0.26	1
vs normal epithelium	lncRNA	pink	no	3	-0.14	-2.3	0.035	0
vs normal epithelium	lncRNA	red	no	3	-0.70	-2.6	0.014	0
vs normal epithelium	lncRNA	tan	no	3	-0.29	-2.6	0.014	0
vs normal epithelium	lncRNA	brown	no	3	-0.36	-3.6	0.00072	0
vs normal epithelium	lncRNA	magenta	no	3	-1.53	-7.5	2.5e-13	0
vs normal epithelium	protein-coding	yellow	no	3	0.70	22.3	3e-108	3
vs normal epithelium	protein-coding	tan	no	3	0.58	10.8	3.4e-26	3
vs normal epithelium	protein-coding	salmon	no	3	0.99	10.1	2e-23	3
vs normal epithelium	protein-coding	red	no	3	0.38	8.7	1.2e-17	3

Contrast	Network	Module	Prognostic	Cohorts	Mean median log2FC	Stouffer z	FDR	Cohorts above null
vs normal epithelium	protein-coding	brown	no	3	0.04	-0.4	0.71	1
vs normal epithelium	protein-coding	pink	no	3	0.06	-0.6	0.61	1
vs normal epithelium	protein-coding	black	no	3	0.15	-1.1	0.32	1
vs normal epithelium	protein-coding	blue	no	3	0.03	-3.0	0.0044	1
vs normal epithelium	protein-coding	turquoise	no	3	-0.02	-7.4	5e-13	0
vs normal epithelium	protein-coding	magenta	no	3	-0.37	-10.4	1.6e-24	0
vs normal epithelium	protein-coding	greenyellow	no	3	-0.95	-20.1	9.5e-89	0
vs microenvironment	lncRNA	blue	yes	4	4.39	37.4	4.7e-305	4
vs microenvironment	lncRNA	greenyellow	yes	4	1.67	7.3	5.1e-13	4
vs microenvironment	lncRNA	turquoise	yes	4	-0.12	-4.5	1e-05	0
vs microenvironment	protein-coding	green	yes	4	2.93	103.1	0	4
vs microenvironment	protein-coding	purple	yes	4	-0.37	-5.4	9.1e-08	0
vs microenvironment	lncRNA	lightcyan	no	4	5.18	16.7	1.8e-62	4
vs microenvironment	lncRNA	salmon	no	4	2.95	8.5	2.8e-17	4
vs microenvironment	lncRNA	pink	no	4	0.66	3.1	0.0028	4
vs microenvironment	lncRNA	tan	no	4	0.76	2.8	0.0068	4
vs microenvironment	lncRNA	cyan	no	4	0.97	2.3	0.027	4
vs microenvironment	lncRNA	purple	no	4	0.05	-1.1	0.26	1
vs microenvironment	lncRNA	brown	no	4	0.12	-2.2	0.034	1
vs microenvironment	lncRNA	magenta	no	4	-0.08	-2.2	0.031	0

Contrast	Network	Module	Prognostic	Cohorts	Mean median log2FC	Stouffer z	FDR	Cohorts above null
vs microenvironment	lncRNA	midnightblue	no	4	-0.26	-2.3	0.027	0
vs microenvironment	lncRNA	green	no	4	-0.70	-4.5	9.4e-06	0
vs microenvironment	lncRNA	black	no	4	-0.94	-7.9	4.4e-15	0
vs microenvironment	lncRNA	red	no	4	-1.58	-18.2	1.2e-73	0
vs microenvironment	lncRNA	yellow	no	4	-3.74	-35.7	4.2e-278	0
vs microenvironment	protein-coding	brown	no	4	0.38	17.6	1e-68	4
vs microenvironment	protein-coding	tan	no	4	0.36	4.2	3.8e-05	3
vs microenvironment	protein-coding	magenta	no	4	-0.14	-0.5	0.63	2
vs microenvironment	protein-coding	salmon	no	4	-0.56	-5.3	2.1e-07	1
vs microenvironment	protein-coding	red	no	4	-0.42	-7.3	4.1e-13	1
vs microenvironment	protein-coding	turquoise	no	4	-0.27	-10.0	2.7e-23	0
vs microenvironment	protein-coding	pink	no	4	-0.69	-11.6	8e-31	0
vs microenvironment	protein-coding	greenyellow	no	4	-1.24	-19.2	2.2e-81	0
vs microenvironment	protein-coding	black	no	4	-1.59	-37.7	0	0
vs microenvironment	protein-coding	blue	no	4	-1.56	-67.7	0	0
vs microenvironment	protein-coding	yellow	no	4	-2.98	-127.4	0	0

Note. Cohorts: GSE159115 (discovery cohort of the single-cell pipeline), Li 2022, GSE222703 and GSE207493 (microenvironment contrast only, no normal kidney). Contrasts are donor-level pseudobulk edgeR quasi-likelihood fits from the outcome-blind single-cell pipeline. The null for each module is 2,000 random same-size sets from the same gene class of the network universe,

841 among genes tested in that cohort, matched on deciles of log10 mean counts per million. The
842 empirical p is two-sided with a floor of 1/2,000. Members up or down are the percentage of
843 tested members with an upstream false discovery rate below 0.05 over all genes. The combined
844 false discovery rate is controlled across the 29 modules within each contrast. Combined z scores
845 overstate precision because genes within a module are correlated. Member coverage in at least
846 one cohort is in 36_sc_module_gene_coverage.tsv. Against normal epithelium it is 25%
847 for greenyellow, 44% for blue, 18% for turquoise, 97% for green and 74% for purple. Source
848 tables: 36_sc_module_localisation_per_dataset.tsv,
849 36_sc_module_localisation_combined.tsv,
850 36_sc_module_lncRNA_pooled_meta_summary.tsv.
851 ---

852 **Supplementary Table 20 | The locked prediction model**

853 **(a) Coefficients**

Term	Clinical comparator, coefficient (HR)	Clinical combined, coefficient (HR)	Augmented comparator, coefficient (HR)	Augmented combined, coefficient (HR)
Age, standardised	0.4011 (1.49)	0.4095 (1.51)	0.3936 (1.48)	0.3927 (1.48)
Male	-0.0171 (0.98)	-0.1383 (0.87)	0.0361 (1.04)	-0.0926 (0.91)
T category, 1 to 4	0.1418 (1.15)	0.1203 (1.13)	0.1274 (1.14)	0.1009 (1.11)
Node positive	0.8099 (2.25)	0.7823 (2.19)	0.6734 (1.96)	0.6604 (1.94)
M1	1.0394 (2.83)	1.0747 (2.93)	1.0343 (2.81)	1.0890 (2.97)
Ordinal grade, 1 to 4	0.3966 (1.49)	0.2886 (1.33)	0.4979 (1.65)	0.3532 (1.42)
ESTIMATE stromal, standardised	--	--	-0.0148 (0.99)	-0.0976 (0.91)
ESTIMATE immune, standardised	--	--	-0.0785 (0.92)	-0.1072 (0.90)
Non-feature fraction, standardised	--	--	0.2930 (1.34)	0.3141 (1.37)
Multimapping fraction, standardised	--	--	0.0661 (1.07)	0.0977 (1.10)

Term	Clinical comparator, coefficient (HR)	Clinical combined, coefficient (HR)	Augmented comparator, coefficient (HR)	Augmented combined, coefficient (HR)
log10 assigned reads, standardised	--	--	-0.0179 (0.98)	-0.0026 (1.00)
Protein-coding black eigengene	--	-0.0589 (0.94)	--	--
Protein-coding blue eigengene	--	-0.0096 (0.99)	--	--
Protein-coding green eigengene	--	-0.2774 (0.76)	--	-0.3285 (0.72)
Protein-coding pink eigengene	--	-0.0558 (0.95)	--	-0.1263 (0.88)
Protein-coding yellow eigengene	--	-0.1255 (0.88)	--	--
lncRNA black eigengene	--	0.0398 (1.04)	--	0.0729 (1.08)
lncRNA blue eigengene	--	-0.1171 (0.89)	--	-0.0486 (0.95)
lncRNA midnightblue eigengene	--	-0.0202 (0.98)	--	--
lncRNA turquoise eigengene	--	0.1982 (1.22)	--	0.1106 (1.12)

854 *Note.* Fitted on the 493 TCGA-KIRC patients of the locked prediction set, with 159 events.

855 Source: `22_locked_model_coefficients.tsv`. Comparator terms are unpenalised. The 29

856 module eigengenes are offered to an elastic net (mixing parameter 0.5) with the penalty at the

857 minimum of ten-fold cross-validated partial likelihood. The chosen penalty is 0.0254 for the

858 clinical arm (9 module scores retained) and 0.0220 for the augmented arm (5 retained).

859 Coefficients are per standard deviation of the standardised covariate or module score. Neither

860 arm retains lncRNA greenyellow, the lncRNA module with the strongest external estimate. A

861 separate full-cohort reporting fit in `05_final_model_coefficients.tsv` retains the same

862 nine module scores at a slightly different penalty, with every coefficient different. Validation,

863 calibration, Brier scores, decision curves and the endpoint analyses all use the model tabulated

864 here.

865 **(b) Module score definitions and retention**

Module score	Genes	Variance explained by PC1	Score scale	Retained, clinical arm	Retained, augmented arm
protein-coding black	323	0.4347	11.849	yes	no
protein-coding blue	994	0.3936	19.779	yes	no
protein-coding brown	988	0.3815	19.414	no	no
protein-coding green	670	0.3624	15.582	yes	yes
protein-coding greenyellow	164	0.4506	8.597	no	no
protein-coding magenta	227	0.4646	10.270	no	no
protein-coding pink	270	0.4521	11.048	yes	yes
protein-coding purple	216	0.4467	9.823	no	no
protein-coding red	473	0.3849	13.494	no	no
protein-coding salmon	63	0.4601	5.384	no	no
protein-coding tan	152	0.4813	8.553	no	no
protein-coding turquoise	2,405	0.3649	29.622	no	no
protein-coding yellow	986	0.4009	19.881	yes	no
lncRNA black	179	0.3408	7.810	yes	yes
lncRNA blue	246	0.2765	8.247	yes	yes
lncRNA brown	219	0.2654	7.624	no	no
lncRNA cyan	54	0.2912	3.966	no	no
lncRNA green	186	0.2614	6.972	no	no
lncRNA greenyellow	108	0.2505	5.201	no	no
lncRNA lightcyan	44	0.3138	3.716	no	no
lncRNA magenta	122	0.3070	6.120	no	no
lncRNA midnightblue	53	0.2893	3.916	yes	no
lncRNA pink	174	0.3057	7.294	no	no
lncRNA purple	110	0.3255	5.983	no	no
lncRNA red	186	0.2994	7.462	no	no
lncRNA salmon	72	0.2992	4.642	no	no
lncRNA tan	93	0.2983	5.267	no	no
lncRNA turquoise	438	0.2655	10.784	yes	yes
lncRNA yellow	192	0.3065	7.671	no	no

866 *Note.* Source: 22_module_score_constants.tsv. A module eigengene is the first principal

867 component of the module's gene-standardised residualised expression, sign-aligned to mean

module expression and divided by the score scale to give unit variance in discovery. The score centre is numerically zero in every row (of order 1e-16). Thirteen protein-coding and 16 lncRNA modules give 29 scores. The per-gene loadings, gene means and gene standard deviations that define each score are in Supplementary Data 2.

(c) Standardisation constants of the comparator covariates, discovery cohort

Covariate	Source	Mean	SD
Age, years	clinical record	61.1754	12.0387
ESTIMATE stromal score	ESTIMATE	842.4403	476.0470
ESTIMATE immune score	ESTIMATE	1683.9339	640.6478
Non-feature read fraction, %	STAR summary	5.1418	2.5660
Multimapping fraction, %	STAR summary	7.8333	1.7663
log10 assigned reads	STAR summary	7.7592	0.1078

Note. Source: 22_locked_model_standardisation.tsv. These constants define the discovery-fixed arm of the eigengene standardisation. They are not applied to the CPTAC-3 comparator covariates, which are z-scored within CPTAC-3 in both arms. Because of the protocol difference, the CPTAC-3 non-feature fraction lies far outside the discovery range.

(d) Discovery baseline survival

Comparator	Model	Mean discovery linear predictor	S0 at 1, 3, 5 years	S at the mean linear predictor, 1, 3, 5 years	Apparent C
clinical	comparator only	1.4891	0.9838 / 0.9551 / 0.9240	0.9300 / 0.8156 / 0.7043	0.766
clinical	combined	1.0812	0.9777 / 0.9367 / 0.8907	0.9356 / 0.8247 / 0.7108	0.787
augmented	comparator only	1.7604	0.9885 / 0.9675 / 0.9436	0.9352 / 0.8250 / 0.7134	0.775
augmented	combined	1.2453	0.9825 / 0.9491 / 0.9102	0.9405 / 0.8342 / 0.7211	0.792

Note. Source: 22_locked_model_baseline.tsv, 493 patients and 159 events. Absolute risk at time t is 1 minus $\exp(-H_0(t) \text{ times } \exp(LP - \text{mean } LP))$, where H_0 is the Breslow baseline

cumulative hazard estimated in discovery and LP is the locked linear predictor. No clinical test is proposed. The model is reported in full for evaluation, not for use. Its primary arm standardises the eigengenes within the scored cohort, so it cannot score a single patient. The discovery-fixed arm can score a single patient, at the cost of the two-fold over-prediction in panel (b) of Supplementary Table 21.

Supplementary Table 21 | External validation: discrimination, standardisation, calibration, Brier scores and clinical utility

(a) Concordance index of the locked model

Cohort	Comparator	Model	Standardisation	n	Events	C (95% CI)
TCGA-KIRC (apparent)	clinical	comparator only	cohort	493	159	0.766 (0.729 to 0.804)
TCGA-KIRC (apparent)	clinical	+ module eigengenes	cohort	493	159	0.787 (0.752 to 0.822)
CPTAC-3	clinical	comparator only	cohort	199	32	0.750 (0.657 to 0.843)
CPTAC-3	clinical	+ module eigengenes	cohort	199	32	0.759 (0.671 to 0.847)
CPTAC-3	clinical	comparator only	discovery-fixed	199	32	0.750 (0.657 to 0.843)
CPTAC-3	clinical	+ module eigengenes	discovery-fixed	199	32	0.754 (0.663 to 0.844)
TCGA-KIRC (apparent)	augmented	comparator only	cohort	493	159	0.775 (0.738 to 0.812)
TCGA-KIRC (apparent)	augmented	+ module eigengenes	cohort	493	159	0.792 (0.757 to 0.826)
CPTAC-3	augmented	comparator only	cohort	199	32	0.730 (0.642 to 0.818)
CPTAC-3	augmented	+ module eigengenes	cohort	199	32	0.745 (0.650 to 0.839)

Note. Source: 09_validation_cindex.tsv. The two TCGA-KIRC rows per comparator are apparent, scoring the locked model on the patients it was fitted to. Cohort standardisation, the primary arm, standardises the module eigengenes within the scored cohort. Discovery-fixed

891 standardisation applies the discovery gene-level and score-level constants. In both arms the
892 clinical, ESTIMATE and STAR covariates are standardised within CPTAC-3, and CPTAC-3
893 expression is residualised on CPTAC-3's own STAR metrics, so the discovery-fixed arm fixes
894 only the eigengene constants.

895 **(b) Calibration**

Cohort	Comparator	Standardisation	Model	Calibration slope (95% CI)
TCGA-KIRC (apparent)	clinical	cohort	comparator	1.00 (0.83 to 1.17)
TCGA-KIRC (apparent)	clinical	cohort	comparator + modules	1.05 (0.89 to 1.22)
CPTAC-3	clinical	cohort	comparator	0.97 (0.60 to 1.34)
CPTAC-3	clinical	cohort	comparator + modules	0.89 (0.58 to 1.20)
CPTAC-3	clinical	discovery-fixed	comparator	0.97 (0.61 to 1.34)
CPTAC-3	clinical	discovery-fixed	comparator + modules	0.89 (0.58 to 1.20)
TCGA-KIRC (apparent)	augmented	cohort	comparator	1.00 (0.84 to 1.16)
TCGA-KIRC (apparent)	augmented	cohort	comparator + modules	1.04 (0.88 to 1.19)
CPTAC-3	augmented	cohort	comparator	0.90 (0.51 to 1.28)
CPTAC-3	augmented	cohort	comparator + modules	0.84 (0.51 to 1.16)

896 Observed against expected risk at each horizon:

Cohort	Comparator	Standardisation	Model	1 year, expected (O/E)	3 years, expected (O/E)	5 years, expected (O/E)
TCGA-KIRC	clinical	cohort	comparator	0.094 (1.00)	0.225 (1.03)	0.335 (1.08)
TCGA-KIRC	clinical	cohort	+ modules	0.095 (0.99)	0.226 (1.02)	0.337 (1.08)
CPTAC-3	clinical	cohort	comparator	0.079 (0.71)	0.192 (0.75)	0.292 (0.90)
CPTAC-3	clinical	cohort	+ modules	0.083 (0.68)	0.194 (0.74)	0.295 (0.89)
CPTAC-3	clinical	discovery-fixed	comparator	0.079 (0.71)	0.191 (0.76)	0.290 (0.91)
CPTAC-3	clinical	discovery-fixed	+ modules	0.115 (0.49)	0.260 (0.55)	0.384 (0.68)
CPTAC-3	augmented	cohort	comparator	0.073 (0.77)	0.182 (0.79)	0.283 (0.93)

Cohort	Comparator	Standardisation	Model	1 year, expected (O/E)	3 years, expected (O/E)	5 years, expected (O/E)
CPTAC-3	augmented	cohort	+ modules	0.075 (0.75)	0.184 (0.79)	0.284 (0.93)

Note. Source: 12_calibration_summary.tsv. The calibration slope, the coefficient of the linear predictor in a Cox model of the outcome on that predictor, does not depend on the horizon, so one slope is reported per arm. Observed risk is the cohort-wide Kaplan-Meier estimate at the horizon: 0.094, 0.231 and 0.363 in TCGA-KIRC and 0.056, 0.144 and 0.263 in CPTAC-3. Patients at risk at 1, 3 and 5 years number 408, 275 and 141 in TCGA-KIRC and 144, 95 and 23 in CPTAC-3. The discovery TCGA-KIRC rows are apparent, and the comparator slope of exactly 1.00 there is an artefact of scoring a model on its training data. Both models over-predict risk in CPTAC-3, a lower-risk cohort, and under discovery-fixed standardisation the combined model over-predicts roughly two-fold.

(c) Inverse-probability-of-censoring-weighted Brier scores, lower is better

Cohort	Comparator	Standardisation	Years	Comparator	Comparator + modules	Kaplan-Meier null
TCGA-KIRC	clinical	cohort	1	0.0715	0.0675	0.0850
TCGA-KIRC	clinical	cohort	3	0.1443	0.1386	0.1779
TCGA-KIRC	clinical	cohort	5	0.1935	0.1795	0.2319
CPTAC-3	clinical	cohort	1	0.0578	0.0607	0.0536
CPTAC-3	clinical	cohort	3	0.0995	0.0979	0.1260
CPTAC-3	clinical	cohort	5	0.1392	0.1455	0.1950
CPTAC-3	clinical	discovery-fixed	1	0.0576	0.0671	0.0564
CPTAC-3	clinical	discovery-fixed	3	0.0994	0.1134	0.1370
CPTAC-3	clinical	discovery-fixed	5	0.1394	0.1625	0.2086
TCGA-KIRC	augmented	cohort	1	0.0711	0.0669	0.0850
TCGA-KIRC	augmented	cohort	3	0.1421	0.1385	0.1779
TCGA-KIRC	augmented	cohort	5	0.1826	0.1697	0.2321
CPTAC-3	augmented	cohort	1	0.0564	0.0574	0.0533
CPTAC-3	augmented	cohort	3	0.1073	0.1041	0.1251

Cohort	Comparator	Standardisation	Years	Comparator	Comparator + Kaplan-Meier modules	Comparator + null
CPTAC-3	augmented	cohort	5	0.1384	0.1417	0.1944

907 *Note.* TCGA-KIRC, 493 patients and 159 events. CPTAC-3, 199 patients and 32 events. Source:
908 `12_brier_scores.tsv`. The null model assigns every patient the cohort Kaplan-Meier risk at
909 the horizon. The TCGA-KIRC rows are apparent. In CPTAC-3 the combined model is worse
910 than the comparator at 1 and 5 years under cohort standardisation and at every horizon under
911 discovery-fixed standardisation. At 1 year both models are worse than the null.

912 **(d) Decision curves in CPTAC-3, baseline hazard fixed in discovery**

Comparator	Standardisation	Years	Threshold window	Thresholds	Median net-benefit gain	Positive thresholds	Bootstrap 95% CI	p
clinical	cohort	3	0.10 to 0.40	31	-0.0037	9	-0.0223 to 0.0061	0.42
clinical	cohort	3	0.05 to 0.30	26	-0.0005	11	-0.0162 to 0.0061	0.87
clinical	cohort	5	0.10 to 0.40	31	0.0026	19	-0.0183 to 0.0219	0.85
clinical	cohort	5	0.05 to 0.30	26	0.0016	16	-0.0075 to 0.0133	0.74
clinical	discovery-fixed	3	0.10 to 0.40	31	-0.0268	3	-0.0421 to 0.0004	0.048
clinical	discovery-fixed	3	0.05 to 0.30	26	-0.0147	5	-0.0322 to 0.0010	0.10
clinical	discovery-fixed	5	0.10 to 0.40	31	-0.0200	2	-0.0570 to 0.0104	0.22
clinical	discovery-fixed	5	0.05 to 0.30	26	-0.0071	2	-0.0286 to 0.0024	0.26
augmented	cohort	3	0.10 to 0.40	31	0.0043	19	-0.0083 to 0.0147	0.50
augmented	cohort	3	0.05 to 0.30	26	-0.0007	12	-0.0125 to 0.0087	0.80
augmented	cohort	5	0.10 to 0.40	31	-0.0026	14	-0.0217 to 0.0174	0.84
augmented	cohort	5	0.05 to 0.30	26	-0.0009	11	-0.0127 to 0.0112	0.82

913 *Note.* CPTAC-3, 199 patients and 32 events. Source:
 914 `12_validation_decision_curve_summary.tsv`. The median gain is the median, across
 915 the threshold window, of the net-benefit difference between the combined model and the
 916 comparator, with a paired patient bootstrap interval over 500 resamples. Every decision curve
 917 uses the discovery baseline hazard, and no arm re-estimates it in CPTAC-3. The only interval
 918 excluding zero does so in the harmful direction.

919 **(e) Drop-one-module sensitivity of the external increment, CPTAC-3**

Comparator and standardisation	Model	C	Increment	95% CI	p
clinical, cohort	all retained modules, 9	0.759	0.0087	-0.0331 to 0.0478	0.68
clinical, cohort	minus protein-coding black	0.757	0.0068	-0.0357 to 0.0466	0.77
clinical, cohort	minus protein-coding blue	0.758	0.0081	-0.0320 to 0.0465	0.71
clinical, cohort	minus protein-coding green	0.749	-0.0013	-0.0339 to 0.0320	0.96
clinical, cohort	minus protein-coding pink	0.755	0.0050	-0.0368 to 0.0438	0.84
clinical, cohort	minus protein-coding yellow	0.746	-0.0042	-0.0480 to 0.0370	0.82
clinical, cohort	minus lncRNA black	0.758	0.0084	-0.0304 to 0.0456	0.66
clinical, cohort	minus lncRNA blue	0.761	0.0113	-0.0258 to 0.0471	0.56
clinical, cohort	minus lncRNA midnightblue	0.758	0.0081	-0.0336 to 0.0464	0.70
clinical, cohort	minus lncRNA turquoise	0.754	0.0037	-0.0361 to 0.0447	0.87
clinical, cohort	minus all lncRNA modules	0.760	0.0105	-0.0192 to 0.0409	0.49
clinical, discovery-fixed	all retained modules	0.754	0.0039	-0.0383 to 0.0433	0.88
clinical, discovery-fixed	minus lncRNA blue	0.762	0.0121	-0.0255 to 0.0478	0.53
clinical, discovery-fixed	minus lncRNA turquoise	0.751	0.0010	-0.0411 to 0.0447	0.98
clinical, discovery-fixed	minus all lncRNA modules	0.762	0.0115	-0.0176 to 0.0419	0.43
augmented, cohort	all retained modules, 5	0.745	0.0147	-0.0192 to 0.0478	0.37
augmented, cohort	minus protein-coding green	0.728	-0.0021	-0.0256 to 0.0208	0.87
augmented, cohort	minus protein-coding pink	0.741	0.0105	-0.0269 to 0.0463	0.58
augmented, cohort	minus lncRNA black	0.750	0.0197	-0.0130 to 0.0518	0.23
augmented, cohort	minus lncRNA blue	0.745	0.0152	-0.0167 to 0.0459	0.33
augmented, cohort	minus lncRNA turquoise	0.744	0.0142	-0.0242 to 0.0503	0.44
augmented, cohort	minus all lncRNA modules	0.743	0.0126	-0.0227 to 0.0461	0.46

Note. CPTAC-3, 199 patients and 32 events, paired patient bootstrap over 2,000 resamples. Source: `20_bootstrap_delta_cindex.tsv`, which also holds the remaining drop-one rows under discovery-fixed standardisation. The comparator concordance index is fixed at 0.750 for the clinical arm and 0.730 for the augmented arm. No single module carries the external increment. Removing every lncRNA module leaves the increment as large or larger, so the small external increment is protein-coding. Every interval spans zero.

(f) Agreement between the two standardisations, per module eigengene, in CPTAC-3

Module score	Pearson r	Spearman rho	Mean under discovery-fixed standardisation	SD under discovery-fixed standardisation
protein-coding black	0.9999	0.9998	-0.06	0.95
protein-coding blue	0.9998	0.9996	-0.14	0.88
protein-coding brown	0.9997	0.9992	-1.47	0.45
protein-coding green	0.9999	0.9998	0.23	0.98
protein-coding greenyellow	0.9996	0.9992	2.27	0.78
protein-coding magenta	0.9997	0.9993	1.57	0.88
protein-coding pink	0.9979	0.9968	3.09	0.66
protein-coding purple	0.9999	0.9995	0.32	0.94
protein-coding red	0.9991	0.9981	1.01	0.80
protein-coding salmon	0.9999	0.9995	-0.66	0.94
protein-coding tan	0.9997	0.9995	-1.07	1.07
protein-coding turquoise	0.9993	0.9983	2.87	0.45
protein-coding yellow	0.9999	0.9998	0.66	0.89
lncRNA black	0.9974	0.9959	5.35	0.90
lncRNA blue	0.9988	0.9975	2.37	1.24
lncRNA brown	0.9977	0.9957	1.95	0.72
lncRNA cyan	0.9961	0.9954	3.85	1.19
lncRNA green	0.9972	0.9958	11.65	0.90
lncRNA greenyellow	0.9973	0.9964	1.43	1.12
lncRNA lightcyan	0.9860	0.9828	2.32	1.45
lncRNA magenta	0.9973	0.9929	2.41	0.98
lncRNA midnightblue	0.9864	0.9734	6.63	0.61

Module score	Pearson r	Spearman rho	Mean under discovery-fixed standardisation	SD under discovery-fixed standardisation
lncRNA pink	0.9978	0.9969	1.68	0.89
lncRNA purple	0.9933	0.9918	1.35	0.49
lncRNA red	0.9986	0.9986	2.59	1.30
lncRNA salmon	0.9930	0.9933	3.92	1.29
lncRNA tan	0.9968	0.9950	2.79	1.06
lncRNA turquoise	0.9978	0.9971	4.64	0.83
lncRNA yellow	0.9989	0.9985	3.00	1.26

Note. CPTAC-3, 199 patients. Source: `20_score_spread_by_standardisation.tsv`. The two standardisations rank patients almost identically (Pearson correlation at least 0.986 for every module), but the discovery constants displace the CPTAC-3 means by up to 11.7 standard deviations. This offset is why discovery-fixed standardisation barely changes the concordance index in panel (a) yet destroys calibration in panel (b).

(g) External association estimates in CPTAC-3 under the two standardisations

Module	HR, cohort standardisation (95% CI)	FDR	HR, discovery-fixed standardisation (95% CI)	FDR
Protein-coding green	0.73 (0.51 to 1.04)	0.14	0.73 (0.52 to 1.04)	0.14
Protein-coding purple	1.52 (1.08 to 2.14)	0.042	1.52 (1.08 to 2.14)	0.042
lncRNA blue	0.78 (0.55 to 1.11)	0.21	0.78 (0.54 to 1.11)	0.20
lncRNA greenyellow	1.59 (1.12 to 2.25)	0.042	1.59 (1.12 to 2.25)	0.042
lncRNA turquoise	1.11 (0.81 to 1.53)	0.51	1.12 (0.81 to 1.54)	0.49

Note. CPTAC-3, 199 patients and 32 events, twelve-parameter principal specification, module membership and eigengene loadings fixed in discovery. Source: `20_replication_unitmatched.tsv`. Benjamini-Hochberg control is across the five modules within each arm. The two standardisations differ by at most 0.006 in the hazard ratio, for lncRNA turquoise. This is the association analysis of Table 2b of the main text, not the fixed prediction rule of panels (a) to (f).

939 **Supplementary Table 22 | Specimen handling time in CPTAC-3**

940 **(a) Spearman correlation of handling time with the alignment metrics, the projected**

941 **lncRNA axis and tumour severity**

Sample set	Handling time	Variable	Expected direction	n	Spearman rho (95% CI)	p	FDR
CPTAC-3 tumours	excision to freezing	Non-feature fraction	positive	175	-0.02 (-0.16 to 0.12)	0.79	
CPTAC-3 tumours	excision to freezing	Projected lncRNA axis	positive	175	-0.09 (-0.23 to 0.06)	0.43	
CPTAC-3 tumours	excision to freezing	Multimapping fraction	--	175	-0.04 (-0.18 to 0.10)	0.72	
CPTAC-3 tumours	excision to freezing	log10 assigned reads	--	175	-0.04 (-0.19 to 0.10)	0.72	
CPTAC-3 tumours	excision to freezing	T category	positive	174	0.14 (-0.02 to 0.28)	0.06	0.14
CPTAC-3 tumours	excision to freezing	Ordinal stage	positive	175	0.15 (0.00 to 0.29)	0.04	0.14
CPTAC-3 tumours	excision to freezing	Ordinal grade	--	175	0.32 (0.18 to 0.44)	1.9e-05	1.3e-04
CPTAC-3 tumours	clamping to freezing	Non-feature fraction	positive	35	0.16 (-0.20 to 0.50)	0.36	0.82
CPTAC-3 tumours	clamping to freezing	Projected lncRNA axis	positive	35	-0.26 (-0.56 to 0.07)	0.13	0.82
CPTAC-3 tumours	clamping to freezing	Multimapping fraction	--	35	-0.08 (-0.43 to 0.28)	0.66	0.82
CPTAC-3 tumours	clamping to freezing	log10 assigned reads	--	35	0.08 (-0.27 to 0.38)	0.67	0.82
CPTAC-3 tumours	clamping to freezing	T category	positive	35	-0.07 (-0.39 to 0.26)	0.70	0.82
CPTAC-3 tumours	clamping to freezing	Ordinal stage	positive	35	-0.11 (-0.42 to 0.23)	0.54	0.82
CPTAC-3 tumours	clamping to freezing	Ordinal grade	--	35	0.00 (-0.28 to 0.30)	0.99	0.99
CPTAC-3 normal tissue	excision to freezing	Non-feature fraction	positive	168	-0.10 (-0.25 to 0.06)	0.21	0.41
CPTAC-3 normal tissue	excision to freezing	Projected lncRNA axis	positive	168	0.01 (-0.14 to 0.16)	0.93	0.93
CPTAC-3 normal tissue	excision to freezing	Multimapping fraction	--	168	0.07 (-0.09 to 0.21)	0.38	0.51
CPTAC-3 normal tissue	excision to freezing	log10 assigned reads	--	168	0.24 (0.09 to 0.38)	0.002	0.008

(b) Handling time and overall survival, CPTAC-3 tumours

Model	n	Events	HR per SD (95% CI)	p	Events per variable
excision-to-freezing time alone	175	23	1.45 (0.96 to 2.19)	0.08	23.0
+ age, sex, T category, M1, ordinal grade	174	23	1.18 (0.76 to 1.83)	0.45	3.8

Note. Handling times are the Genomic Data Commons sample fields for the time from excision and from clamping to freezing, in minutes, taken from the sample whose library was sequenced. Excision to freezing is recorded for 175 of the 200 validation tumours (median 16, range 3 to 30, the CPTAC protocol cap) and all 168 normal-tissue libraries, and clamping to freezing for 35 tumours. Intervals are from 2,000 bootstrap resamples, and Benjamini-Hochberg control is within each sample set and handling-time variable. Median excision-to-freezing time is 12, 14, 17 and 21 minutes for grades 1 to 4. The 25 tumours without a recorded time account for 9 of the 32 deaths, so the survival estimates in panel (b) rest on 23 events. Source tables:

45_cptac_handling_time_correlations.tsv,
 45_cptac_handling_time_survival.tsv,
 45_cptac_handling_time_per_sample.tsv.

Supplementary Data

Five Supplementary Data items, supplied as tab-separated files, accompany this Supplementary Information.

Supplementary Data 1 | Case list and file manifest

958 `SupplementaryData1_case_list.tsv`, 1,549 rows and 25 columns, one row per STAR-
959 Counts expression file in the GDC manifest of each of the four projects. The 25 columns are, in
960 groups:

- 961 • `cohort` and `role`, the project and its part in the study.
- 962 • `patient`, `sample_barcode` and `sample_type`, the submitter identifiers and the GDC
963 sample type.
- 964 • `file_id`, `file_name`, `gdc_md5sum`, `gdc_file_size`, `gdc_data_release`,
965 `gdc_metadata_fetched`, `gdc_state`, `gdc_workflow_version`,
966 `gdc_created_datetime` and `gdc_updated_datetime`, the GDC record of the file.
- 967 • `local_file_present`, `local_file_size`, `local_file_date`, `local_md5sum`
968 and `local_md5_matches_gdc`, the local copy and its verification.
- 969 • `assigned_reads`, the STAR library size used throughout the pipeline.
- 970 • `retained_in_analysed_cohort` and `exclusion_reason`, whether the file supplied
971 the expression profile of one patient in an analysed cohort and, if not, why not.
- 972 • `in_adjusted_models` and `in_prediction_set`, the two downstream analysis sets.

973 `SupplementaryData1_column_definitions.tsv` is a 25-row data dictionary giving the
974 type, definition and applicable cohorts of every column.

975 All metadata were read from GDC Data Release 46.0 of 10 August 2026. Local copies exist for
976 1,491 of the 1,549 files, and all 1,491 md5 checksums match the GDC record. The other 58 files,
977 the subtype normal libraries and one additional new primary, were never downloaded.

Cohort	Role	Files listed	Patients retained	In adjusted models	In the prediction set
TCGA-KIRC	discovery	614	528	520	493
CPTAC-3	validation	521	200	199	199
TCGA-KIRP	subtype	323	269	263	not applicable

Cohort	Role	Files listed	Patients retained	In adjusted models	In the prediction set
TCGA-KICH	subtype	91	65	65	not applicable
All cohorts	--	1,549	1,062	1,047	692

Exclusions by reason, TCGA-KIRC: non-primary sample 73, failed library 5, duplicate aliquot 4, non-positive or missing survival time 4. CPTAC-3: non-primary sample 168, duplicate aliquot 92 files from 40 patients, non-clear-cell histology 38, non-positive or missing survival time 23. TCGA-KIRP: non-primary sample 33, non-positive or missing survival time 3, missing age or sex 18. TCGA-KICH: non-primary sample 25, non-positive or missing survival time 1. The all-cohorts patient total of 1,150 in the source table is de-duplicated, because one patient can appear in more than one row. The file count and the retained-patient count are simple sums. Source: 31_supplementary_data1_summary.tsv.

Supplementary Data 2 | Module gene lists, module membership and eigengene loadings

Three files. 02_lncRNA_module_genes.tsv (3,442 rows) and 02_mRNA_module_genes.tsv (12,000 rows) give one row per analysed gene. Their columns are gene_id (versioned GENCODE v36 Ensembl identifier), gene_name, module (WGCNA colour label, grey for genes assigned to no module) and kME (correlation of the gene with its own module eigengene). 22_module_loadings.tsv gives one row per gene per module, 10,407 rows over the 29 non-grey modules. Its columns are module (mRNA_ME<colour> or lnc_ME<colour>), gene_id, gene_name, loading (element of the first principal component of the module's gene-standardised residualised discovery expression), gene_center (discovery mean of the gene on the log2(FPKM + 1) scale) and gene_scale (its discovery standard deviation). To score a new

998 sample, each gene is standardised with its `gene_center` and `gene_scale`, the inner product is
999 taken with `loading`, and the result is divided by the `score_scale` in Supplementary Table
1000 20b. The three files fully specify every module eigengene.

1001 **Supplementary Data 3 | Survival results for all 29 modules**

1002 `03_all_modules_principal.tsv`, 29 rows, one per non-grey module of either network
1003 under the principal specification. The columns are:

- 1004 • `biotype`, `module`, `n_genes`, `n` and `events`.
- 1005 • `HR`, `lo` and `hi`, the hazard ratio per standard deviation with its 95% confidence limits,
1006 and `p`.
- 1007 • `fdr_within`, the Benjamini-Hochberg false discovery rate within the module's own
1008 network of 13 or 16 tests, and `fdr_joint`, the same across all 29 modules of both
1009 networks.
- 1010 • `ph_p_ME` and `ph_p_global`, the proportional-hazards `p` values for the eigengene term
1011 and for the global test.

1012 Five modules reach a 5% false discovery rate under either family. This file is also the source of
1013 Table 2 of the main text.

1014 **Supplementary Data 4 | Gene Ontology enrichment of the protein-coding modules**

1015 `06_GO_enrichment_all_modules.tsv`, 472 rows, one per enriched biological-process term
1016 per module. The columns are `module`, `ID`, `Description`, `GeneRatio`, `BgRatio`, `pvalue`,
1017 `p.adjust`, `Count` and `geneID`, the slash-separated list of module genes annotated to the term.
1018 `clusterProfiler::enrichGO` was run for every module significant under the continuity

1019 specification, against a universe of the 12,000 protein-coding genes in the network, with
1020 Benjamini-Hochberg adjustment and cut-offs of 0.05 for p and 0.10 for q. Three modules return
1021 enriched terms: the catabolic module (229 terms), the proliferation module (232) and the splicing
1022 module (11).

1023 **Supplementary Data 5 | Pan-cancer file manifest**

1024 `38_pancancer_file_manifest.tsv`, 11,505 rows, one per STAR-Counts file of the 33
1025 TCGA projects at Data Release 46.0, with the columns `file_id`, `project`, `file_name`, `md5`,
1026 `sample_type`, `group` and `reason` (retained, failed library, duplicate library of a retained
1027 patient, or not a primary tumour or solid-tissue normal). Every retained file was verified against
1028 its md5 checksum.